Class Field Theory

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Chapter 1

Local Fields: Basics

A **local field** is a finite extension of either $\mathbb{F}_p((t))$ (the field of formal Laurent series with coefficients in the finite field $\mathbb{F}_p$) or $\mathbb{Q}_p$ (the field of p-adic numbers). They are arithmetically interesting topological fields, behaving somewhere between $\mathbb{Q}$ and $\mathbb{R}$, so that we will be able to simultaneously talk about limits and prime elements for the very same ring! On the other hand, in terms of arithmetic complexity (more precisely, in terms of the complexity of their absolute Galois groups), local fields lie strictly between finite fields (simpler) and number fields (harder). That is exactly what makes the study of local fields so attractive: it allows us to obtain information about global fields (such as $\mathbb{Q}$ and $\mathbb{F}_p(t)$) at a much cheaper price. For instance, finding solutions to polynomial equations over $\mathbb{F}_p((t))$ or $\mathbb{Q}_p$ can often be reduced to the much simpler task of finding solutions to their reductions modulo t or p.

1.1 The starting point: $\mathbb{F}_p((t))$ and $\mathbb{Q}_p$

Let p be a prime and let $\mathbb{F}_p$ be the finite field with p elements. Consider the ring $\mathbb{F}_p[\![t]\!]$ of formal power series with coefficients in $\mathbb{F}_p$:

$$a_0 + a_1 t + a_2 t^2 + \cdots, \qquad a_i \in \mathbb{F}_p$$

Addition and multiplication are performed in the usual way as with polynomials. For instance, one has

$$(1-t)\cdot(1+t+t^2+t^3+\cdots) = 1$$

The principal ideal (t) is a maximal ideal of $\mathbb{F}_p[\![t]\!]$ with residue field $\mathbb{F}_p[\![t]\!]/(t) \cong \mathbb{F}_p$. Also the group of units of $\mathbb{F}_p[\![t]\!]$ is given by

$$\boxed{\mathbb{F}_p[\![t]\!]^\times = \left\{a_0 + a_1 t + a_2 t^2 + \cdots \in F_p[\![t]\!] \;\middle|\; a_0 \neq 0\right\}}$$

In fact, if

$$(a_0 + a_1 t + a_2 t^2 + \cdots)(b_0 + b_1 t + b_2 t^2 + \cdots) = 1$$

has a solution in the b_i's, one must have $a_0 b_0 = 1$ and hence $a_0 \neq 0$, and conversely if $a_0 \neq 0$ then one can recursively set

$$\begin{cases} b_0 &= a_0^{-1} \text{ and} \\ b_n &= -a_0^{-1}(a_n b_0 + a_{n-1} b_1 + \cdots + a_1 b_{n-1}) \quad \text{for } n \geqslant 1. \end{cases}$$

In other words, the complement of the maximal ideal (t), namely the set of power series $a_0 + a_1 t + a_2 t^2 + \cdots$ with nonzero constant term $a_0 \neq 0$, consists solely of units, and therefore (t) is the unique maximal ideal of $\mathbb{F}_p[\![t]\!]$, which is thus a **local ring**. The ring $\mathbb{F}_p[\![t]\!]$ is also a UFD and, even better, a **discrete valuation ring** (dvr for short; see the appendix for the definition, and specially theorem 5.2.7), as every nonzero element $f \in \mathbb{F}_p[\![t]\!]$ admits a rather simple prime factorisation

$$\boxed{f = \underbrace{t^n}_{\substack{\text{power of} \\ \text{uniformiser } t}} \times \underbrace{(a_n + a_{n+1} t + a_{n+2} t^2 + \cdots)}_{\text{unit in } \mathbb{F}_p[\![t]\!]}, \quad a_n \neq 0}$$

for some $n \geqslant 0$. As a consequence, the nonzero elements of the fraction field

$$\mathbb{F}_p((t)) \overset{\text{def}}{=} \operatorname{Frac} \mathbb{F}_p[\![t]\!] = \mathbb{F}_p[\![t]\!]\left[\frac{1}{t}\right]$$

also admit a similar factorisation with $n \in \mathbb{Z}$, and thus elements of $\mathbb{F}_p((t))$ can be identified with "Laurent power series" $f = \sum_{i \geqslant i_0} a_i t^i$, $i_0 \in \mathbb{Z}$. The **discrete valuation** v on $\mathbb{F}_p((t))$ associated to the dvr $\mathbb{F}_p[\![t]\!]$ is given by

$$\boxed{\begin{aligned} v(f) &= n \in \mathbb{Z} \text{ such that } f \text{ has prime factorisation } f = t^n \cdot u,\ u \in \mathbb{F}_p[\![t]\!]^\times \\ &= \min\{n \in \mathbb{Z} \mid a_n \neq 0\} \end{aligned}}$$

for $f = \sum_{i \geqslant i_0} a_i t^i \in \mathbb{F}_p((t))^\times$, and $v(0) = \infty$ if $f = 0$. The discretely valued field $\mathbb{F}_p((t))$ is our first example of a **local field**.
Being a discretely valued field, $\mathbb{F}_p((t))$ is also a normed field with norm given by (see appendix)

$$|f|_v = 2^{-v(f)} \qquad \text{for} \quad f \in \mathbb{F}_p((t))$$

(Here 2 denotes your favourite real > 1). One has the following rule of the thumb for the topology defined by $|-|_v$: the power series $a_0 + a_1 t + a_2 t^2 + \cdots$ is convergent, and thus its general term should approach 0:

$$\boxed{\lim_{n \to \infty} t^n = 0}$$

This limit is clear from the definition of $|-|_v$. For example, for any $f \in \mathbb{F}_p((t))$ one has the following "derivative rule"

$$\lim_{n \to \infty} \frac{(f + t^n)^5 - f^5}{t^n} = 5 f^4$$

Observe that $\mathbb{F}_p((t))$ contains $\mathbb{F}_p(t) \overset{\text{def}}{=} \operatorname{Frac} \mathbb{F}_p[t]$ as a dense subfield since any power series $f = \sum_{i \geqslant i_0} a_i t^i$ can be written as a limit of rational functions (for instance, of the "truncations" $\sum_{i_0 \leqslant i \leqslant n} a_i t^i$ of f). This is analogous to the fact that $\mathbb{R}$ contains $\mathbb{Q}$ as a dense subfield as any real number is as a limit of rational numbers.

Much of the intuition from Analysis can be borrowed for the study of $\mathbb{F}_p((t))$ (and other local fields). The good news is that, thanks to the **strong triangle inequality** (see appendix)

$$|f+g|_v \leqslant \max\{|f|_v, |g|_v\},$$

Analysis on $\mathbb{F}_p((t))$ turns out to be **much easier** compared to $\mathbb{R}$ or $\mathbb{C}$! For instance, one has the following amazing

Lemma 1.1.1 (Calculus Student's Psychedelic Dream)**.** *Let $f_n \in \mathbb{F}_p((t))$, $n \geqslant 0$. Then*

$$f_0 + f_1 + f_2 + f_3 + \cdots \textit{ converges in } \mathbb{F}_p((t)) \iff \lim_{n\to\infty} f_n = 0.$$

Proof. ($\Longrightarrow$) is clear as it holds for any metric space. Conversely, assume that $\lim_{n\to\infty} f_n = 0$, which amounts to the real limit $\lim_{n\to\infty} v(f_n) = \infty$. Simply put, for a fixed n there are only finitely many initial terms in $f_0 + f_1 + f_2 + \cdots$ contributing with a non-zero coefficient of t^n, thus $f_0 + f_1 + f_2 + \cdots$ gives a well-defined element $F \in \mathbb{F}_p((t))$. A routine check (using the definitions) now shows that such F is indeed the v-limit of the partial sums $f_0 + \cdots + f_m$. $\square$

The above lemma is quite useful in explicit computations. For instance, to find the multiplicative inverse of $1 + t + t^2$, say, we just have to apply the usual formula for the sum of a geometric progression:

$$\frac{1}{1+t+t^2} = 1 - (t+t^2) + (t+t^2)^2 - (t+t^2)^3 + \cdots$$

(a convergent series as its general term has valuation $v\big((t+t^2)^n\big) = n \to \infty$.)

A very important consequence of the Psychedelic Dream lemma is that $\mathbb{F}_p((t))$ is actually a **complete discretely valued field**: every Cauchy sequence in $\mathbb{F}_p((t))$ converges. In fact, if $\{f_n\}_{n\geqslant 0}$ is Cauchy then $\lim_{n\to\infty}(f_{n+1} - f_n) = 0$ and therefore

$$\lim_{n\to\infty} f_n = f_0 + \sum_{n\geqslant 0}(f_{n+1} - f_n)$$

converges, as desired.

Remark 1.1.2. *Conversely, any complete discretely valued field K with valuation v satisfies the Calculus Student's Psychedelic Dream: if $\lim_{n\to\infty} f_n = 0$ then to show that $\sum_{n\geqslant 0} f_n$ converges, by completeness we must show that the partial sums $\sum_{0\leqslant n\leqslant N} f_n$ form a Cauchy sequence. But this is clear:*

$$\Big|\sum_{M\leqslant n\leqslant N} f_n\Big|_v \leqslant \max\big\{|f_n|_v \;\big|\; M \leqslant n \leqslant N\big\}$$

can be made arbitrarily small by choosing M sufficiently large as $\lim_{n\to\infty} f_n = 0$.

Before leaving the realm of $\mathbb{F}_p((t))$, we wish to give an alternative but rather useful description of $\mathbb{F}_p[\![t]\!]$ as a **projective limit** of the finite discrete rings $\mathbb{F}_p[t]/(t^n)$ (check the appendix if you are unfamiliar with limits). Namely, there is an algebraic and topological isomorphism

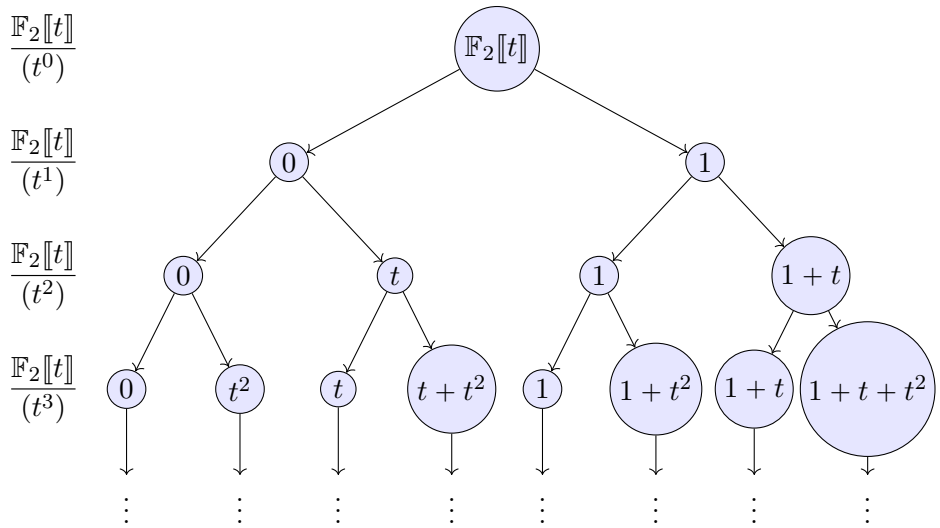
$$\boxed{\mathbb{F}_p[\![t]\!] \simeq \varprojlim_{n\in\mathbb{N}} \frac{\mathbb{F}_p[t]}{(t^n)} \overset{\text{def}}{=} \left\{ (f_n) \in \prod_{n\in\mathbb{N}} \frac{\mathbb{F}_p[t]}{(t^n)} \;\middle|\; f_m = f_n \bmod t^m \text{ for all } n \geqslant m \right\}}$$

given by

$$a_0 + a_1 t + a_2 t^2 + \cdots \mapsto (a_0 \bmod t,\ a_0 + a_1 t \bmod t^2,\ a_0 + a_1 t + a_2 t^2 \bmod t^3, \ldots)$$

Under this isomorphism the projection maps $\phi_n \colon \mathbb{F}_p[\![t]\!] \to \mathbb{F}_p[t]/(t^n)$ become just the "truncations" given by $a_0 + a_1 t + \cdots \mapsto a_0 + \cdots + a_{n-1}t^{n-1} \bmod t^n$. A good way to picture this limit (or any other projective limit for the matter) is to think of it as the set of "infinite paths in an $\mathbb{F}_p[t]/(t^n)$-layered rooted tree". For instance, for $p = 2$, consider the following tree whose vertices in the n-th level are the elements of $\mathbb{F}_2[t]/(t^n)$, and where we connect an element to its image in the previous level. Then the series $a_0 + a_1 t + a_2 t^2 + \cdots \in \mathbb{F}_2[\![t]\!]$ will correspond to the path on the tree given by

$$(a_0 \bmod t,\ a_0 + a_1 t \bmod t^2,\ a_0 + a_1 t + a_2 t^2 \bmod t^3, \ldots) \in \varprojlim_{n\in\mathbb{N}} \mathbb{F}_2[t]/(t^n).$$



$\mathbb{F}_2[\![t]\!]$ as a projective limit

Observe that $\prod_{n\in\mathbb{N}} \mathbb{F}_p[t]/(t^n)$ is compact by Tychonoff's theorem, and hence $\mathbb{F}_p[\![t]\!]$, being a closed subspace of this product, is also compact (this can also be proven directly by the original description of $\mathbb{F}_p[\![t]\!]$, try!). Therefore any element $f \in \mathbb{F}_p((t))$ has a compact neighbourhood $f + \mathbb{F}_p[\![t]\!] = \{g \in \mathbb{F}_p((t)) \mid |g - f|_v < 2\}$. We have shown that

Lemma 1.1.3. *$\mathbb{F}_p((t))$ is a locally compact complete discretely valued field with finite residue field $\mathbb{F}_p$ (imagine how this mouthful would look like written in German!).*

Enough about $\mathbb{F}_p((t))$ for now. Next we introduce the second main example of a local field. By analogy with the above, we define the **ring of p-adic integers** $\mathbb{Z}_p$ as the projective limit of the finite discrete rings $\mathbb{Z}/(p^n)$:

$$\boxed{\mathbb{Z}_p \stackrel{\text{def}}{=} \varprojlim_{n\in\mathbb{N}} \frac{\mathbb{Z}}{(p^n)} = \left\{ (f_n) \in \prod_{n\in\mathbb{N}} \frac{\mathbb{Z}}{(p^n)} \;\middle|\; f_m = f_n \bmod p^m \text{ for all } n \geqslant m \right\}}$$

We may choose unique integers F_n with $0 \leqslant F_n < p^n$ representing $f_n \in \mathbb{Z}/(p^n)$. Writing F_n in base p

$$F_n = a_0 + a_1 p + a_2 p^2 + \cdots + a_{n-1}p^{n-1}, \qquad 0 \leqslant a_i < p$$

we find that for $m \leqslant n$ we must have $F_m = a_0 + a_1 p + \cdots + a_{m-1}p^{m-1}$, i.e., one obtains F_m by "truncating" F_n. Hence a p-adic integer corresponds **uniquely** to a sequence of integers of the form

$$(a_0,\ a_0 + a_1 p,\ a_0 + a_1 p + a_2 p^2, \ldots) \qquad 0 \leqslant a_i < p$$

which is usually written as an "infinite base p representation"

$$a_0 + a_1 p + a_2 p^2 + \cdots \qquad \text{with} \quad 0 \leqslant a_i < p$$

Computations with such p-adic numbers can be done as we did before for $\mathbb{F}_p[\![t]\!]$, except that one has to pay attention to the "carry 1". For instance in $\mathbb{Z}_2$ one has that

$$1 + 2 + 2^2 + 2^3 + \cdots = \frac{1}{1-2} = -1$$

by the usual formula for the sum of a geometric series! From a different perspective, adding 1 to $1 + 2 + 2^2 + 2^3 + \cdots$ we obtain

$$\begin{aligned} 1 + (1 + 2 + 2^2 + 2^3 + \cdots) &= 2 + 2 + 2^2 + 2^3 + \cdots \\ &= 2^2 + 2^2 + 2^3 + \cdots \\ &= 2^3 + 2^3 + \cdots \\ &= \cdots = 0 \end{aligned}$$

We obtain an "infinite" sequence of "carry 1's" and the end result is zero! But wait, is that licit or sheer nonsense? Going back to the original definition, everything becomes clear:

$$\begin{gathered} 1 \in \mathbb{Z}_2 \text{ corresponds to } (1 \bmod 2, 1 \bmod 2^2, 1 \bmod 2^3, \ldots) \\ 1 + 2 + 2^2 + \cdots \in \mathbb{Z}_2 \text{ corresponds to } (1 \bmod 2, 1 + 2 \bmod 2^2, 1 + 2 + 2^2 \bmod 2^3, \ldots), \end{gathered}$$

hence their sum (done componentwise in each ring $\mathbb{Z}/(p^n)$) is indeed zero. The punchline is: **if a computation with the "infinite base p representation" works modulo p^n for all n then it works**. To keep our mental sanity, most of the time we shall work with the much more

intuitive "base p representation" of a p-adic integer, leaving it to the sceptical reader the chore of translating the statements back into the projective limit definition of $\mathbb{Z}_p$.

In the same way that $\mathbb{F}_p[t] \subset \mathbb{F}_p[\![t]\!]$ we have that a containment $\mathbb{Z} \subset \mathbb{Z}_p$ given by the "diagonal embedding"

$$\begin{aligned} \mathbb{Z} &\hookrightarrow \mathbb{Z}_p \\ a &\mapsto (a \bmod p^n)_{n\in\mathbb{N}} \end{aligned}$$

(simply put, a finite base p representation is a special case of an infinite one). Also, as with $\mathbb{F}_p[\![t]\!]$, there is a very simple description of the units of $\mathbb{Z}_p$ as those p-adic integers with "non-zero constant term":

$$\boxed{\mathbb{Z}_p^\times = \left\{a_0 + a_1 p + a_2 p^2 + \cdots \in \mathbb{Z}_p \;\middle|\; a_0 \neq 0, \quad 0 \leqslant a_i < p\right\}}$$

This is clear since $a_0 + a_1 p + \cdots + a_{n-1}p^{n-1} \bmod p^n$, $0 \leqslant a_i < p$, is invertible in $\mathbb{Z}/(p^n)$ if and only if $a_0 \neq 0$. Therefore $\mathbb{Z}_p^\times$ is precisely the complement of the principal ideal

$$(p) = p\mathbb{Z}_p = \{a_1 p + a_2 p^2 + \cdots \mid 0 \leqslant a_i < p\},$$

which is maximal since $\mathbb{Z}_p/(p) = \mathbb{F}_p$. To sum up, $\mathbb{Z}_p$ is also a local ring with finite residue field $\mathbb{F}_p$. And it is also a discrete valuation ring, since any element $f \in \mathbb{Z}_p$ has a prime factorisation

$$\boxed{f = \underbrace{p^n}_{\substack{\text{power of}\\ \text{uniformiser } p}} \times \underbrace{(a_n + a_{n+1}p^{n+1} + a_{n+2}p^{n+2} + \cdots)}_{\text{unit in } \mathbb{Z}_p} \qquad a_n \neq 0}$$

for some $n \geqslant 0$, with $0 \leqslant a_i < p$. Hence elements of the field of fractions

$$\mathbb{Q}_p \overset{\text{def}}{=} \operatorname{Frac} \mathbb{Z}_p = \mathbb{Z}_p\left[\frac{1}{p}\right]$$

can be written as "Laurent power series" $\sum_{i\geqslant i_0} a_i p^i$, $0 \leqslant a_i < p$, and the valuation on $\mathbb{Q}_p$ associated to the dvr $\mathbb{Z}_p$ (called **p-adic valuation**) is given by

$$\boxed{\begin{aligned} v(f) &= n \in \mathbb{Z} \text{ such that } f \text{ has prime factorisation } f = p^n \cdot u,\ u \in \mathbb{Z}_p^\times \\ &= \min\{n \in \mathbb{Z} \mid a_n \neq 0\} \end{aligned}}$$

for a nonzero element $f = \sum_{i\geqslant i_0} a_i p^i$, $0 \leqslant a_i < p$.

The valuation defines a norm $|f|_p \overset{\text{def}}{=} p^{-v(f)}$, and the topology on $\mathbb{Q}_p$ is now expressed by the rule of the thumb

$$\boxed{\lim_{n\to\infty} p^n = 0}$$

It is easy to check that this v-topology on $\mathbb{Z}_p$ coincides with the induced topology of $\mathbb{Z}_p$ as a subspace of the compact product $\prod_{n\in\mathbb{N}} \mathbb{Z}/(p^n)$. Hence $\mathbb{Z}_p$ is **compact** and thus $\mathbb{Q}_p$ is **locally compact**. And since any element of $\mathbb{Q}_p$ is a limit of rational numbers (for instance of the truncations of its base p expansion), we have that $\mathbb{Q}$ is contained as a dense subfield of $\mathbb{Q}_p$. Finally, Calculus Student's Psychedelic Dream holds in $\mathbb{Q}_p$ and $\mathbb{Q}_p$ is a complete, virtually by the very same proofs for $\mathbb{F}_p((t))$. To sum up,

Lemma 1.1.4. *$\mathbb{Q}_p$ is a locally compact complete discretely valued field with finite residue field $\mathbb{F}_p$.*

$\mathbb{Q}_p$ is our second main example of a local field. Note that while $\operatorname{char} \mathbb{F}_p((t)) = p$ we have that $\operatorname{char} \mathbb{Q}_p = 0$ (since $\mathbb{F}_p \subset \mathbb{F}_p((t))$ and $\mathbb{Q} \subset \mathbb{Q}_p$).

1.2 Hensel's lemma and applications

Hensel's lemma is the most important tool in the study of local fields. In a nutshell, it reduces the task of finding roots to polynomial equations over $\mathbb{F}_p((t))$ or $\mathbb{Q}_p$ to that over the residue field $\mathbb{F}_p$. For a single polynomial of one variable, the punchline is

$$\boxed{\text{Hensel: “Simple roots in the residue field lift”}}$$

Or, in the general case of multiple polynomials in several variables,

$$\boxed{\text{Hensel: “Smooth points in the residue field lift”}}$$

Here are precise statements:

Lemma 1.2.1 (Hensel)**.** *Let K be a complete valued field with valuation ring A. Let $\mathfrak{m}$ be the maximal ideal of A and $k = A/\mathfrak{m}$ be its residue field. Let $m \leqslant n$ and consider polynomials $f_1, \ldots, f_m \in A[x_1, \ldots, x_n]$. Write $\bar{f}_i$ for the image of f_i in $A[x_1, \ldots, x_n]/\mathfrak{m}A[x_1, \ldots, x_n] = k[x_1, \ldots, x_n]$. Suppose that we are given a* **smooth point** $\mathbf{c} = (c_1, \ldots, c_n) \in k^n$ *in the variety cut out by the system of equations $\bar{f}_i(\mathbf{x}) = 0$, i.e.,*

(i) $\bar{f}_i(\mathbf{c}) = 0$ *for all i and*

(ii) $m = \operatorname{rank}\left(\frac{\partial \bar{f}_i}{\partial x_j}(\mathbf{c})\right)_{\substack{1\leqslant i\leqslant m\\ 1\leqslant j\leqslant n}}$

(in other words the codimension of the variety equals the rank of the Jacobian matrix)

Then $\mathbf{c}$ lifts to a point $\mathbf{a} = (a_1, \ldots, a_n) \in A^n$ of the the variety cut out by the system of equations $f_i(\mathbf{x}) = 0$, i.e., $f_i(\mathbf{a}) = 0$ for all $1 \leqslant i \leqslant m$ and $c_j = a_j \bmod \mathfrak{m}$ for all $1 \leqslant j \leqslant n$.

Applying the above with $n = m = 1$, we immediately get the following

Corollary 1.2.2. *Let $(A, \mathfrak{m}, k)$ as above. Let $f(x) \in A[x]$, and $\overline{f}(x) \in k[x]$ be its reduction modulo $\mathfrak{m}$. If $c \in k$ is a simple root of $\overline{f}(x)$, i.e., $\overline{f}(c) = 0$ but $\overline{f}'(c) \neq 0$, then c lifts to a root $a \in A$ of $f(x)$, i.e., $f(a) = 0$ with $c = a \bmod \mathfrak{m}$.*

Before delving into the proof, let us give the general idea, which is based on the classical **Newton's method** for numerically finding the roots of a polynomial equation (see picture below). For instance, in the case $n = m = 1$, we construct a sequence of elements $c_0, c_1, c_2, \dots \in A$, converging to a root of $f(x)$. First we pick an arbitrary lift $c_0 \in A$ of $c \in k$. The Taylor expansion of $f(x)$ around $x = c_0$ is

$$f(x) = f(c_0) + f'(c_0)(x - c_0) + \text{higher order terms}$$

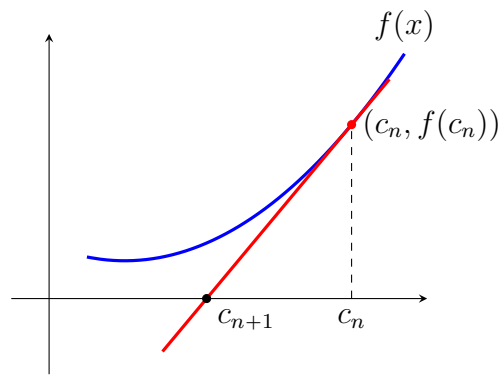
Newton's method defines the next approximation $c_1 \in A$ to a root of $f(x)$ by setting the linear part of this expression to 0:

$$0 = f(c_0) + f(c_0) + f'(c_0)(c_1 - c_0) \iff c_1 = c_0 - \frac{f(c_0)}{f'(c_0)}$$

This very same expression also gives a better approximation to a root of $f(x)$: since $\overline{f}(c) = 0$ and $\overline{f}'(c) \neq 0$, we have that $f(c_0) \in \mathfrak{m}$ and $f'(c_0) \notin \mathfrak{m}$ is a unit in A, so by the above expression defining c_1 we have $c_1 \equiv c_0 \pmod{\mathfrak{m}}$. Moreover

$$\begin{aligned} f(c_1) &= f(c_1) - \big(f(c_0) + f'(c_0)(c_1 - c_0)\big) \\ &= \text{sum of Taylor terms divisible by } (c_1 - c_0)^2 \equiv 0 \pmod{\mathfrak{m}^2} \end{aligned}$$

so now $f(c_1)$ is $\mathfrak{m}$-adically closer to 0 than $f(c_0)$. Now repeat to get $c_2, c_3, \dots$



Newton's method: $c_{n+1} = c_n - \dfrac{f(c_n)}{f'(c_n)}$

Proof of lemma 1.2.1. As with Newton's method, we inductively construct a sequence of "approximate" roots $\mathbf{c}_i = (c_{i1}, \dots, c_{in}) \in (A/\mathfrak{m}^i)^n$, $i = 1, 2, \dots$, such that

(a) the point $\mathbf{c}_i \in (A/\mathfrak{m}^i)^n$ belongs to the variety cut out by the f_r, that is, $f_r(\mathbf{c}_i) = 0$ for all r (here and in the following we will indistinctively write f_r for its image in $A[x_1, \dots, x_n]/\mathfrak{m}^i A[x_1, \dots, x_n]$ as it will be clear from the context in which ring we are working);

(b) for all $i \leqslant j$, we have that $\mathbf{c}_i = \mathbf{c}_j \bmod \mathfrak{m}^i$ (which of course means that $c_{it} = c_{jt} \bmod \mathfrak{m}^i$ holds for each coordinate, $1 \leqslant t \leqslant n$)

The sequence $\mathbf{c}_i$ will then define a point $\mathbf{a} \in A^n = \varprojlim (A/\mathfrak{m}^i)^n$ in the variety cut out by the f_r. We begin by setting $\mathbf{c}_1 = \mathbf{c}$. For $i \geqslant 2$, let $\tilde{\mathbf{c}}_{i-1} \in (A/\mathfrak{m}^i)^n$ be any lift of $\mathbf{c}_{i-1} \in (A/\mathfrak{m}^{i-1})^n$ and consider the system in $\mathbf{y}_i = (y_1, \ldots, y_n)$

$$\begin{pmatrix} \frac{\partial f_1}{\partial x_1}(\tilde{\mathbf{c}}_{i-1}) & \ldots & \frac{\partial f_1}{\partial x_n}(\tilde{\mathbf{c}}_{i-1}) \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1}(\tilde{\mathbf{c}}_{i-1}) & \ldots & \frac{\partial f_m}{\partial x_n}(\tilde{\mathbf{c}}_{i-1}) \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} f_1(\tilde{\mathbf{c}}_{i-1}) \\ \vdots \\ f_m(\tilde{\mathbf{c}}_{i-1}) \end{pmatrix} \tag{$*$}$$

Modulo $\mathfrak{m}$ the Jacobian matrix has full rank m and thus by relabelling the x_j we may assume that the first m columns are, modulo $\mathfrak{m}$, linearly independent over k. Hence the determinant of the $m \times m$ minor given by the first m columns is a unit in $A/\mathfrak{m}^i$, and therefore we may set $y_{m+1} = \cdots = y_n = 0$ and solve for the first m variables. Now define

$$\mathbf{c}_i = \tilde{\mathbf{c}}_{i-1} - \mathbf{y}_i$$

Notice that when $m = n = 1$ this reduces to the formula of Newton's method.

Now we need to show that the $\mathbf{c}_i$'s satisfy our requirements (a) and (b). Assume by induction that $f_r(\mathbf{c}_{i-1}) = 0$ for all r. Since $\tilde{\mathbf{c}}_{i-1}$ lifts $\mathbf{c}_{i-1}$, to show (b) we have to show that $\mathbf{y}_i \bmod \mathfrak{m}^{i-1} = (0, \ldots, 0)$. But this is clear since $y_{m+1} = \cdots = y_n = 0$ and modulo $\mathfrak{m}^{i-1}$ the system $(*)$ is homogeneous and non-singular in the first m variables. This proves that $\mathbf{c}_{i-1} = \mathbf{c}_i \pmod{\mathfrak{m}^{i-1}}$. Finally, to show (a), from Taylor's formula we have that

$$\begin{aligned} \begin{pmatrix} f_1(\mathbf{c}_i) \\ \vdots \\ f_m(\mathbf{c}_i) \end{pmatrix} &= \begin{pmatrix} f_1(\tilde{\mathbf{c}}_i - \mathbf{y}_i) \\ \vdots \\ f_m(\tilde{\mathbf{c}}_i - \mathbf{y}_i) \end{pmatrix} \\ &= \begin{pmatrix} f_1(\tilde{\mathbf{c}}_i) \\ \vdots \\ f_m(\tilde{\mathbf{c}}_i) \end{pmatrix} - \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(\tilde{\mathbf{c}}_{i-1}) & \ldots & \frac{\partial f_1}{\partial x_n}(\tilde{\mathbf{c}}_{i-1}) \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1}(\tilde{\mathbf{c}}_{i-1}) & \ldots & \frac{\partial f_m}{\partial x_n}(\tilde{\mathbf{c}}_{i-1}) \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} + q(\mathbf{y}) \\ &= q(\mathbf{y}) \qquad \text{by } (*) \end{aligned}$$

where $q(\mathbf{y})$ is a vector whose components are sums of forms in $y_1, \ldots, y_n$ of degree at least 2. But $\mathbf{y}_i \bmod \mathfrak{m}^{i-1} = (0, \ldots, 0)$ and since $i \geqslant 2$ this implies that $q(\mathbf{y})$ vanishes in $(A/\mathfrak{m}^i)^n$. $\square$

Remark 1.2.3. *Given some extra hypotheses, we can also lift some "singular points". We have the following stronger version of Hensel's lemma, due to Tougeron, and whose proof is a variant of the above (left as an exercise for the reader, of course!)*

With the above notation, suppose that there is a point $\mathbf{a}_0 \in A^n$ such that

$$f_r(\mathbf{a}_0) \equiv 0 \pmod{\mathfrak{m} \cdot \delta^2}$$

for all $r = 1, \ldots, m$, where $\delta \in A$ denotes the determinant of the $m \times m$ minor $\left(\frac{\partial f_i}{\partial x_j}(\mathbf{a}_0)\right)_{1 \leqslant i,j \leqslant m}$ of the Jacobian. Then there exists a point $\mathbf{a} \in A^n$ in the variety cut out by the f_r with $\mathbf{a}_0 \equiv \mathbf{a} \pmod{\mathfrak{m} \cdot \delta^2}$.

1.2.1 Hensel in action

Let us see how we can use Hensel's lemma in practice.

Example 1.2.4 (Squares in Local Fields)**.** *Let K be either $\mathbb{Q}_p$ or $\mathbb{F}_p((t))$ with p an odd prime. Denote by A its valuation ring and let π be a uniformiser. Let $a \in A$ be a nonzero element and write it $a = \pi^n u$ with $u \in A^\times$. Then*

$$a \text{ is a square in } A \iff n \text{ is even and } u \bmod \pi \text{ is a square in } \mathbb{F}_p.$$

$\implies$ is clear: any square $a = \pi^n u$ has even valuation n with u a square in A, and thus $u \bmod \pi$ must be a square in $\mathbb{F}_p$ as well. Conversely, considering the polynomial $f(x) = x^2 - u$, if there exists $v_0 \in \mathbb{F}_p$ such that $u \bmod \pi = v_0^2$ then $f(v_0) = 0$ and $f'(v_0) = 2v_0 \neq 0$ in $\mathbb{F}_p$ (remember that $p \neq 2$). Hence by Hensel's lemma we may lift v_0 to a root of $f(x)$ in A, showing that u is a square. Therefore a is a square too since n is even.
From the above, one immediately obtains a group isomorphism

$$K^\times/(K^\times)^2 = \{1, u, \pi, u\pi \bmod (K^\times)^2\} \cong \mathbb{Z}/2 \times \mathbb{Z}/2$$

where $u \bmod \pi$ is not a quadratic residue in $\mathbb{F}_p$. In particular, we conclude that there are exactly 3 quadratic extensions of K (in some fixed algebraic closure of K). This is in stark contrast with a global field, such as $\mathbb{Q}$ or $\mathbb{F}_p[t]$, which have infinitely many non-isomorphic quadratic extensions (obtained, for instance, by adjoining the square roots of different prime elements).
A more careful analysis (exercise!) shows that the group $\mathbb{Q}_2^\times/(\mathbb{Q}_2^\times)^2$ is isomorphic to $\mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/2$, with generators given by the images of $-1, 2, 3$.

Next we give an application of Hensel's lemma to quadratic forms. First we need some background in quadratic forms over finite fields.

Example 1.2.5 (Quadratic Forms over Finite Fields)**.** *Let q be a power of an odd prime. Then any quadratic form with at least 3 variables over $\mathbb{F}_q$ is* **isotropic***, i.e., it represents zero non-trivially. In fact, we may assume that the quadratic form is of the form $ax^2 + by^2 + z^2$ with $a, b \neq 0$. Since $\mathbb{F}_q^\times$ is cyclic, there are exactly $(q-1)/2$ squares in $\mathbb{F}_q^\times$ and therefore the sets*

$$\{ax^2 \mid x \in \mathbb{F}_q\} \qquad \text{and} \qquad \{-by^2 - 1 \mid y \in \mathbb{F}_q\}$$

have both cardinality $(q+1)/2$. Hence they must intersect non-trivially, yielding a nontrivial solution of $ax^2 + by^2 + z^2 = 0$. On the other hand, there exist **anisotropic** *quadratic forms in 2 variables: just take any non-square $u \in \mathbb{F}_q$, and consider the quadratic form $x^2 - uy^2$.*

Now we use Hensel's lemma to derive a similar result over a local field.

Example 1.2.6 (Quadratic Forms over Local Fields)**.** *Let K be either $\mathbb{Q}_p$ or $\mathbb{F}_p((t))$ with p an odd prime. Let A be its valuation ring and π be a uniformiser. Let u be a non-square unit so that $\{1, u, \pi, u\pi\}$ are representatives of the elements in $K^\times/(K^\times)^2$. We claim that the quadratic form*

$$\phi(w, x, y, z) = w^2 - u \cdot x^2 - \pi \cdot y^2 + u\pi \cdot z^2$$

is anisotropic. In fact, suppose that $\phi(w, x, y, z) = 0$ has a nontrivial solution (w_0, x_0, y_0, z_0). Multiplying this solution by a convenient power of π we may assume that $w_0, x_0, y_0, z_0 \in A$ and at least one of them is a unit. Now we have two cases. If either w_0 or x_0 is a unit, then reducing $\bmod \pi$, we

obtain that the quadratic form $w^2 - \bar{u}x^2$ over $\mathbb{F}_p$ is isotropic, a contradiction. On the other hand, if both w_0 and x_0 are multiples of π, say $w_0 = \pi w_0'$ and $x_0 = \pi x_0'$ with $w_0', x_0' \in A$, then $\pi({w_0'}^2 - u{x_0'}^2) - (y_0^2 - uz_0^2) = 0$. But now either y_0 or z_0 is a unit, and we get a contradiction as in the previous case.

Now we show that every quadratic form $\phi(x_1, \ldots, x_n) = a_1x_1^2 + \cdots + a_nx_n^2$ over K in $n \geqslant 5$ variables is isotropic. Since $\{1, u, \pi, u\pi\}$ represent the elements in $K^\times/(K^\times)^2$, in order to show that ϕ is isotropic we may assume that each $a_i \in \{1, u, \pi, u\pi\}$. Hence we may write $\phi = \psi_1 + \pi \cdot \psi_2$ where ψ_1 and ψ_2 are quadratic forms whose coefficients are units in A. Hence it is enough to show that any quadratic form in 3 variables whose coefficients are units is isotropic.

Wlog let $\psi(x, y, z) = ux^2 + vy^2 + z^2$ be such a quadratic form where $u, v \in A^\times$ are arbitrary units. By the previous example, we have that the congruence $\psi(x, y, z) \equiv 0 \pmod{\pi}$ has a nontrivial solution $(x_0, y_0, z_0) \not\equiv (0, 0, 0) \pmod{\pi}$. But then

$$\left(\frac{\partial\psi}{\partial x}(x_0), \frac{\partial\psi}{\partial y}(y_0), \frac{\partial\psi}{\partial z}(z_0)\right) = (2x_0, 2y_0, 2z_0) \not\equiv (0, 0, 0) \pmod{\pi}$$

and therefore Hensel's lemma implies that $\psi(x, y, z) = 0$ has a nontrivial solution, as required.

A more careful analysis (left to the reader, of course!) shows that for $K = \mathbb{Q}_2$ the result still holds: every quadratic form over $\mathbb{Q}_2$ in 5 or more variables is isotropic.

Example 1.2.7 (Quaternion Algebras)**.** *In the preceding example, the fact that $\phi(w, x, y, z) = w^2 - u \cdot x^2 - \pi \cdot y^2 + u\pi \cdot z^2$ is anisotropic allows us to construct a nontrivial quaternion algebra over the local field K: just take*

$$\left(\frac{u, \pi}{K}\right) \stackrel{\text{def}}{=} \{a + bi + cj + dij \mid a, b, c, d \in K, \quad i^2 = u, \quad j^2 = \pi, \quad ij = -ji\}$$

This is a division algebra since the reduced norm of $a + bi + cj + dij$ is

$$\phi(a, b, c, d) = (a + bi + cj + dij)(a - bi - cj - dij) = a^2 - ub^2 - \pi c^2 + u\pi d^2$$

which is never zero for $a + bi + cj + dij \neq 0$ as we have seen. Therefore we have that any $a + bi + cj + dij \neq 0$ is invertible:

$$(a + bi + cj + dij)^{-1} = \frac{a - bi - cj - dij}{\phi(a, b, c, d)}$$

Later we will see that this the only nontrivial quaternion algebra over K up to isomorphism. Again, this is in stark contrast with the global field case, for which there are always infinitely many non-isomorphic quaternion algebras (but is close to the real case: there is just one nontrivial quaternion algebra over $\mathbb{R}$).

Example 1.2.8 (Roots of Unity)**.** *Here we show that the group of roots of unity of $\mathbb{Q}_p$ has order $p - 1$ for p odd. Consider $f(x) = x^{p-1} - 1 \in \mathbb{Z}_p[x]$. Since the image $\bar{f}(x) \in \mathbb{F}_p[x]$ of $f(x)$ splits completely and $\bar{f}'(r_0) = -r_0^{p-2} \neq 0$ for any root $r_0 \in \mathbb{F}_p$ of $\bar{f}(x) = 0$ (namely any $r_0 \in \mathbb{F}_p^\times$), by Hensel's lemma each element of $\mathbb{F}_p^\times$ lifts to a root of $f(x)$ (these are called* **Teichmüller lifts***), and hence $f(x)$ splits completely in $\mathbb{Z}_p[x]$. Hence $\mathbb{Q}_p$ contains all the $(p - 1)$-th roots of unity.*

Next we show that $\mathbb{Q}_p$ does not contain any primitive n-th root of unity with $p \mid n$. It is enough to show that the cyclotomic polynomial $g(x) = x^{p-1} + x^{p-2} + \cdots + 1$ is irreducible over $\mathbb{Q}_p[x]$, but this follows from the usual combination of Gauß' lemma and Eisenstein's criterion applied to $g(x+1)$.

Finally, assume that $p \nmid n$ and that there exists a primitive n-th root of unity $\zeta \in \mathbb{Q}_p$. We show that $n \mid (p-1)$. First, observe that ζ has valuation 0 and hence $\zeta \in \mathbb{Z}_p^\times$. Second, we claim that reduction modulo p is injective when restricted to the subgroup of $\mathbb{Z}_p^\times$ generated by ζ. In fact, if $\zeta^i \equiv 1 \pmod p$ but $\zeta^i \neq 1$ then we would get a contradiction $0 = \zeta^{i(n-1)} + \zeta^{i(n-2)} + \cdots + \zeta^i + 1 \equiv n \pmod p$. Therefore $\zeta \bmod p$ has order n in $\mathbb{F}_p^\times$. But by "Fermat's little theorem" we also have $\zeta^{p-1} \equiv 1 \pmod p$, and it follows that $n \mid (p-1)$.

A variation of the above argument (again left to the reader) shows that ± 1 are the only roots of unity in $\mathbb{Q}_2$.

Example 1.2.9 (Automorphisms of $\mathbb{Q}_p$)**.** *It is easy to show that any* **continuous** *field automorphism ϕ of $\mathbb{Q}_p$ has to be the identity: since ϕ restricts to the identity on $\mathbb{Q}$, if $(a_n)_{n\geqslant 1}$ is a sequence of rational numbers converging to any given $a \in \mathbb{Q}_p$ we have that $\phi(a) = \lim_{n\to\infty} \phi(a_n) = \lim_{n\to\infty} a_n = a$. However it is not so easy to show that* **any** *field automorphism of $\mathbb{Q}_p$ is in fact trivial. To prove that, we first show that $a \in \mathbb{Z}_p^\times$ if and only if the equation $x^n = a^{p-1}$ can be solved in $\mathbb{Q}_p$ for infinitely many $n \geqslant 1$. In fact, if a is a unit then $a^{p-1} \equiv 1 \pmod p$ and Hensel's lemma shows that $x_0 = 1 \bmod p$ lifts to a solution of $x^n = a^{p-1}$ for all n not divisible by p. Conversely, denote by v the p-adic valuation. Since $x^n = a^{p-1}$ implies that $n \mid (p-1) \cdot v(a)$, if this equation can be solved for infinitely many n then necessarily $v(a) = 0$.*

Hence if ϕ is a field automorphism of $\mathbb{Q}_p$ it must take units to units. Writing an element $a \in \mathbb{Q}_p^\times$ as $a = p^n u$ with $u \in \mathbb{Z}_p^\times$ and using the fact that ϕ restricts to the identity on $\mathbb{Q}$, we have that $\phi(a) = p^n u'$ for some unit u'. Hence ϕ preserves the valuation and is therefore continuous, and so it has to be the identity.

1.2.2 Hensel's lemma and separable factorisations

Finally we show how one may refine Hensel's lemma in order not only to lift roots but also factorisations. The punchline is:

$$\boxed{\text{Hensel: "Separable factorisations in the residue field lift"}}$$

Lemma 1.2.10 (Hensel, revisited)**.** *Let K be a complete valued field with valuation ring A. Let $\mathfrak{m}$ be the maximal ideal and $k = A/\mathfrak{m}$ be the residue field of A. Denote the image of a polynomial $p \in A[x]$ in $A[x]/\mathfrak{m}A[x] = k[x]$ by $\bar{p}$. Let $f(x) \in A[x]$ with $\bar{f} \neq 0$ and suppose that $\bar{f}$ factors as*

$$\bar{f}(x) = g_0(x)h_0(x) \qquad \textit{with} \qquad \gcd\big(g_0(x), h_0(x)\big) = 1$$

Then there exist polynomials $g(x), h(x) \in A[x]$ with $\deg g(x) = \deg g_0(x)$ lifting the above factorisation: $f(x) = g(x)h(x)$ with $\bar{g}(x) = g_0(x)$ and $\bar{h}(x) = h_0(x)$.

Proof. Write $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$, $g(x) = b_r x^r + b_{r-1}x^{r-1} + \cdots + b_0$ and $h(x) = c_s x^s + c_{s-1}x^{s-1} + \cdots + c_0$ where the b_i, c_i are indeterminates and $r = \deg g_0(x)$ and $s = n - r$. Expanding $f(x) = g(x)h(x)$ we obtain $n+1$ equations $\sum_{i+j=d} b_i c_j = a_d$, $0 \leqslant d \leqslant n$, in the $(r+1) + (s+1) = n+2$

variables b_i, c_j. The Jacobian matrix of this system is the $(n+1) \times (n+2)$ matrix

$$\begin{pmatrix} b_0 & 0 & 0 & \cdots & 0 & c_0 & 0 & 0 & \cdots & 0 \\ b_1 & b_0 & 0 & \cdots & 0 & c_1 & c_0 & 0 & \cdots & 0 \\ b_2 & b_1 & b_0 & \cdots & 0 & c_2 & c_1 & c_0 & \cdots & 0 \\ \vdots & & & & & \vdots & & & & \\ 0 & 0 & 0 & \cdots & b_0 & 0 & 0 & 0 & \cdots & \vdots \\ & & & & \vdots & & & & & \\ 0 & 0 & 0 & \cdots & b_r & 0 & 0 & 0 & \cdots & c_s \end{pmatrix}$$

Now the coefficients of $g_0(x)$ and $h_0(x)$ give a solution to this system modulo $\mathfrak{m}$, which is a smooth point since the rank of Jacobian matrix evaluated at this point is $n+1$: the determinant of the first $n+1$ columns is non-zero since b_r assumes a non-zero value (recall that $r = \deg g_0(x)$) and the $n \times n$ matrix obtained by suppressing the r-th column and the last line is the resultant of the relatively prime polynomials $g_0(x)$ and $h_0(x)$. $\square$

Corollary 1.2.11. *Keep the above notation and let v be the valuation of K. If*

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

is an irreducible polynomial in $K[x]$ then

$$\min_{0 \leqslant i \leqslant n} v(a_i) = \min\{v(a_0), v(a_n)\}$$

In particular, if $a_n = 1$ and $a_0 \in A$ then $a_i \in A$ for all i.

Proof. Choose the smallest integer i for which $v(a_i)$ is minimal and let us show that $0 < i < n$ yields a contradiction. We have that $p(x) \stackrel{\text{def}}{=} a_i^{-1} \cdot f(x) \in A[x]$ is such that $\bar{p} \neq 0$ in $k[x]$. But then we have a factorisation $\bar{p}(x) = x^i g_0(x)$ with $g_0(0) = 1$ and hence x^i and $g_0(x)$ are relatively prime. By Hensel's lemma $p(x)$ factors in $A[x]$ non-trivially (since $0 < i < n$), which is impossible since $f(x) = a_i \cdot p(x)$ is irreducible in $K[x]$. $\square$

1.3 Local fields in general

By now you may be wondering: what is a local field after all?

Definition 1.3.1. *A finite field extension L of either $\mathbb{F}_p((t))$ or $\mathbb{Q}_p$ is called a* **local field**.

Example 1.3.2. *Let q be a power of a prime p. Then $\mathbb{F}_q((t))$ is a local field. By example 1.2.4 there are exactly 3 local fields of degree 2 over $\mathbb{Q}_p$ when p is odd, and 7 for $p = 2$.*

Example 1.3.3. *Although we won't cover global fields in these notes, it is instructive to show how local fields arise from them. For instance, take the ring of Gaußian integers $\mathbb{Z}[i]$ and the maximal ideal (3), with residue field $\mathbb{Z}[i]/(3) \cong \mathbb{F}_9$. Then*

$$A = \varprojlim_{n \in \mathbb{N}} \frac{\mathbb{Z}[i]}{(3^n)} = \left\{ (a_n) \in \prod_{n \in \mathbb{N}} \frac{\mathbb{Z}[i]}{(3^n)} \;\middle|\; a_m = a_n \bmod 3^m \text{ for all } n \geqslant m \right\}$$

is a local domain with a principal maximal ideal (3)*. Moreover* $\mathbb{Z}[i] \subseteq A$ *by "diagonal embedding"* $a \mapsto (a, a, \ldots)$*. The unit group of* A *is* $A^\times = A - (3)$*, i.e. the subset of* A *consisting of tuples* $(a_1, a_2, \ldots)$ *with* $a_1 \neq 0 \bmod 3$*, and for* $f \in A$ *one has a prime factorisation* $f = 3^n u$ *with* $u \in A^\times$ *so that setting* $w(f) = n$ *defines a valuation* w *on the fraction field* $K = \operatorname{Frac} A$*. As with* $\mathbb{Q}_p$*, this makes* K *into a complete valued field. Moreover, choosing a set of representatives* S *of* $\mathbb{Z}[i]/(3) \cong \mathbb{F}_9$*, for instance* $S = \{a + bi \mid 0 \leqslant a, b < 3\}$*, we may uniquely write each element of* K *as a convergent power series* $\sum_{n \geqslant n_0} s_n 3^n$ *with* $s_n \in S$*.*
We now show that K *is a local field, actually a quadratic extension of* $\mathbb{Q}_3$*. First observe that* A *contains a copy of* $\mathbb{Z}_3$ *since* $\mathbb{Z}/3^n$ *is a subring of* $\mathbb{Z}[i]/(3^n)$ *for all* n*. Hence* $K \supseteq \mathbb{Q}_3$*. Besides* $A = \mathbb{Z}_3[i]$ *(recall that* $\mathbb{Z}[i] \subseteq A$*). But* $i^2 = -1$ *and* $i \notin \mathbb{Q}_3$ *since the image of* $x^2 + 1$ *is irreducible in* $\mathbb{F}_3[x]$*, so* $x^2 + 1$ *is irreducible in* $\mathbb{Z}_3[x]$ *and hence in* $\mathbb{Q}_3[x]$ *by Gauß' lemma. Therefore* $K = \mathbb{Q}_3(i)$ *is the quadratic extension of* $\mathbb{Q}_3$ *generated by a root of* $x^2 + 1$*.*

The reason why we spent so much time looking at the two basic examples of the previous section is that virtually all the difficulty in the study of a general local field is already present in those two particular instances. This is no accident: we now show that a general local field L is structurally similar to $K = \mathbb{F}_p((t))$ or $K = \mathbb{Q}_p$ in that it is a complete discretely valued field. The first step is to show how to extend the valuation v of K to a valuation w of L. In fact, there is no wiggle room: the extension is **unique**.

Valuative Highlander's Philosophy: "there can be only one [valuation]".

Granting this result, we can easily "guess" a formula for w. For simplicity, assume that L is Galois over K (the general case can be reduced to this one by considering the Galois closure of L). Then for any $\sigma \in \operatorname{Gal}(L/K)$ we have that $w \circ \sigma = w$ since both are valuations extending v. But then for $x \in L$ we have that

$$w\big(N_{L/K}(x)\big) = w\Big(\prod_\sigma \sigma x\Big) = [L : K] \cdot w(x) \implies w(x) = \frac{1}{[L : K]} \cdot v\big(N_{L/K}(x)\big)$$

Our approach in proving the theorem below will be the opposite one: we will use the "guessed" formula to show the existence of an extension.

Theorem 1.3.4 (Valuative Highlander's Philosophy)**.** *Let* K *be a complete discretely valued field with valuation* v*, and* L *be a finite extension of* K*. Then there is a unique valuation* w *on* L *extending* v*. It is given by*

$$\boxed{w(x) = \frac{1}{[L : K]} \cdot v\big(N_{L/K}(x)\big) \qquad \text{for } x \in L}$$

Moreover L *is complete with respect to* w*.*

Proof. Let A be the valuation ring of K and let B be the integral closure of A in L (check the appendix for a list of basic results about integral extensions). We first show that

$$B = \{x \in L \mid N_{L/K}(x) \in A\} = \{x \in L \mid v\big(N_{L/K}(x)\big) \geqslant 0\}$$

Since A is a UFD, it is normal, and hence $N_{L/K}(b) \in A$ for $b \in B$, i.e., $B \subseteq \{x \in L \mid N_{L/K}(x) \in A\}$. To prove the opposite inclusion, take $x \in L$ with $N_{L/K}(x) \in A$ and let $f(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_0 \in K[t]$ be its minimal polynomial. Since $N_{L/K}(x) = \pm a_0^m$ for some $m > 0$ we must have $v(a_0) \geqslant 0 \iff a_0 \in A$. By corollary 1.2.11 we conclude that $f(t) \in A[t]$, i.e., that $x \in B$, as required.

Next we show that the above formula for w defines a valuation on L (with valuation ring equal to B). Clearly $w(xy) = w(x) + w(y)$ and $w(x) = \infty \iff x = 0$, so we just have to show that $w(x+y) \geqslant \min\{w(x), w(y)\}$ for $x, y \in L$. First observe that we already know the special case $w(x) \geqslant 0 \implies w(1+x) \geqslant 0$. In fact:

$$w(x) = \frac{v\big(N_{L/K}(x)\big)}{[L:K]} \geqslant 0 \implies x \in B \implies 1 + x \in B$$
$$\implies w(1+x) = \frac{v\big(N_{L/K}(1+x)\big)}{[L:K]} \geqslant 0$$

The general case now follows easily: wlog $w(x) \geqslant w(y)$ and hence $w(x/y) \geqslant 0 \implies w(1 + x/y) \geqslant 0$, i.e., $w(x+y) \geqslant w(y) = \min\{w(x), w(y)\}$.
Now we show that w is unique. Suppose that w' is another valuation of L extending v. Since w and w' are distinct but agree on K, they must be inequivalent and hence there exists an element $b \in L$ such that $w(b) \geqslant 0$ but $w'(b) < 0$ (see appendix). Then $b \in B$. Let $f(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_0$ be its minimal polynomial over K. Since A is normal, $f(t) \in A[t]$. But since $w'(b) < 0$ we obtain a contradiction:

$$0 \leqslant v(a_0) = w'(a_0) = w'(-b^n - a_{n-1}b^{n-1} \cdots - a_1 b) = w'(b^n) < 0$$

Finally we show that L is complete with respect to w. Let $\omega_1, \ldots, \omega_n$ be a basis of L over K. Let $(x_i)_{i \geqslant 1}$ be a Cauchy sequence in L and write x_i in terms of the chosen basis: $x_i = y_{i1}\omega_1 + \cdots + y_{in}\omega_n$, $y_{ij} \in K$. Since K is complete and L is finite dimensional over K, we have that all norms on L are equivalent (exercise!) and hence using for instance the sup norm we conclude that, for each j, $(y_{ij})_{i \geqslant 1}$ is a Cauchy sequence in K. Hence this sequence converges to an element $y_j \in K$ and $(x_i)_{i \geqslant 1}$ converges to $y_1\omega_1 + \cdots + y_n\omega_n$ in L, completing the proof that L is complete! $\square$

Thanks to the Valuative Highlander's Philosophy, the results of the previous section hold mutatis mutandis for any local field, for their proofs were based solely on properties of general complete valued fields. In particular we still have at our disposal Calculus Student's Psychedelic Dream and Hensel's lemma and all their wonderful consequences in our more general setting.

Definition 1.3.5. *Let K be a local field with valuation v. Let L be a finite field extension of K and w be the unique extension of v to L. Let π and Π be uniformisers of K and L, and denote by k and l the residue fields of K and L respectively. We define the* **ramification degree** *$e_{L/K}$ of the extension $L \supseteq K$ to be the index of the value group of v in the value group of w:*

$$\boxed{e_{L/K} = [w(L^\times) : v(K^\times)]}$$

In other words, we have the factorisation $\pi = u\Pi^{e_{L/K}}$ for $u \in L$ with $w(u) = 0$.
Observe that since w restricts to v in K we may view k as a subfield of l. We define the **inertia degree** *$f_{L/K}$ of the extension $L \supseteq K$ to be the degree of the corresponding extension of residue fields:*

$$\boxed{f_{L/K} = [l : k]}$$

Since index of groups and degree of extensions are multiplicative, the same holds for the ramification and the inertia degrees: given finite extensions $M \supseteq L \supseteq K$ of local fields we have that

$$\boxed{e_{M/K} = e_{M/L} \cdot e_{L/K} \qquad \text{and} \qquad f_{M/K} = f_{M/L} \cdot f_{L/K}}$$

Definition 1.3.6. *A finite extension of local fields $L \supseteq K$ is* **unramified** *if its ramification degree is $e_{L/K} = 1$. In other words, $L \supseteq K$ is unramified if a uniformiser of K is still a uniformiser of L. On the other hand, if the ramification degree is as large as possible, i.e., $e_{L/K} = [L : K]$, then the extension is said to be* **totally ramified**.

Example 1.3.7. *Let e and f be positive integers and write $K = \mathbb{F}_p((t))$ and $L = \mathbb{F}_{p^f}((t))(t^{1/e})$ where $t^{1/e}$ denotes an e-th root of t (in some algebraic closure of K). Let w be the unique valuation on L extending the valuation v on K. Then $w(t^{1/e}) = 1/e$ and using the fact that L is basically a "power series ring" in $t^{1/e}$ it is easy to show that the valuation ring of w is $\mathbb{F}_{p^f}[\![t]\!][t^{1/e}]$ with residue field $\mathbb{F}_{p^f}$. Therefore $L \supseteq K$ has inertia degree f and ramification degree e. Observe that $L \supseteq K$ breaks into two parts: an* **unramified extension** *$M \supseteq K$ where $M = \mathbb{F}_{p^f}((t))$ and a totally ramified extension $L \supseteq M$.*

$$\begin{tikzcd}
L \arrow[r, equal] \arrow[d, dash, "e", "\textit{totally ramified}"'] & \mathbb{F}_{p^f}((t^{1/e})) \\
M \arrow[r, equal] \arrow[d, dash, "f", "\textit{unramified}"'] & \mathbb{F}_{p^f}((t)) \\
K \arrow[r, equal] & \mathbb{F}_p((t))
\end{tikzcd}$$

Later we will show that any extension of local fields admits such a decomposition.

Example 1.3.8. *Let p be an odd prime and $u \in \mathbb{F}_p$ be a non-square. By example 1.2.4, there are exactly 3 quadratic extensions of $K = \mathbb{F}_p((t))$, namely*

$$L_1 = K(\sqrt{u}), \qquad L_2 = K(\sqrt{t}), \qquad L_3 = K(\sqrt{ut}).$$

Since $\mathbb{F}_p(\sqrt{u}) = \mathbb{F}_{p^2}$ we have that $L_1 = \mathbb{F}_{p^2}((t))$ and t is a uniformiser for both K and L_1, that is, $L_1 \supseteq K$ is unramified with inertia degree $f_{L_1/K} = [\mathbb{F}_{p^2} : \mathbb{F}_p] = 2$. On the other hand, if w_2 and w_3 are the extensions of the valuation v of K to L_2 and L_3 respectively we must have that $w_2(\sqrt{t}) = w_3(\sqrt{ut}) = 1/2$ since $w_2(t) = w_3(ut) = 1$, showing that both $L_2 \supseteq K$ and $L_3 \supseteq K$ are totally ramified. By the proof of the Valuative Highlander's Philosophy, the valuation rings of w_2 and w_3 are the integral closures of $\mathbb{F}_p[\![t]\!]$ in L_2 and L_3, which can easily be shown to be $\mathbb{F}_p[\![t]\!](\sqrt{t})$ and $\mathbb{F}_p[\![t]\!](\sqrt{ut})$ respectively and thus can be viewed as the rings of "power series" in the uniformisers $\sqrt{t}$ and $\sqrt{ut}$ with residue field $\mathbb{F}_p$. Therefore the inertia degrees of $L_2 \supseteq K$ and $L_3 \supseteq K$ are both 1.

A similar computation shows that $\mathbb{Q}_p$ has exactly 1 unramified quadratic extension and 2 totally ramified quadratic extensions. For $\mathbb{Q}_2$, of its 7 quadratic extensions, exactly 1 is unramified and the other 6 are totally ramified.

Observe that for each extension in the last example the product of the ramification and inertia degrees was always 2, the degree of the extension. Magic? Coincidence? Or would it be

Theorem 1.3.9. *Let $L \supseteq K$ be a degree n extension of complete discretely valued fields. Let v be the valuation of K and w be its unique extension to L. Denote by A and B the valuation rings of v and w, π and Π be uniformisers of A and B, and $k = A/(\pi)$ and $l = B/(\Pi)$ be their residue fields respectively.*

Then the ramification and inertia degrees e and f of $L \supseteq K$ are finite. Moreover B is the integral closure of A in L and, as an A-module, it is free of rank n. A basis is given by $\{\omega_i \Pi^j \mid 1 \leqslant i \leqslant f, 0 \leqslant j < e\}$, where $\omega_1, \ldots, \omega_f \in B$ are representatives of a basis of l over k. In particular we have the relation

$$\boxed{ef = n}$$

Proof. We normalise $v(\pi) = w(\pi) = 1$ and $w(\Pi) = 1/e$. We have already seen in the proof of the Valuative Highlander's Philosophy 1.3.4 that B is the integral closure of A in L. Also it is clear from the explicit formula for w that $e \leqslant n$. Now we show that f is also finite and that $ef \leqslant n$. Let $\omega_1, \ldots, \omega_r \in B$ be elements whose images in l are linearly independent over k. It is enough to show that

$$\{\omega_i \Pi^j \in L \mid 1 \leqslant i \leqslant r, 0 \leqslant j < e\} \text{ is linearly independent over } K.$$

Suppose not. Given any dependency relation $\sum_{ij} a_{ij}\omega_i \Pi^j = 0$ with $a_{ij} \in K$, we may assume that all $a_{ij} \in A$ and that at least one of them is a unit, by clearing out the denominators. Let $j_0 \in \mathbb{N}$ be the smallest integer such that $a_{ij_0} \in A^\times$ for some i. Then $\sum_i a_{ij_0}\omega_i \not\equiv 0 \pmod{\Pi}$ by the linear independence of the ω_i mod Π, and we claim that

$$w\Big(\sum_i a_{ij}\omega_i \Pi^j\Big) > w\Big(\sum_i a_{ij_0}\omega_i \Pi^{j_0}\Big) = \frac{j_0}{e} \qquad \text{for all } j \neq j_0$$

This is clear for $j > j_0$ and for $j < j_0$ one has that $\sum_i a_{ij}\omega_i \in \pi B$ and therefore $w(\sum_i a_{ij}\omega_i \Pi^j) \geqslant 1 > j_0/e$. From the strong triangle inequality, $w(\sum_{ij} a_{ij}\omega_i \Pi^j) = j_0/e$ and hence $\sum_{ij} a_{ij}\omega_i \Pi^j$ cannot be zero, a contradiction.

Now pick $\omega_1, \ldots, \omega_f \in B$ so that their images form a basis of $l = B/(\Pi)$ over $k = A/\mathfrak{m}$. We next show that B is a free A-module of rank ef with basis

$$\{\omega_i \Pi^j \mid 1 \leqslant i \leqslant f, 0 \leqslant j < e\}.$$

This set is already known to be linearly independent over A, so we must now show that it generates B over A. Pick $b \in B$; then it suffices to show that

$$b \equiv c_0 \pmod{\pi B} \quad \text{for some } c_0 \text{ in the } A\text{-span of the } \omega_i \Pi^j$$

Indeed, granting this fact, the rest follows: write $b = c_0 + b_1\pi$ for some $b_1 \in B$; by the same token $b_1 = c_1 + b_2\pi$ with c_1 in the A-span of the $\omega_i \Pi^j$, and so $b = c_0 + c_1\pi + b_2\pi^2$; inductively we can write $b = c_0 + c_1\pi + c_2\pi^2 + \cdots + c_n\pi^n + b_{n+1}\pi^{n+1}$ where the "error term" $b_{n+1}\pi^{n+1}$ approaches zero. By the Calculus Student's Psychedelic Dream, we have that $b = c_0 + c_1\pi + c_2\pi^2 + \cdots$, which is in the A-span of the $\omega_i \Pi^j$.

We have thus to show that $B/\pi B$ is generated over $k = A/(\pi)$ by the images of the $\omega_i \Pi^j$. Given $b \in B$, we may write $b = \Pi^j u$ for $j = e \cdot w(b)$ and some unit $u \in B^\times$, and wlog we may assume that $b \in B - \pi B$ so that $0 \leqslant j < e$. Now we can find $a_{ij} \in A$ such that $u \equiv \sum_i a_{ij}\omega_i \pmod{\Pi}$ and therefore $b = \sum_i a_{ij}\omega_i \Pi^j + b'$ with $w(b') > w(b)$. If $w(b') \geqslant 1 \iff b' \in \pi B$ then we are done. Otherwise we repeat the procedure with b'. Since the valuations of the "tails" are increasing, we must eventually stop.

Now the $\omega_i \Pi^j$ also form a basis of L over K: we have already shown linearly independence, and any $c \in L$ is in the K-span of this set, as $c\pi^m \in B$ for m sufficiently large. (In short, if B is free over A of rank ef then $L = B \otimes_A K$ is free over K with the same rank). □

Remark 1.3.10. *When L is separable over K, the above is a particular case of the well-known formula $n = \sum_i e_i f_i$ for extensions of Dedekind domains (see any Number Theory book in the bibliography), except that our situation is much simpler since there is just one prime to deal with. Then only difficulty is then to show that, in the above notation, B is finitely generated over A.*

1.3.1 Filtration of the unit group

Now we introduce some important notation that will be used throughout.

Definition 1.3.11. *Let K be a local field with valuation v. For $i \geqslant 1$ write*

$$\begin{aligned}
O_K &\overset{\text{def}}{=} \textit{valuation ring of } v = \{x \in K \mid v(x) \geqslant 0\} \\
U_K &\overset{\text{def}}{=} \textit{group of units of } O_K = \{x \in O_K \mid v(x) = 0\} \\
\mathfrak{m}_K &\overset{\text{def}}{=} \textit{maximal ideal of } O_K = \{x \in O_K \mid v(x) > 0\} \\
U_K^{(i)} &\overset{\text{def}}{=} 1 + \mathfrak{m}_K^i = \textit{closed ball } \{x \in K \mid |x-1|_v \leqslant 2^{-i}\} \textit{ centred at } 1 \\
&\phantom{\overset{\text{def}}{=}} = \textit{open ball } \{x \in K \mid |x-1|_v < 2^{-i+1}\} \textit{ centred at } 1
\end{aligned}$$

We also extend the last definition to $i = 0$ by setting $U_K^{(0)} = U_K$. Observe that the $U_K^{(i)}$ are subgroups of U_K and that their translates $a \cdot U_K^{(i)}$, $a \in U_K$, form a topological basis of U_K.

Fix a uniformiser π and let $k = O_K/\mathfrak{m}_K$ be the (finite) residue field of K. Denote by k^+ (respectively $k^\times$) the additive (respectively multiplicative) group of k. For U_K we have a filtration

$$U_K = U_K^{(0)} \supseteq U_K^{(1)} \supseteq U_K^{(2)} \supseteq U_K^{(3)} \supseteq \cdots$$

with quotients

$$\boxed{\frac{U_K}{U_K^{(1)}} = k^\times \qquad (u \bmod U_K^{(1)} \mapsto u \bmod \mathfrak{m}_K)}$$

and

$$\boxed{\frac{U_K^{(i)}}{U_K^{(i+1)}} \cong k^+ \qquad (u = 1 + a\pi^{i+1} \bmod U_K^{(i+1)} \mapsto a \bmod \mathfrak{m}_K)}$$

The first isomorphism is canonical, but the second depends on the choice of the uniformiser π. A more canonical way to write it would be as the isomorphism $U_K^{(i)}/U_K^{(i+1)} = \mathfrak{m}_K^i/\mathfrak{m}_K^{i+1}$ given by $u \bmod U_K^{(i+1)} \mapsto u - 1 \bmod \mathfrak{m}_K^{i+1}$.
Just to make sure we understand the notation above, take for instance $K = \mathbb{Q}_p$ with uniformiser $\pi = p$. Then

$$\begin{aligned}
O_{\mathbb{Q}_p} &= \mathbb{Z}_p, \quad \mathfrak{m}_{\mathbb{Q}_p} = (p), \quad U_{\mathbb{Q}_p} = \mathbb{Z}_p^\times = \mathbb{Z}_p - (p) \\
U_{\mathbb{Q}_p}^{(i)} &= \{1 + a_i p^i + a_{i+1} p^{i+1} + \cdots \mid 0 \leqslant a_j < p.\}
\end{aligned}$$

and we have isomorphisms ($0 \leqslant a_i < p$)

$$\begin{aligned}
U_{\mathbb{Q}_p}/U_{\mathbb{Q}_p}^{(1)} &= k^\times \\
a_0 + a_1 p + a_2 p^2 + \cdots \bmod U_{\mathbb{Q}_p}^{(1)} &\mapsto a_0 \bmod p
\end{aligned}$$

and

$$U_{\mathbb{Q}_p}^{(i)}/U_{\mathbb{Q}_p}^{(i+1)} \cong k^+$$
$$1 + a_i p^i + a_{i+1}p^{i+1} + \cdots \bmod U_{\mathbb{Q}_p}^{(i+1)} \mapsto a_i \bmod p$$

It is easy to check that we have an isomorphism, both algebraic and topological,

$$U_K = \varprojlim_{i \geqslant 1} \frac{U_K}{U_K^{(i)}}$$

and hence U_K is a **profinite group** (that is, a projective limit of finite groups). In particular, we have that U_K is compact.

1.4 Structure of group of units

In this section we describe the structure of the multiplicative group $K^\times$ of a local field K. First of all the valuation v on K gives rise to an exact sequence

$$\begin{tikzcd} 1 \arrow[r] & U_K \arrow[r] & K^\times \arrow[r, shift left, "v"] & \mathbb{Z} \arrow[r] \arrow[l, bend left, "s"] & 0 \end{tikzcd}$$

which admits a (non-canonical) splitting $s\colon \mathbb{Z} \to K^\times$, $s(n) = \pi^n$, given by a choice of a uniformiser π. Hence we have a isomorphism

$$K^\times \cong U_K \times \mathbb{Z} \qquad \text{(non-canonical)}$$

and we are left to describe the structure of U_K.
Let $\mathbb{F}_q$ be the residue field of K, where q is a power of a prime p. We have an exact sequence

$$\begin{tikzcd} 1 \arrow[r] & U_K^{(1)} \arrow[r] & U_K \arrow[r, shift left, "reduction"] & \mathbb{F}_q^\times \arrow[r] \arrow[l, bend left, "\text{Teichmüller}"] & 1 \end{tikzcd}$$

Here $U_K \to \mathbb{F}_q^\times$ is just the reduction modulo $\mathfrak{m}_K$. This sequence splits **canonically**: in fact, by Hensel's lemma each element in $\mathbb{F}_q^\times$ lifts to a **uniquely** determined $(q-1)$-th root of unity in K (the so-called **Teichmüller lifts**, see example 1.2.8). Hence we may write

$$U_K = \mu_{q-1} \times U_K^{(1)} \qquad \text{(canonical)}$$

where μ_{q-1} denotes the subgroup of $(q-1)$-th roots of unity in $K^\times$. We have thus reduced the problem to finding the structure of $U_K^{(1)}$.
First observe that $U_K^{(1)}$ is a continuous $\mathbb{Z}_p$-module, with $\mathbb{Z}_p$ acting by exponentiation. In fact, given $u \in U_K^{(1)}$ and $a \in \mathbb{Z}_p$ we may define

$$u^a \stackrel{\text{def}}{=} \lim_{n\to\infty} u^{a_n}$$

where $(a_n)_{n\geqslant 1}$ is any sequence of integers converging to a. This makes sense because

$$U_K = \varprojlim_{i\geqslant 1} \frac{U_K^{(1)}}{U_K^{(i+1)}} \qquad \text{and} \qquad \mathbb{Z}_p = \varprojlim_{i\geqslant 1} \frac{\mathbb{Z}}{(q^i)}$$

and from the isomorphism $U_K^{(i)}/U_K^{(i+1)} \cong \mathbb{F}_q^+$ we conclude that $U_K^{(1)}/U_K^{(i+1)}$ is a finite group of cardinality q^i, i.e., a $\mathbb{Z}/(q^i)$-module. Hence writing $u = (u_1, u_2, \ldots)$, $u_i \in U_K^{(1)}/U_K^{(i+1)}$, and $a = (a_1, a_2, \ldots)$, $a_i \in \mathbb{Z}/(q^i)$, we have that $u^a = (u_1^{a_1}, u_2^{a_2}, \ldots)$ under the above isomorphisms.
From now on we will concentrate on the case $\operatorname{char} K = 0$ and leave the positive characteristic case as an exercise. We show using "p-adic Lie Theory" that $U_K^{(1)}$ is in fact finitely generated as a $\mathbb{Z}_p$-module. Since $\mathbb{Z}_p$ is a PID, $U_K^{(1)}$ will break into a torsion part (roots of unity) and a free part, and then we will be left to compute the rank of this free part. In order to "Lie-nearise" $U_K^{(1)}$, we make use of the p-adic logarithmic and exponential functions:

Lemma 1.4.1. *Let K be a local field of char 0 with valuation v and residue field of char p. Consider the power series*

$$\log(1+x) \overset{\text{def}}{=} x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$
$$\exp x \overset{\text{def}}{=} 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

Then $\log(1+x)$ converges for all x with $v(x) > 0$ while $\exp x$ converges for all x with $v(x) > v(p)/(p-1)$. Hence for $i \gg 0$ sufficiently large we have a continuous isomorphisms of $\mathbb{Z}_p$-modules

$$U_K^{(i)} \underset{\exp}{\overset{\log}{\rightleftarrows}} \mathfrak{m}_K^i$$

Notice that while $U_K^{(i)}$ is a multiplicative $\mathbb{Z}_p$-module with action given by exponentiation, $\mathfrak{m}_K^i$ is an additive $\mathbb{Z}_p$-module with action given by multiplication.

Proof. We apply the Calculus Student's Psychedelic Dream. Let $n \geqslant 1$ and write $n = p^k m$ with $p \nmid m$. We have that

$$v(n) = v(p)\cdot k \leqslant v(p)\cdot \log_p n \quad \text{and}$$
$$v(n!) = v(p)\cdot\left(\left\lfloor\frac{n}{p}\right\rfloor + \left\lfloor\frac{n}{p^2}\right\rfloor + \left\lfloor\frac{n}{p^3}\right\rfloor + \cdots\right) \leqslant v(p)\cdot\frac{n/p}{1-1/p} = \frac{v(p)\cdot n}{p-1}$$

Therefore

$$v(x) > 0 \implies v\left(\frac{x^n}{n}\right) = n\cdot v(x) - v(n) \geqslant n - v(p)\cdot\log_p n \to \infty$$
$$v(x) > v(p)/(p-1) \implies v\left(\frac{x^n}{n!}\right) = n\cdot v(x) - v(n!) \geqslant n\cdot\left(v(x) - \frac{v(p)}{p-1}\right) \to \infty$$

as $n \to \infty$. The convergence of $\log(1+x)$ and $\exp x$ now follows.

Finally log and exp are inverses of each other as formal power series in $\mathbb{Q}[\![x]\!]$ (since this holds over $\mathbb{R}$), and it is easily seen that they are morphisms of $\mathbb{Z}_p$-modules, thus for $i \gg 0$ sufficiently large (so as to ensure convergence) $\log\colon U_K^{(i)} \to \mathfrak{m}_K^i$ and $\exp\colon \mathfrak{m}_K^i \to U_K^{(i)}$ define continuous isomorphisms of $\mathbb{Z}_p$-modules. $\square$

The lemma shows that $U_K^{(i)} \cong \mathfrak{m}_K^i$ for i sufficiently large. But we have an isomorphism $O_K \cong \mathfrak{m}_K^i$ of $\mathbb{Z}_p$-modules given by multiplication by π^i. On the other hand we know that O_K is free over $\mathbb{Z}_p$ of rank $[K : \mathbb{Q}_p]$ by theorem 1.3.9, and so the same holds for $U_K^{(i)}$.
Now since $U_K^{(i)}/U_K^{(i+1)} = k^+$ for $i \geqslant 1$, we have that $[U_K^{(1)} : U_K^{(i)}]$ is finite for all $i \geqslant 1$. Therefore $U_K^{(1)}$ contains a finite index $\mathbb{Z}_p$-submodule which is free of rank $[K : \mathbb{Q}_p]$. This proves that $U_K^{(1)}$ is finitely generated as a $\mathbb{Z}_p$-module. The free part of $U_K^{(1)}$ has rank $[K : \mathbb{Q}_p]$ and its torsion part is a cyclic p-group $\mathbb{Z}_p/p^r = \mathbb{Z}/p^r$ for some r (it is cyclic because it is isomorphic to a torsion subgroup of $K^\times$). Putting everything together, we have just shown

Theorem 1.4.2 (Structure of group of units)**.** *Let K be a local field with residue field $\mathbb{F}_q$ where q is a power of a prime p. If* $\operatorname{char} K = 0$ *then there is a non-canonical isomorphism, both algebraic and topological,*

$$\boxed{K^\times \cong \mathbb{Z} \times \mu_K \times \mathbb{Z}_p^{[K:\mathbb{Q}_p]}}$$

where μ_K is the finite cyclic group of roots of units in $K^\times$ with $|\mu_K| = (q-1)p^r$ for some r.
If $\operatorname{char} K = p$ *then there is a non-canonical isomorphism, both algebraic and topological,*

$$\boxed{K^\times \cong \mathbb{Z} \times \mu_K \times \mathbb{Z}_p^{\mathbb{N}}}$$

where μ_K is the finite cyclic group of roots of units in $K^\times$ with $|\mu_K| = (q-1)$.

We skip the positive characteristic case, since it is a bit more involved. But we indicate how to construct the isomorphism $U_K^{(1)} \cong \mathbb{Z}_p^{\mathbb{N}}$. Let $\omega_1, \ldots, \omega_f$ be a basis of $\mathbb{F}_q$ over $\mathbb{F}_p$. For any n not divisible by p define a continuous morphism $g_n\colon \mathbb{Z}_p^f \to U_K^{(n)}$ by

$$g_n(a_1, \ldots, a_f) = \prod_{1 \leqslant i \leqslant f} (1 + \omega_i t^n)^{a_i}$$

The required isomorphism $g\colon \mathbb{Z}_p^{\mathbb{N}} \to U_K^{(1)}$ is then given by the convergent product

$$g = \prod_{p \nmid n} g_n\colon \prod_{p \nmid n} \mathbb{Z}_p^f \to U_K^{(1)}$$

The necessary verifications are left as an exercise to the reader (for the lazy one, the answer can be found in the excellent book by Neukirch [9], page 140).

1.5 Extensions of Local Fields

We conclude this chapter with a study of extension of local fields, in particular unramified ones. We begin with the important

Theorem 1.5.1. *Let K be a local field, and $L \supseteq K$ be a finite extension. Let $l \supseteq k$ be the corresponding residue field extension, and write $q = |l|$.*

(i) L contains the splitting field M of $x^q - x$ over K: one has

$$x^q - x = x \prod_{\omega \in l^\times} (x - \omega)$$

where $\omega \in \mu_{q-1}(L)$ is the Teichmüller lift of $\overline{\omega} \in l^\times$ to $L^\times$. Moreover $M \supseteq K$ is unramified with $[M : K] = [l : k] = f_{L/K}$, and $L \supseteq M$ is totally ramified with $[L : M] = e_{L/K}$:

$$\begin{array}{c} L \\ {\scriptstyle \textit{totally ramified}} \Big| {\scriptstyle e_{L/K}} \\ M \\ {\scriptstyle \textit{unramified}} \Big| {\scriptstyle f_{L/K}} \\ K \end{array}$$

In particular, $L \supseteq K$ is unramified if and only if $L = M$, i.e. $L \supseteq K$ is generated by the Teichmüller lifts of the elements in $l^\times$ (so that all unramified extensions are **cyclotomic***).*

(ii) $M \supseteq K$ is the maximal unramified subextension contained in $L \supseteq K$.

(iii) If $L \supseteq K$ is unramified it is Galois and there is a natural isomorphism

$$\begin{aligned} \mathrm{Gal}(L/K) &\xrightarrow{\approx} \mathrm{Gal}(l/k) \\ \sigma &\mapsto \overline{\sigma} \end{aligned}$$

where $\overline{\sigma} \colon l \to l$ is the k-automorphism induced by σ on residue fields:

$$\overline{\sigma}(a \bmod \mathfrak{m}_L) = \sigma(a) \bmod \mathfrak{m}_L \qquad (a \in \mathcal{O}_L)$$

Proof. (i) By Lagrange's theorem, the roots of $x^{q-1} - 1 \in k[x]$ are exactly the elements of $l^\times$, so $x^{q-1} - 1$ and thus $x^q - x$ are separable, hence the factorisation of $x^q - x$ follows immediately from Hensel's lemma.

Next we show that $[M : K] = [l : k]$. As $l^\times$ is a cyclic group, we can choose a generator $\overline{\omega} \in l^\times$ with Teichmüller lift $\omega \in \mu_{q-1}(L) \subset L^\times$. Then to prove $[M : K] = [l : k]$ is the same as to prove $[K(\omega) : K] = [k(\overline{\omega}) : k]$, i.e., to prove that the minimal polynomials of $\overline{\omega}$ and ω over k and K have the same degree.

Let $f(x) \in O_K[x]$ be the minimal polynomial of ω over K ($f(x)$ lies in $O_K[x]$ as ω is integral over the normal domain O_K), and let $\overline{f}(x) \in k[x]$ be its reduction. We will actually show that $\overline{f}(x)$ is the minimal polynomial of $\overline{\omega}$ over k. In fact, note that

- $f(\omega) = 0 \implies \overline{f}(\overline{\omega}) = 0$, and thus $\overline{f}(x)$ will be the minimal polynomial of $\overline{\omega}$ over k if it is irreducible in $k[x]$.
- $f(x) \mid x^q - x$ in $O_K[x]$, thus $\overline{f}(x) \mid x^{q-1} - 1$ in $k[x]$. Since $x^{q-1} - 1$ is separable (its roots are the $q-1$ elements in $l^\times$), so is $\overline{f}(x)$.
- $\overline{f}(x)$ is irreducible in $k[x]$ by Hensel's lemma 1.2.10, as any (hypothetical) non-trivial factorisation of $\overline{f}(x)$ lifts to a non-trivial factorisation of the irreducible polynomial $f(x) \in O_K[x]$.

This completes the proof of $[M : K] = [l : k] = f_{L/K}$. Since the residue field of M is a subfield of l containing $\overline{\omega}$, it must coincide with l, therefore $f_{M/K} = f_{L/M}$ and $f_{L/M} = 1$. Also, $e_{M/K} = 1$ and $e_{L/M} = e_{L/K} = [L : M]$, by theorem 1.3.9). In short: $M \supseteq K$ is unramified of degree $f_{L/K}$ while $L \supseteq M$ is totally ramified of degree $e_{L/K}$. The rest of (i) is clear.

(ii) Suppose that $L \supseteq N \supseteq K$ is such that $N \supseteq K$ is unramified. The residue field of N is contained in l, hence its Teichmüller lifts are a subset of those of l. As $N \supseteq K$ is generated by Teichmüller lifts, N is contained in M.

(iii) $L \supseteq K$ is clearly Galois, being the splitting field of the separable polynomial $x^q - x$. Since an element of either $\mathrm{Gal}(L/K)$ or $\mathrm{Gal}(l/k)$ is completely determined by its actions on the roots of $x^q - x$, and the reduction map is a bijection on such roots, it follows that $\mathrm{Gal}(L/K) \xrightarrow{\approx} \mathrm{Gal}(l/k)$ is an isomorphism.

□

Compare the picture in (i) with the one in example 1.3.7. Moreover, note that the decomposition of $L \supseteq K$ into a tower $L \supseteq M \supseteq K$ is analogous to the decomposition of an arbitrary finite field extension of positive characteristic into a separable extension followed by a purely inseparable one. Also, taking the splitting field of all polynomials of the form $x^{q^n} - x$, $n \geqslant 1$, where $q = |k|$, we obtain an immediate, but very important

Corollary 1.5.2 (Unramified Highlander's Philosophy)**.** *There is a bijection between finite unramified extensions of K (in some separable closure K_{sep}) and finite extensions of $\mathbb{F}_q$, which associates to each finite extension of L of K its residue field l. In particular, there is* **exactly one** *unramified extension of each degree, and they are all cyclic extensions (i.e. Galois extensions with cyclic Galois group).*

Definition 1.5.3. *Let $L \supseteq K$ be an unramified extension. The unique automorphism in $\mathrm{Frob}_{L/K} \in \mathrm{Gal}(L/K)$ lifting the Frobenius automorphism $x \mapsto x^q$ of the residue field extension $l \supset k$ is also called* **Frobenius automorphism** *of $L \supseteq K$.*

Example 1.5.4. *The unramified extension of degree n over $\mathbb{F}_p((t))$ is just $\mathbb{F}_{p^n}((t))$. For p odd, the unramified extension over $\mathbb{Q}_p$ of degree 2 is $\mathbb{Q}_p(\sqrt{u})$, where u is a non-square in $\mathbb{Z}_p^\times$ (see example 1.3.8). The Frobenius map is $\sqrt{u} \mapsto -\sqrt{u}$ (there is no other choice).*
The degree 3 unramified extension L of $\mathbb{Q}_5$ is given by $\mathbb{Q}_5(\theta)$ where θ is a root of $f(x) = x^3 + 3x^2 - 1$. This follows from the fact that its reduction $\bar{f}(x) \in \mathbb{F}_5[x]$ is irreducible, hence $f(x)$ is also irreducible; on the other hand, the residue field l of L contains the image $\bar{\theta} \in l$ of $\theta \in O_L$ (θ is integral over $\mathbb{Z}_5$), hence $[l : \mathbb{F}_5] \geqslant [L : \mathbb{Q}_5] = 3$, and we thus must have equality, showing that L is unramified over $\mathbb{Q}_3$. The Frobenius automorphism ϕ is characterised by $\phi(\theta) \equiv \theta^5 \pmod 5$. A straightforward computation shows that $-\theta^2 - 3\theta - 1$ is another root of $f(x)$ and that ϕ is explicitly given by $\theta \mapsto -\theta^2 - 3\theta - 1$.

Next we concentrate on Galois extensions. We begin with a

Definition 1.5.5. *Let $L \supseteq K$ be a Galois extension of local fields with $G = \mathrm{Gal}(L/K)$. Note that since any $\sigma \in G$ preserves the valuation of L, $\sigma(O_L) \subseteq O_L$ and $\sigma(\mathfrak{m}_L^{i+1}) \subseteq \mathfrak{m}_L^{i+1}$, hence σ acts on $O_L/\mathfrak{m}_L^{i+1}$ for $i \geqslant 0$. We define the i-th* **higher ramification group** *to be the subgroup G_i of G consisting of those automorphims $\sigma \in G$ having trivial action on $O_L/\mathfrak{m}_L^{i+1}$. The group G_0 is called* **inertia group**.

The higher ramification groups give a filtration

$$G_{-1} \overset{\text{def}}{=} G \supseteq G_0 \supseteq G_1 \supseteq G_2 \supseteq G_3 \supseteq \cdots$$

of the Galois group G. Observe that this filtration is **exhaustive** (i.e. G_i is trivial for i sufficiently large): if $\omega_1, \ldots, \omega_n \in O_L$ is a basis of L over K, if $\sigma \neq 1$ then $\sigma(\omega_j) \neq \omega_j$ for some j and hence $\sigma(\omega_j) \not\equiv \omega_j \pmod{\mathfrak{m}_L^{i+1}} \iff \sigma \notin G_i$ for i sufficiently large. Also observe that $G_{i+1} \trianglelefteq G_i$ for all i: for $\tau \in G_i$, $\sigma \in G_{i+1}$ and $b \in O_L$, we have that $\sigma\big(\tau^{-1}(b)\big) \equiv \tau^{-1}(b) \pmod{\mathfrak{m}_L^{i+2}}$ and hence $\tau\sigma\tau^{-1}(b) \equiv b \pmod{\mathfrak{m}_L^{i+2}}$, i.e., $\tau\sigma\tau^{-1} \in G_{i+1}$.
We study the beginning of this filtration a bit closer.

Theorem 1.5.6 (Galois Group and Maximal Unramified Extension)**.** *Let $L \supseteq K$ be a Galois extension of local fields with $G = \mathrm{Gal}(L/K)$ and let $l \supseteq k$ be the corresponding extension of residue fields. Let G_0 be the inertia group of this extension. Denote by $\bar{\sigma} \in \mathrm{Gal}(l/k)$ the automorphism induced by $\sigma \in G$ on $l = O_L/\mathfrak{m}_L$. Then*

(i) the map $\sigma \mapsto \bar{\sigma}$ induces an isomorphism between G/G_0 and $\mathrm{Gal}(l/k)$. In particular we have that $|G_0| = e_{L/K}$.

(ii) the field $M \overset{\text{def}}{=} L^{G_0}$ is the maximal unramified extension of K contained in L. In particular, M is Galois over K with cyclic Galois group $G/G_0 = \mathrm{Gal}(l/k)$.

We have thus the following picture:

$$\begin{tikzcd}
L \arrow[d, dash, "\textit{totally ramified}"', "e"] \arrow[r, dash, "G_0"] & l \arrow[d, equal] \\
M \arrow[d, dash, "\textit{unramified}"', "f"] \arrow[r, dash, "G/G_0 = \mathrm{Gal}(l/k)"] & l \arrow[d, dash] \\
K \arrow[r, dash] & k
\end{tikzcd}$$

Proof. Item (i) is clear: the projection $G \twoheadrightarrow \mathrm{Gal}(M/K)$ is surjective by Galois theory, and by theorem 1.5.1 the map $\sigma \mapsto \overline{\sigma}$ induces an isomorphism $\mathrm{Gal}(M/K) \xrightarrow{\approx} \mathrm{Gal}(l/k)$. Observe that, by definition, G_0 is the kernel of the composition $G \twoheadrightarrow \mathrm{Gal}(M/K) \xrightarrow{\approx} \mathrm{Gal}(l/k)$.
Finally, to prove (ii), let $M = L^{G_0}$ and just apply (i) to the Galois extension $L \supseteq M$. We then conclude that M has residue field l and hence that $f_{M/K} = f_{L/K}$, proving that M is the maximal unramified extension of K contained in L. $\square$

We close this section showing that, in a local field, you will never have trouble solving equations by radicals!

Theorem 1.5.7. *Every Galois extension $L \supseteq K$ of local fields is solvable.*

Proof. Although quite impressive, this theorem has a simple proof. Write $G = \mathrm{Gal}(L/K)$, let G_i denote the i-th higher ramification group, and l and k be the residue fields of L and K respectively. Since $G/G_0 = \mathrm{Gal}(l/k)$ is cyclic, in order to show that G is solvable it is enough to show that G_0 is solvable. The key idea is then to construct, for $i \geqslant 0$, injective morphisms $f_i \colon G_i/G_{i+1} \hookrightarrow U_L^{(i)}/U_L^{(i+1)}$.
Let Π be a uniformiser of L. Define $f_i \colon G_i/G_{i+1} \to U_L^{(i)}/U_L^{(i+1)}$ by

$$\boxed{f(\sigma G_{i+1}) = \frac{\sigma(\Pi)}{\Pi} \bmod U_L^{(i+1)} \qquad \text{for } \sigma \in G_i}$$

First we show that this map is well-defined: since $\sigma \in G_i$ preserves the valuation, $\sigma(\Pi) = u\Pi$ for some unit $u \in O_L^\times$; but we also have $\sigma(\Pi) \equiv \Pi \pmod{\Pi^{i+1}}$ and therefore $u \equiv 1 \pmod{\Pi^i}$, i.e., $u = \sigma(\Pi)/\Pi \in U_L^{(i)}$. Replacing i by $i+1$ shows that $\sigma(\Pi)/\Pi \in U_L^{(i+1)}$ whenever $\sigma \in G_{i+1}$.
Next we show that the definition of f_i does not depend on the choice of Π: replacing Π by another uniformiser $u\Pi$, $u \in O_L^\times$, alters f_i by $\sigma(u)/u \bmod U_L^{(i+1)}$. But $\sigma \in G_i$ and thus $\sigma(u) \equiv u \pmod{\mathfrak{m}_L^{i+1}}$ showing that $\sigma(u)/u \in U_L^{(i+1)}$.
We now check that f_i is group morphism. Let $\sigma, \tau \in G_i$. Since $\tau(\Pi)$ is also a uniformiser, we have that

$$\begin{aligned} f_i(\sigma\tau G_{i+1}) &= \frac{\sigma\tau(\Pi)}{\Pi} \bmod U_L^{(i+1)} = \frac{\sigma\big(\tau(\Pi)\big)}{\big(\tau(\Pi)\big)} \cdot \frac{\tau(\Pi)}{\Pi} \bmod U_L^{(i+1)} \\ &= f_i(\sigma G_{i+1}) f_i(\tau G_{i+1}) \end{aligned}$$

Finally, we show that f_i is injective. Let $\sigma \in G_i$ and suppose that $u \overset{\text{def}}{=} \sigma(\Pi)/\Pi$ satisfies $u \in U_L^{(i+1)} \iff u \equiv 1 \pmod{\Pi^{i+1}}$. Then $\sigma(\Pi) \equiv \Pi \pmod{\Pi^{i+2}}$, which implies that $\sigma \in G_{i+1}$. In fact, we have that $\sigma \in G_0$ and that L is totally ramified over $M \overset{\text{def}}{=} L^{G_0}$ by 1.5.6. Hence we may choose Π to be a $|G_0|$-th root of a uniformiser of M and by theorem 1.3.9 we have that O_L is generated over O_M by the powers of Π. Therefore $\sigma(\Pi) \equiv \Pi \pmod{\Pi^{i+2}}$ implies that $\sigma(b) \equiv b \pmod{\Pi^{i+2}}$ for all $b \in O_L$, as claimed. □

1.6 Exercises

1.6.1. *Show that $\mathbb{F}_p((t))$ and $\mathbb{Q}_p$ are uncountable.*

1.6.2. *Show that the p-adic number $f = \sum_i a_i p^i \in \mathbb{Q}_p$ belongs to $\mathbb{Q}$ if and only if its p-base expansion is periodic. Find the 5-base expansion for $2/3$ and $1/10$ in $\mathbb{Q}_5$.*

1.6.3 (Multiplicative Calculus Student's Psychedelic Dream)**.** *Let K be a local field and consider elements $f_n \in K^\times$, $n \geqslant 0$. Show that the infinite product $\prod_{n\geqslant 0} f_n$ converges to an element of $K^\times$ if and only if $\lim_{n\to\infty} f_n = 1$.*

1.6.4. *Show that $\mathbb{F}_p((t))$ is the completion of $\mathbb{F}_p(t)$ with respect to the valuation given by the prime t of $\mathbb{F}_p[t]$. Similarly show that $\mathbb{Q}_p$ is the completion of $\mathbb{Q}$ with respect to the valuation given by the prime p of $\mathbb{Z}$.*

1.6.5. *Over which $\mathbb{Q}_p$ is the quadratic form $3x^2 + 7y^2 - 15z^2$ anisotropic?*

1.6.6. *Give an example of two totally ramified extensions of some local field K whose compositum is not totally ramified over K.*

1.6.7 (Base change)**.** *Let K be a local field and let $\mathbb{F}_q$ be its residue field. If K' be an arbitrary finite extension of K. If $L' = LK'$ is the compositum of L and K' (in some separable closure K_{sep}) then $L' \supseteq K'$ is also unramified.*

1.6.8. *Show that the splitting field of $x^{p^n} - x$ is the unramified extension of $\mathbb{Q}_p$ of degree n.*

1.6.9. *Find the Galois groups of $x^5 + x + 1$ over $\mathbb{Q}$, $\mathbb{Q}_3$ and $\mathbb{Q}_5$.*

1.6.10. *Let n be a positive integer and p be an odd prime. Is D_n the Galois group of some Galois extension of $\mathbb{Q}_p$? (I write D_n for the dihedral group with $2n$ elements with the sole purpose of confusing my audience!)*

Chapter 2

Galois Cohomology in a Nutshell

In this chapter we present for the convenience of the reader a brief summary of all the results from group cohomology that we will need for the proofs of the reciprocity theorems. The methods of this section are inspired in computations of Algebraic Topology, with the starting point being the well-known relation $H_1(X, \mathbb{Z}) = \pi_1(X)^{ab}$ between the first singular homology group of a topological space X and the maximal abelian quotient $\pi_1(X)^{ab}$ of the fundamental group of X. Since Galois extensions are algebraic analogs of covering spaces in Topology, this turns out to be a quite natural point of view (if, of course, you've studied Algebraic Topology before, but don't worry if you haven't, the proofs below are purely algebraic, but it's a good idea to eventually look at the source of inspiration for them).

The main result of this chapter is the famous Tate-Nakayama theorem 2.5.2, which will be one of the conerstones in the proof of the reciprocity law. All that preceeds it is, frankly, just technical tools, albeit important ones. The reader is advised to skim over the results, and go straight to §2.5, coming back as necessary.

2.1 Definitions

In this subsection, G will denote a finite group.

Definition 2.1.1. *A* G**-module** M *is just a left* $\mathbb{Z}[G]$*-module, where* $\mathbb{Z}[G]$ *is the group ring of* G *with integer coefficients. In other words,* M *is an abelian group together with a* G*-action, that is, a map* $G \times M \to M$ *sending* (σ, m) *to an element* $\sigma \cdot m \in M$ *such that, for all* $m, m' \in M$*, and* $\sigma, \sigma' \in G$*,*

(i) $1 \cdot m = m$

(ii) $\sigma \cdot (m + m') = \sigma \cdot m + \sigma \cdot m'$

(iii) $(\sigma\sigma') \cdot m = \sigma \cdot (\sigma' \cdot m)$

A **morphism of** G**-modules** $f \colon M \to N$ *is a map of left* $\mathbb{Z}[G]$*-modules, i.e., a group morphism such that* $f(\sigma m) = \sigma f(m)$ *for all* $\sigma \in G$ *and* $m \in M$*. If* M *is a* G*-module, we write* M^G *for the subgroup of* G*-fixed points:*

$$M^G \stackrel{\text{def}}{=} \{m \in M \mid \sigma \cdot m = m \text{ for all } \sigma \in G\}$$

Example 2.1.2. *Examples of G-modules arising in nature are*

(i) $M = \mathbb{Z}$, $M = \mathbb{Q}$, *or* $M = \mathbb{Q}/\mathbb{Z}$ *where* G *operates trivially:* $\sigma m = m$ *for all* $\sigma \in G$ *and* $m \in M$*;*

(ii) $M = \mathbb{Z}[G]$ *or* $M = I_G$*, where* G *operates by left multiplication. Here* I_G *is the so-called* **augmentation ideal** *of* $\mathbb{Z}[G]$*, defined as the kernel of the* **augmentation map** $\epsilon\colon \mathbb{Z}[G] \to \mathbb{Z}$ *given by*

$$\sum_{\sigma\in G} n_\sigma \sigma \mapsto \sum_{\sigma\in G} n_\sigma$$

Observe that I_G *is a free* $\mathbb{Z}$*-module with basis* $\sigma - 1$, $\sigma \in G$, $\sigma \neq 1$*. If* $\mathbb{Z}$ *is given the trivial action as above, then the augmentation map is a morphism of* G*-modules;*

(iii) if $L \supseteq K$ *is a finite Galois extension with* $G = \mathrm{Gal}(L/K)$*, then* $M = L^+$, $M = L^\times$ *and* $M = \mu_L$ *are all examples of* G*-modules where* G *operates via the Galois action. Here* L^+ *and* $L^\times$ *are the additive and multiplicative groups of* L *and* $\mu_L \subseteq L^\times$ *is the subgroup of roots of 1 contained in* L*.*

(iv) for any G*-module* M *we have a morphism of* G*-modules* $N_G\colon M \to M^G$ *given by*

$$N_G(m) \overset{\text{def}}{=} \sum_{\sigma\in G} \sigma m, \qquad m \in M$$

called **norm map***. In the previous example, when* $M = L^\times$ *the norm map* N_G *coincides with the usual norm of fields* $N_{L/K}\colon L \to K$*. When* $M = L^+$ *then it becomes the trace* $T_{L/K}\colon L \to K$*.*

From now on, unless otherwise stated the abelian groups M of the example will always be given the G-actions above. With this convention, for any G-module M we have an isomorphisms of abelian groups

$$\mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, M) = M^G$$

and

$$\mathbb{Z} \otimes_{\mathbb{Z}[G]} M = \frac{M}{I_G \cdot M}$$

since $\mathbb{Z}[G]/I_G = \mathbb{Z}$ (induced by augmentation).
Now we show how to build new modules from old ones. Let M and N be two G-modules. Then the set

$$\mathrm{Hom}_G(M, N)$$

of all morphisms of G-modules between M and N can be made into a G-module by "conjugation" $(\sigma f)(m) \overset{\text{def}}{=} \sigma f(\sigma^{-1} m)$ for $f \in \mathrm{Hom}_G(M, N)$ and $m \in M$. Similarly, the tensor product of M and N over $\mathbb{Z}$

$$M \otimes N$$

can be made into a G-module by "diagonal action" $\sigma(m \otimes n) \overset{\text{def}}{=} \sigma(m) \otimes \sigma(n)$ for $m \in M$ and $n \in N$. The next definition shows how to "lift" modules from subgroups:

Definition 2.1.3. *Let $H \leqslant G$ be a subgroup and N be an H-module. The* **induced module** *from N is the G-module*

$$\mathrm{Ind}_H^G(N) \stackrel{\text{def}}{=} \mathrm{Hom}_H(G, N)$$

of all H-linear functions $f\colon G \to N$ (i.e. $f(h \cdot g) = h \cdot f(g)$ for all $h \in H$ and $g \in G$) and where the G-action is given by $(\sigma f)(g) = f(g\sigma)$ for all $\sigma, g \in G$. When $H = 1$ we simply write $\mathrm{Ind}^G(N)$. A G-module M is called **induced** *if $M = \mathrm{Ind}^G(N)$ for some abelian group N.*

Another way to "lift" an H-module is via the "base change" $\mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} N$, where the G-action is given by multiplication on the left component (in the tensor product, the left component is viewed as a right $\mathbb{Z}[H]$-module via $m \cdot \sigma \stackrel{\text{def}}{=} \sigma^{-1} \cdot m$ for all $\sigma \in H$ and $m \in \mathbb{Z}[G]$). Base change is related to induced modules via the isomorphism of G-modules

$$\mathrm{Ind}^G(N) \cong \mathbb{Z}[G] \otimes N$$

given by $\phi \mapsto \sum_{g \in G} g \otimes \phi(g^{-1})$. Here N is any abelian group with trivial G-action. Observe that an induced G-module $M = \mathbb{Z}[G] \otimes N$ is also induced as an H-module: if $H\sigma_1, \ldots, H\sigma_n$ are right cosets of H then $M = \mathbb{Z}[H] \otimes N'$ where $N' = \bigoplus_{1 \leqslant i \leqslant n} \sigma_i N$.

Example 2.1.4. *Let $L \supseteq K$ be a finite Galois extension with $G = \mathrm{Gal}(L/K)$. Then L^+ is an induced G-module. In fact, by the normal basis theorem there exists $\omega \in L$ such that $\{\sigma(\omega) \mid \sigma \in G\}$ is a basis of L over K. Then we have a G-isomorphism $K[G] \cong L^+$ given by $\sum_{\sigma \in G} a_\sigma \sigma \mapsto \sum_{\sigma \in G} a_\sigma \sigma(\omega)$. Since $K[G] = \mathbb{Z}[G] \otimes K$ is induced by the above, so is L^+.*

The following lemma will be useful in arguments involving dimension shifting (which we will see later).

Lemma 2.1.5. *Let M be an arbitrary G-module and denote by M_0 the underlying abelian group. Then there is an injective map of G-modules*

$$M \hookrightarrow \mathrm{Ind}^G(M_0)$$

sending m to the function $\phi_m\colon G \to M_0$ given by $\phi_m(\sigma) = \sigma \cdot m$. There is also a surjective map of G-modules

$$\mathrm{Ind}^G(M_0) \twoheadrightarrow M$$

given by $\phi \mapsto \sum_{\sigma \in G} \sigma\phi(\sigma^{-1})$.

Definition 2.1.6. *Let M be a G-module and $i \geqslant 0$. The i-th* **cohomology group** *of M is defined as*

$$H^i(G, M) \stackrel{\text{def}}{=} \mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, M)$$

while the i-th **homology group** *of M is defined as*

$$H_i(G, M) \stackrel{\text{def}}{=} \mathrm{Tor}_i^{\mathbb{Z}[G]}(\mathbb{Z}, M)$$

We briefly recall the definitions of Ext and Tor below, but we warn you that except for low degrees $i = 0, 1$ the computations of these groups are not done directly from their definitions but rather via their functorial properties and some "vanishing theorems" that tell you sufficient conditions under which these groups are trivial.

A G-module I is called **injective** if the functor $\mathrm{Hom}_G(-, I)$ is exact. A G-module P is called **projective** if the functor $\mathrm{Hom}_G(P, -)$. For instance, free $\mathbb{Z}[G]$-modules are projective. It can be shown that any G-module M can be embedded into an injective module and it can also be written as a quotient of a projective module. Hence we can inductively construct an **injective resolution** of M, namely an exact sequence

$$0 \to M \to I^0 \to I^1 \to I^2 \to \cdots$$

where the G-modules I^i are all injective, and similarly one may construct a **projective resolution**

$$\cdots \to P^2 \to P^1 \to P^0 \to M \to 0$$

Now we can define $\mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, M)$ as follows. Choose an injective resolution of M as above and apply the G-fixed point functor $\mathrm{Hom}_G(\mathbb{Z}, -) = (-)^G$ to it. We obtain a complex

$$0 \to M^G \to (I^0)^G \to (I^1)^G \to (I^2)^G \to \cdots$$

Then

$$\mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, M) \stackrel{\text{def}}{=} \frac{\ker\Big((I^i)^G \to (I^{i+1})^G\Big)}{\mathrm{im}\Big((I^{i-1})^G \to (I^i)^G\Big)}$$

(Here we interpret $I^{-1} = 0$). Now an easy but rather tedious computation shows that the groups thus obtained are independent of the choice of the injective resolution of M. Alternatively, one can choose a projective resolution of $\mathbb{Z}$

$$\cdots \to P^2 \to P^1 \to P^0 \to \mathbb{Z} \to 0$$

and apply the functor $\mathrm{Hom}_G(-, M)$ to it, defining

$$\mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, M) \stackrel{\text{def}}{=} \frac{\ker\Big(\mathrm{Hom}_G(P^i, M) \to \mathrm{Hom}_G(P^{i+1}, M)\Big)}{\mathrm{im}\Big(\mathrm{Hom}_G(P^{i-1}, M) \to \mathrm{Hom}_G(P^i, M)\Big)}$$

(Here we interpret $P^{-1} = 0$) Again, this is independent of the chosen projective resolution of $\mathbb{Z}$, and it can be shown that either procedure, via projective resolutions of $\mathbb{Z}$ or injective resolutions of M, yield isomorphic groups.

For $\mathrm{Tor}_i^{\mathbb{Z}[G]}(\mathbb{Z}, M)$, the procedure is similar, except that Hom is replaced by the tensor product and the we use projective resolutions for both entries (in the tensor product $M \otimes_{\mathbb{Z}[G]} N$ we view M as a right $\mathbb{Z}[G]$-module via $m \cdot \sigma \stackrel{\text{def}}{=} \sigma^{-1} \cdot m$ for $\sigma \in G$ and $m \in M$). For instance, if $P_\bullet \to M \to 0$ is a projective resolution of M, applying the functor $\mathbb{Z} \otimes_{\mathbb{Z}[G]} - = -/I_G \cdot -$ we obtain

$$\mathrm{Tor}_i^{\mathbb{Z}[G]}(\mathbb{Z}, M) = \frac{\ker\Big(P_i/I_G \cdot P_i \to P_{i-1}/I_G \cdot P_{i-1}\Big)}{\mathrm{im}\Big(P_{i+1}/I_G \cdot P_{i+1} \to P_i/I_G \cdot P_i\Big)}$$

Example 2.1.7 (Cyclic groups)**.** *Let G be a cyclic group of order n and let σ be a generator of G. Then we have a projective resolution of $\mathbb{Z}$*

$$\cdots \xrightarrow{I} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{I} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{I} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where ϵ is the augmentation map, and I and N denote multiplication by $\sigma - 1$ and $1 + \sigma + \cdots + \sigma^{n-1}$, respectively. Hence we obtain

$$H^i(G, M) = \begin{cases} M^G & \text{if } i = 0 \\ \dfrac{\ker N_G}{I_G \cdot M} & \text{if } i \text{ is odd} \\ \dfrac{M^G}{N_G(M)} & \text{if } i > 0 \text{ is even} \end{cases}$$

where $N_G \colon M \to M^G$ is the **norm map** *and I_G is the augmentation ideal of $\mathbb{Z}[G]$. Using the same projective resolution of $\mathbb{Z}$, for homology we obtain*

$$H_i(G, M) = \begin{cases} \dfrac{M}{I_G \cdot M} & \text{if } i = 0 \\ \dfrac{M^G}{N_G(M)} & \text{if } i \text{ is odd} \\ \dfrac{\ker N_G}{I_G \cdot M} & \text{if } i > 0 \text{ is even} \end{cases}$$

The main functorial property of Tor and Ext, and thus of the homology and cohomology groups, are their **long exact sequences**. For any short exact sequence of G-modules

$$0 \to A \to B \to C \to 0$$

one has long exact sequences

$$\begin{aligned} 0 &\to H^0(G, A) = A^G \to H^0(G, B) = B^G \to H^0(G, C) = C^G \\ &\xrightarrow{\delta^0} H^1(G, A) \to H^1(G, B) \to H^1(G, C) \\ &\xrightarrow{\delta^1} H^2(G, A) \to H^2(G, B) \to H^2(G, C) \xrightarrow{\delta^2} \cdots \end{aligned}$$

and

$$\begin{aligned} \cdots &\xrightarrow{\delta_3} H_2(G, A) \to H_2(G, B) \to H_2(G, C) \\ &\xrightarrow{\delta_2} H_1(G, A) \to H_1(G, B) \to H_1(G, C) \\ &\xrightarrow{\delta_1} H_0(G, A) = \frac{A}{I_G \cdot A} \to H_0(G, B) = \frac{B}{I_G \cdot B} \to H_0(G, C) = \frac{C}{I_G \cdot C} \to 0 \end{aligned}$$

for cohomology and homology respectively. The maps δ^i and δ_i are called **connecting morphisms**. The other maps are the natural ones induced by the maps $A \to B$ and $B \to C$.

Think of the short exact sequence as a way to "decompose" B into simpler modules, a submodule A and a quotient module C. If we know the homology/cohomology of A and C then the long exact sequence allows us to find out the homology/cohomology of B.

Example 2.1.8. *Let ϵ be the augmentation map. From the exact sequence of G-modules*

$$0 \to I_G \to \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

and the fact that $\mathbb{Z}[G]$ is free (and thus has trivial homology), we conclude that the connecting morphisms give isomorphisms $H_p(G, \mathbb{Z}) = H_{p-1}(G, I_G)$ for all $p \geqslant 1$. In particular, for $p = 1$ we have that

$$H_1(G, \mathbb{Z}) = H_0(G, I_G) = \frac{I_G}{I_G^2} = G^{ab}$$

where $G^{ab} \overset{\text{def}}{=} G/[G:G]$ is the maximal abelian quotient of G. The isomorphism $G^{ab} \approx I_G/I_G^2$ is given by $\sigma \cdot [G:G] \mapsto (\sigma - 1) \cdot I_G^2$.

The main vanishing theorem is **Shapiro's lemma**.

Lemma 2.1.9 (Shapiro)**.** *Let $H \subseteq G$ be a subgroup and N be an H-module. Then for all $i \geqslant 0$ we have isomorphisms*

$$H^i(H, N) = H^i(G, \operatorname{Ind}_H^G(N)) \qquad \text{and} \qquad H_i(H, N) = H_i(G, \operatorname{Ind}_H^G(N))$$

In particular, any induced G-module has trivial cohomology and homology.

Proof. (Sketch) For any any G-module M and any H-module N we have a canonical isomorphism $\operatorname{Hom}_G(M, \operatorname{Ind}_H^G N) = \operatorname{Hom}_H(M, N)$ and the functor $\operatorname{Ind}_H^G(-)$ is exact. From these two properties it follows that $\operatorname{Ind}_H^G(-)$ preserves injectives, hence given an injective resolution $0 \to N \to I^\bullet$ of N, we obtain an injective resolution $0 \to \operatorname{Ind}_H^G(N) \to \operatorname{Ind}_H^G(I^\bullet)$ of $\operatorname{Ind}_H^G(N)$. The result follows by applying $\operatorname{Hom}_G(\mathbb{Z}, -)$ and using the fact that $\operatorname{Hom}_G(\mathbb{Z}, \operatorname{Ind}_H^G N) = \operatorname{Hom}_H(\mathbb{Z}, N)$. The proof for homology is similar. $\square$

Corollary 2.1.10. *If $L \supseteq K$ is a finite Galois extension with $G = \operatorname{Gal}(L/K)$ then $H^0(G, L^+) = K^+$ and $H^p(G, L^+) = 0$ for $p \geqslant 1$.*

Definition 2.1.11. *The p-th* **Tate cohomology group** *is defined by*

$$H_T^p(G, M) = \begin{cases} H^p(G, M) & \text{if } p \geqslant 1 \\ \dfrac{M^G}{N_G M} & \text{if } p = 0 \\ \dfrac{\ker N_G}{I_G \cdot M} & \text{if } p = -1 \\ H_{-p-1}(G, M) & \text{if } p \leqslant -2 \end{cases}$$

The importance of the Tate cohomology groups is that it allows us to splice the long exact sequences of homology and cohomology into a single very long one. Given a short exact sequence of G-modules

$$0 \to A \to B \to C \to 0$$

we have a commutative diagram with exact rows

$$\begin{tikzcd}
H_1(G,C) \arrow[r] & H_0(G,A) \arrow[r] \arrow[d, "N_G"] & H_0(G,B) \arrow[r] \arrow[d, "N_G"] & H_0(G,C) \arrow[r] \arrow[d, "N_G"] & 0 \\
0 \arrow[r] & H^0(G,A) \arrow[r] & H^0(G,B) \arrow[r] & H^0(G,C) \arrow[r] & H^1(G,A)
\end{tikzcd}$$

where the vertical arrows are induce by the norm maps. Hence we obtain an exact sequence

$$\begin{aligned}
\cdots \to H_T^{-1}(G,A) \to H_T^{-1}(G,B) \to H_T^{-1}(G,C) \\
\to H_T^0(G,A) \to H_T^0(G,B) \to H_T^0(G,C) \\
\to H_T^1(G,A) \to H_T^1(G,B) \to H_T^1(G,C) \to \cdots
\end{aligned}$$

Example 2.1.12 (Periodicity of Tate cohomology for cyclic groups)**.** *If G is cyclic we have that*

$$H_T^p(G,M) = \begin{cases} \dfrac{M^G}{N_G M} & if p is even \\ \dfrac{\ker N_G}{I_G \cdot M} & if p is odd \end{cases}$$

2.2 Explicit Resolutions

Here we show how to define homology and cohomology via an explicit projective resolution of $\mathbb{Z}$:

Definition 2.2.1. *The* **standard resolution** *of $\mathbb{Z}$ is the projective resolution*

$$\cdots \xrightarrow{d_3} \tilde{C}_2 \xrightarrow{d_2} \tilde{C}_1 \xrightarrow{d_1} \tilde{C}_0 \xrightarrow{d_0} \mathbb{Z} \to 0$$

where $\tilde{C}_p = \mathbb{Z}[G^{p+1}]$ is the free $\mathbb{Z}$-module with basis $(\sigma_0, \sigma_1, \ldots, \sigma_p) \in G^{p+1} = G \times \cdots \times G$ ($p+1$ times) with "diagonal" G-action $s \cdot (\sigma_0, \ldots, \sigma_p) = (s \cdot \sigma_0, \ldots, s \cdot \sigma_p)$, and where $d_{p+1} \colon \tilde{C}_{p+1} \to \tilde{C}_p$ is given by

$$d(\sigma_0, \sigma_1, \ldots, \sigma_{p+1}) = \sum_{0 \leqslant k \leqslant p+1} (-1)^k (\sigma_0, \sigma_1, \ldots, \sigma_{k-1}, \sigma_{k+1}, \ldots, \sigma_{p+1})$$

Applying $\mathrm{Hom}_G(-, M)$ to the standard resolution, we obtain a complex of abelian groups

$$0 \to \tilde{C}^0(G,M) \xrightarrow{\tilde{d}^0} \tilde{C}^1(G,M) \xrightarrow{\tilde{d}^1} \tilde{C}^2(G,M) \xrightarrow{\tilde{d}^2} \cdots$$

where

$$\tilde{C}^p(G,M) \stackrel{\text{def}}{=} \left\{ \text{functions } \tilde{f}\colon G^{p+1} \to M \;\middle|\; \begin{array}{l} \tilde{f}(s\cdot\sigma_0,\ldots,s\cdot\sigma_p) = s\cdot\tilde{f}(\sigma_0,\ldots,\sigma_p) \text{ for all } s\in G \text{ and } (\sigma_0,\ldots,\sigma_p)\in \\ G^{p+1} \end{array} \right\}$$

and $\tilde{d}^p\colon \tilde{C}^p(G,M) \to \tilde{C}^{p+1}(G,M)$ is given by

$$(\tilde{d}^p\tilde{f})(\sigma_0,\ldots,\sigma_{p+1}) = \sum_{0\leqslant k\leqslant p+1} (-1)^k \tilde{f}(\sigma_0,\ldots,\sigma_{k-1},\sigma_{k+1},\ldots,\sigma_{p+1})$$

There is an isomorphism of abelian groups

$$\tilde{C}^p(G,M) \xrightarrow{\approx} C^p(G,M) \stackrel{\text{def}}{=} \{\text{all functions } \phi\colon G^p \to M\}$$

taking $f\in\tilde{C}^p(M)$ to the function in $C^p(G,M)$ given by

$$(\sigma_1,\sigma_2,\ldots,\sigma_p) \mapsto \tilde{f}(1,\sigma_1,\sigma_1\sigma_2,\sigma_1\sigma_2\sigma_3,\ldots,\sigma_1\sigma_2\cdots\sigma_p)$$

The above complex is therefore isomorphic to the following "inhomogeneous" one

$$0 \to C^0(G,M) \xrightarrow{d^0} C^1(G,M) \xrightarrow{d^1} C^2(G,M) \xrightarrow{d^2} \cdots$$

where $d^p\colon C^p(G,M) \to C^{p+1}(G,M)$ is given by

$$\begin{aligned}(d^pf)(\sigma_1,\ldots,\sigma_{p+1}) = \sigma_1\cdot f(\sigma_2,\ldots,\sigma_{p+1}) \\ + \sum_{1\leqslant k\leqslant p} (-1)^k f(\sigma_1,\ldots,\sigma_{k-1},\sigma_k\cdot\sigma_{k+1},\sigma_{k+2},\ldots,\sigma_{p+1}) \\ + (-1)^{p+1} f(\sigma_1,\ldots,\sigma_p)\end{aligned}$$

Hence we obtain an explicit formula

$$\boxed{H^p(G,M) = \frac{\ker d^p}{\operatorname{im} d^{p+1}}}$$

An element of $\ker d^p$ is called a **p-cocycle** while an element of $\operatorname{im} d^{p-1}$ is called a **p-coboundary**. Let

$$0 \to A \to B \to C \to 0$$

be a short exact sequence of G-modules. In terms of cocycles and coboundaries, one has a very explicit description of the **connecting morphism**

$$\delta\colon H^p(G,C) \to H^{p+1}(G,A)$$

as follows: given a p-cocycle $f\colon G^p \to C$ representing an element $\varphi = [f] \in H^p(G,C)$, we may lift it to a function $\widehat{f}\colon G^p \to B$. Then $d^p\widehat{f}$ is in the image of the map $C^{p+1}(G,A) \to C^{p+1}(G,B)$ induced by $A \to B$, so we may view it as a p-cocycle $d^p\widehat{f}\colon G^{p+1} \to A$, and $\delta(\varphi) = [d^p\widehat{f}] \in H^{p+1}(G,A)$.

Similarly, for homology one obtains an inhomogeneous complex

$$\cdots \xrightarrow{d_3} C_2(G,M) \xrightarrow{d_2} C_1(G,M) \xrightarrow{d_1} C_0(G,M) \to 0$$

where $C_p(G,M) = C^p(G,M)$ and $d_p\colon C_p(G,M) \to C_p(G,M)$ is given now by

$$\begin{aligned}(d_p f)(\sigma_1,\dots,\sigma_{p-1}) &= \sum_{\sigma\in G} \sigma^{-1}\cdot f(\sigma,\sigma_1,\dots,\sigma_{p-1}) \\ + \sum_{1\leqslant k\leqslant p-1} (-1)^k \sum_{\sigma\in G} & f(\sigma_1,\dots,\sigma_{k-1},\sigma_k\cdot\sigma,\sigma^{-1},\sigma_{k+1},\dots,\sigma_{p-1}) \\ &+ (-1)^{p+1}\sum_{\sigma\in G} f(\sigma_1,\dots,\sigma_{p-1},\sigma)\end{aligned}$$

Example 2.2.2. *We have that a 1-cocycle is a function $f\colon G\to M$ such that $f(\sigma\tau) = \sigma f(\tau) + f(\sigma)$. It is a coboundary if and only if it has the form $f(\sigma) = \sigma m - m$ for some $m\in M$. In particular, if the G-action on M is trivial, then all 1-coboundaries are trivial and a 1-cocycle is just a group morphism: $H^1(G,M) = \mathrm{Hom}(G,M)$ in this case.*

Example 2.2.3. *Let G be a cyclic group of order n generated by σ. Then one has an exact sequence of G-modules*

$$0\to\mathbb{Z}\to\mathbb{Q}\to\mathbb{Q}/\mathbb{Z}\to 0$$

As we shall see later, $H^i(G,\mathbb{Q}) = 0$ for all $i\geqslant 1$ and hence the connecting maps induce isomorphisms $\delta\colon H^i(G,\mathbb{Q}/\mathbb{Z}) \xrightarrow{\approx} H^{i+1}(G,\mathbb{Z})$ for all $i\geqslant 0$. Then $H^1(G,\mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}(G,\mathbb{Q}/\mathbb{Z})$ is a group of order n, generated by a morphism $f\colon G\to\mathbb{Q}/\mathbb{Z}$ given by $f(\sigma) = \frac{1}{n} \bmod \mathbb{Z}$, and hence $H^2(G,\mathbb{Z})$ is also cyclic of order n, generated by the class of the 2-cocycle

$$\delta f(\sigma^i,\sigma^j) = \begin{cases} 1 & \text{if } i+j\geqslant n \\ 0 & \text{otherwise}\end{cases} \qquad \text{for} \quad 0\leqslant i,j<n$$

which can be computed as described above using the lift $\widehat{f}\colon G\to\mathbb{Q}$ of f given by $\widehat{f}(\sigma^i) = \frac{i}{n}$ for $0\leqslant i<n$.

Now we use the explicit characterisation of cohomology in terms of cocycles to prove a very important vanishing theorem in Galois cohomology:

Theorem 2.2.4 (Hilbert's Satz 90)**.** *Let $L\supseteq K$ be a finite Galois extension with Galois group $G = \mathrm{Gal}(L/K)$. Then*

$$H^1(G,L^\times) = 0$$

Proof. Let $f\colon G\to L^\times$ be a 1-cocycle, and consider the element

$$m \overset{\text{def}}{=} \sum_{\sigma\in G} f(\sigma)\cdot\sigma(a)$$

where $a\in L^\times$ is chosen so that $m\neq 0$, which is possible by Dedekind's independency of characters. Then for every $\tau\in G$ we have that

$$m = \sum_{\sigma\in G} f(\tau\sigma)\cdot\tau\sigma(a) = \sum_{\sigma\in G}\tau\big(f(\sigma)\big)\cdot f(\tau)\cdot\tau\sigma(a) = \tau(m)\cdot f(\tau)$$

That is, $f(\tau)^{-1} = \tau(m)/m$, showing that f is a coboundary. $\square$

2.3 Dimension shifting; Inflation, Restriction, Corestriction

We have seen how changing modules alters the cohomology groups via the long exact sequence. Now we show how changing the group alters the cohomology. Let $f\colon G' \to G$ be a group morphism and A be a G-module. We may view A as a G'-module as well via

$$\sigma' \cdot a \overset{\text{def}}{=} f(\sigma') \cdot a \qquad \text{for} \quad a \in A, \quad \sigma' \in G'$$

We denote this G'-module by f^*A. Clearly A^G is a subgroup of $(f^*A)^{G'}$, hence the inclusion map defines a functorial map $H^0(G, A) \to H^0(G', f^*A)$. By general properties of derived functors, this map in degree 0 extends uniquely for all $p \geqslant 0$ to a functorial map $H^p(G, A) \to H^p(G', f^*A)$, compatible with the connecting maps. We recall the proof of this fact since it relies on a recurrent technique in group homology/cohomology known as *dimension shifting*.

First write an exact sequence of G-modules

$$0 \to A \to I \to B \to 0$$

with I injective (cohomologically trivial would do, for instance an induced module). Since $H^p(G, I) = 0$ for all $p > 0$ we have that the connecting maps give isomorphisms $\delta\colon H^{p-1}(G, B) \xrightarrow[\approx]{} H^p(G, A)$ for all $p > 1$. Now, for $p > 1$, if we already know the map $H^{p-1}(G, -) \to H^{p-1}(G', f^*-)$ in dimension $p - 1$ we may define it in dimension p via the composition

$$H^p(G, A) \xrightarrow[\approx]{\delta^{-1}} H^{p-1}(G, B) \to H^{p-1}(G', f^*B) \xrightarrow{\partial} H^p(G', f^*A)$$

where ∂ denotes the connecting map with respect to the exact sequence of G'-modules

$$0 \to f^*A \to f^*I \to f^*B \to 0$$

For the "base case" $p = 1$ the procedure is similar but we need to use the commutative diagram instead

$$\begin{tikzcd}
0 \arrow[r] & A^G \arrow[r] \arrow[d, hook] & I^G \arrow[r] \arrow[d, hook] & B^G \arrow[r] \arrow[d, hook] & H^1(G, A) \arrow[r] \arrow[d, hook, dashed] & 0 \\
0 \arrow[r] & (f^*A)^{G'} \arrow[r] & (f^*I)^{G'} \arrow[r] & (f^*B)^{G'} \arrow[r] & H^1(G', f^*A) \arrow[r] & \cdots
\end{tikzcd}$$

Similar procedures work for homology and also for Tate cohomology groups. One may consider the more general case where A' is a G'-module and $g\colon A \to A'$ is a group morphism which is compatible with f in the sense that $g(f(\sigma') \cdot a) = \sigma' \cdot g(a)$ for all $a \in A$ and $\sigma' \in G'$. Then the degree 0 map induced by g extends to a map $H^p_T(G, A) \to H^p_T(G', A')$ for all p.

Definition 2.3.1. *Let $H \leqslant G$ be a subgroup and M be a G-module. We define the* **restriction** *map*

$$\operatorname{res}\colon H^p_T(G, M) \to H^p_T(H, M)$$

to be the map induced by the inclusion map $M^G \hookrightarrow M^H$ in degree 0 (take $f\colon H \hookrightarrow G$ to be the inclusion map).

Now suppose that $H \lhd G$ is normal. We define the **inflation**

$$\inf\colon H^p_T(G/H, M^H) \to H^p_T(G, M)$$

to be the map induced by the identity $(M^H)^{G/H} = M^G$ in degree 0 (take $f\colon G \twoheadrightarrow G/H$ to be the quotient map and $g\colon M^H \hookrightarrow M$ to be the inclusion).

For $p \geqslant 0$, the restriction and inflation maps have a very simple description in terms of the standard resolution. Given a p-cocycle $f\colon G^p \to M$, $\operatorname{res}([f])$ is represented by the restriction $f\colon H^p \to M$ of f to H^p. On the other hand, for a p-cocycle $f\colon (G/H)^p \to M^H$ we have that $\inf([f])$ is given by the p-cocycle $\tilde{f}\colon G^p \to M$ given by the composition

$$G^p \twoheadrightarrow (G/H)^p \xrightarrow{f} M^H \hookrightarrow M$$

where the unlabelled maps are the natural ones.

Theorem 2.3.2 (Inflation-restriction sequence)**.** *Let M be a G-module, and H be a normal subgroup of G. Suppose that $H^p(H, M) = 0$ for $p = 1, \ldots, q-1$. Then*

$$0 \to H^q(G/H, M^H) \xrightarrow{\inf} H^q(G, M) \xrightarrow{\operatorname{res}} H^q(H, M)$$

is exact.

Proof. The result follows easily for $p = 1$ by direct computation with cocycles. The general case follows by dimension shifting. Suppose that $q \geqslant 2$ and consider an exact sequence of G-modules (see 2.1.5)

$$0 \to M \to \operatorname{Ind}^G(M_0) \to N \to 0 \tag{$*$}$$

Note that the middle term is induced also as an H-module. Since $H^1(H, M) = 0$ by hypothesis we have that the sequence of G/H-modules

$$0 \to M^H \to \left(\operatorname{Ind}^G(M_0)\right)^H \to N^H \to 0 \tag{$**$}$$

is still exact. The middle term $\left(\operatorname{Ind}^G(M_0)\right)^H = \operatorname{Ind}^{G/H}(M_0)$ is induced as a G/H-module, and we have a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & H^q(G/H, M^H) \arrow[r, "\inf"] & H^q(G, M) \arrow[r, "\operatorname{res}"] & H^q(H, M) \\
H^{q-1}(G/H, N^H) \arrow[u, "\approx"] \arrow[r, "\inf"] & H^{q-1}(G, N) \arrow[u, "\approx"] \arrow[r, "\operatorname{res}"] & H^{q-1}(H, N) \arrow[u, "\approx"] &
\end{tikzcd}$$

where the vertical arrows are the connecting maps associated to $(*)$ and $(**)$, which are isomorphisms since the middle terms of $(*)$ and $(**)$ are cohomologically trivial. Since $H^p(H, N) = H^{p+1}(H, M) = 0$ for $p = 1, 2, \ldots, q-2$ we have that the bottom row is exact by induction on q. Hence the top row is also exact. $\square$

Remark 2.3.3. *For the expert (but then you shouldn't be reading this!): the previous result follows directly from the Hochschild-Serre spectral sequence*

$$H^p(G/H, H^q(H, M)) \implies H^{p+q}(G, M)$$

Remark 2.3.4. *Using the inflation map one can define cohomology for profinite groups as well. A group is* **profinite** *if it is the projective limit of finite groups. For instance, for any field k, its* **absolute Galois group** $G_k \overset{\text{def}}{=} \mathrm{Gal}(k_{sep}/k)$ *is profinite. If $G = \varprojlim_{i\in I} G_i$, giving the discrete topology to the finite groups G_i, G is made into a topological space as well, namely the projective limit of the topological spaces G_i. Since the product of discrete groups is compact by Tychonoff's theorem and G is a closed subgroup of $\prod_{i\in I} G_i$ we have that G is compact. Now let M a continuous G-module, namely a G-module for which the action $G \times M \to M$ is continuous. This means that for every $m \in M$ the orbit $G \cdot m$ is finite. Then we can define the p-th cohomology group as*

$$H^p(G, M) \overset{\text{def}}{=} \varprojlim_{H} H^p(G/H, M^H)$$

where H runs over all open normal subgroups of G and the transition maps are given by inflation.

Let us go back to the finite case. Let $H \leqslant G$ be a subgroup and M be a G-module. We define the **norm map** $N_{G/H} \colon M^H \to M^G$ via

$$N_{G/H}(m) \overset{\text{def}}{=} \sum_{\sigma\in S} \sigma(m)$$

where S is a set of left cosets representatives of G/H. Clearly this does not depend on the choice of S and the above sum is G-invariant. Now dimension shifting allows us to extend this map to all other dimensions using an exact sequence of G-modules (see 2.1.5)

$$0 \to M \to \mathrm{Ind}^G(M_0) \to N \to 0$$

and the fact that the middle term is also induced as an H-module.

Definition 2.3.5. *The map*

$$\mathrm{cor} \colon H^p_T(H, M) \to H^p_T(G, M)$$

induced by the norm map $N_{G/H} \colon M^H \to M^G$ in degree 0 is called **corestriction**.

Theorem 2.3.6 (Restriction-Corestriction)**.** *Let $H \leqslant G$ be a subgroup and M be a G-module. Then the composition*

$$H^p_T(G, M) \overset{\text{res}}{\to} H^p_T(H, M) \overset{\text{cor}}{\to} H^p_T(G, M)$$

equals multiplication by $[G : H]$. In particular, $H^p_T(G, M)$ is killed by $|G|$.

Proof. For $p = 0$ we have that the above composition is

$$\frac{M^G}{N_G M} \to \frac{M^H}{N_H M} \overset{N_{G/H}}{\to} \frac{M^G}{N_G M}$$

where the first map is the natural one. This composition is clearly multiplication by $[G : H]$. The general case now follows easily by dimension shifting. $\square$

Example 2.3.7. *Let G_p be any p-Sylow subgroup of G. Then* $\mathrm{res}\colon H^p(G,M) \to H^p(G_p,M)$ *is injective on the p-primary components of these groups. In fact,* $\mathrm{cor} \circ \mathrm{res}$ *is multiplication by $[G:G_p]$, which is prime to p, and thus is an automorphism on these p-primary components.*

Example 2.3.8. *We have that $H^p_T(G,\mathbb{Q}) = 0$ for all p. In fact, multiplication by $|G|$ is an automorphism of $\mathbb{Q}$. Since $H^p(G,-)$ is a functor, this implies that multiplication by $|G|$ is also an automorphism of $H^p_T(G,\mathbb{Q})$ which must then be zero by the theorem.*

2.4 Cup product

Definition 2.4.1. *There exists a unique family of maps*

$$H^p_T(G,A) \otimes H^q_T(G,B) \xrightarrow{\cup} H^{p+q}_T(G, A\otimes B)$$

(tensor products over $\mathbb{Z}$) called **cup products**, *which are characterised by the following properties:*

(i) the cup product is a morphism of bifunctors (a.k.a natural transformations) in the pair (A,B);

(ii) for $p=q=0$ the cup product is induced by the natural map $A^G \otimes B^G \to (A\otimes B)^G$;

(iii) the cup product is compatible with connection morphisms: if

$$0 \to A' \to A \to A'' \to 0$$

is an exact sequence of G-modules and B is a G-module such that

$$0 \to A'\otimes B \to A\otimes B \to A''\otimes B \to 0$$

is exact then the diagram

$$\begin{tikzcd}
H^p_T(G,A'') \otimes H^q_T(G,B) \arrow[r, "\cup"] \arrow[d, "\delta^p\otimes id"] & H^{p+q}_T(G, A''\otimes B) \arrow[d, "\delta^{p+q}"] \\
H^{p+1}_T(G,A') \otimes H^q_T(G,B) \arrow[r, "\cup"] & H^{p+q+1}_T(G, A'\otimes B)
\end{tikzcd}$$

commutes. On the other hand, if

$$0 \to B' \to B \to B'' \to 0$$

is an exact sequence of G-modules and A is a G-module such that

$$0 \to A\otimes B' \to A\otimes B \to A\otimes B'' \to 0$$

is exact then the diagram

$$\begin{tikzcd}
H^p_T(G,A) \otimes H^q_T(G,B'') \arrow[r, "\cup"] \arrow[d, "id\otimes\delta^q"] & H^{p+q}_T(G, A\otimes B'') \arrow[d, "(-1)^p\cdot\delta^{p+q}"] \\
H^p_T(G,A) \otimes H^{q+1}_T(G,B') \arrow[r, "\cup"] & H^{p+q+1}_T(G, A\otimes B')
\end{tikzcd}$$

commutes.

Uniqueness of this family is easily proven by dimension shifting, while existence is given by the explicit formulas in cocycles. First we extend the standard resolution by setting $\tilde{C}_{-p} = \tilde{C}^*_{p-1}$ for $p \geqslant 1$, where $\tilde{C}^*_{p-1}$ is the dual of $\tilde{C}_{p-1} = \mathbb{Z}[G^p]$, namely the free $\mathbb{Z}$-module with basis given by functions $(\sigma_1^*, \ldots, \sigma_p^*)$, which send $(\sigma_1, \ldots, \sigma_p)$ to $1 \in \mathbb{Z}$ and every other basis element of $\tilde{C}_{p-1}$ to $0 \in \mathbb{Z}$. The boundary map $d_{-p} \colon \tilde{C}_{-p} \to \tilde{C}_{-p-1}$ is given by

$$d_{-p}(\sigma_1^*, \ldots, \sigma_p^*) = \sum_{s \in G} \sum_{0 \leqslant i \leqslant q} (-1)^i (\sigma_1^*, \ldots, \sigma_i^*, s^*, \sigma_{i+1}^*, \ldots, \sigma_p^*)$$

and $d_0 \colon \tilde{C}_0 \to \tilde{C}_{-1}$ is given by $d_0(\sigma_0) = \sum_{s \in G}(s^*)$. One may then compute the Tate cohomology groups by applying $\mathrm{Hom}_G(-, M)$ to this sequence and computing the cohomology of the resulting complex.
Now define $\phi_{p,q} \colon \tilde{C}_{p+q} \to \tilde{C}_p \otimes \tilde{C}_q$ as follows:

(i) For $p, q \geqslant 0$

$$\phi_{p,q}(\sigma_0, \ldots, \sigma_{p+q}) = (\sigma_0, \ldots, \sigma_p) \otimes (\sigma_p, \ldots, \sigma_{p+q})$$

(ii) For $p, q \geqslant 1$

$$\phi_{-p,-q}(\sigma_1^*, \ldots, \sigma_{p+q}^*) = (\sigma_1^*, \ldots, \sigma_p^*) \otimes (\sigma_{p+1}^*, \ldots, \sigma_{p+q}^*)$$

(iii) For $p \geqslant 0$, $q \geqslant 1$

$$\begin{aligned}
\phi_{p,-p-q}(\sigma_1^*, \ldots, \sigma_q^*) &= \sum (\sigma_1, s_1, \ldots, s_p) \otimes (s_p^*, \ldots, s_1^*, \sigma_1^*, \ldots, \sigma_q^*) \\
\phi_{-p-q,p}(\sigma_1^*, \ldots, \sigma_q^*) &= \sum (\sigma_1^*, \ldots, \sigma_q^*, s_1^*, \ldots, s_p^*) \otimes (s_p, \ldots, s_1, \sigma_q) \\
\phi_{p+q,-q}(\sigma_0, \ldots, \sigma_p) &= \sum (\sigma_0, \ldots, \sigma_p, s_1, \ldots, s_q) \otimes (s_q^*, \ldots, s_1^*) \\
\phi_{-q,p+q}(\sigma_0, \ldots, \sigma_p) &= \sum (s_1^*, \ldots, s_q^*) \otimes (s_q, \ldots, s_1, \sigma_0, \ldots, \sigma_p)
\end{aligned}$$

where the sum runs over all $(s_1, \ldots, s_p) \in G^p$.

Then a straightforward but long check shows that $(\epsilon \otimes \epsilon) \circ \phi_{0,0} = \epsilon$ (here $\epsilon \colon \tilde{C}_0 \to \mathbb{Z}$ denotes the augmentation map) and that

$$\phi_{p,q} \circ d = (d \otimes 1) \circ \phi_{p+1,q} + (-1)^p (1 \otimes d) \circ \phi_{p,q+1}$$

for all $p, q \in \mathbb{Z}$ (we omitted the indices of the coboundary maps d for notational clarity). Now given $f \in \mathrm{Hom}_G(\tilde{C}_p, A)$ and $g \in \mathrm{Hom}_G(\tilde{C}_q, B)$ we define $f \cup g \in \mathrm{Hom}_G(\tilde{C}_{p+q}, A \otimes B)$ by

$$f \cup g = (f \otimes g) \circ \phi_{p,q}$$

and now it is easy to check that if f and g are cocycles, so is $f \cup g$, and that its class depends only on the classes of f and g. A lengthy but easy check shows that the pairing $\cup$ so defined has the desired properties of the cup product.

Example 2.4.2. *Given elements of $H^1(G,M)$ and $H^1(G,N)$ represented by (inhomogeneous) 1-cochains $f\colon G \to M$ and $g\colon G \to N$ then their cup product is the class of the 2-cochain*

$$(f \cup g)(\sigma_1, \sigma_2) = f(\sigma_1) \otimes \sigma_1 g(\sigma_2)$$

in $H^2(G, M \otimes N)$.

The following lemma can be easily proved by dimension shifting, and is left as an exercise for the reader since I'm running out of time and energy:

Lemma 2.4.3. *Let $H \leqslant G$ be a subgroup. We have*

(i) $(a \cup b) \cup c = a \cup (b \cup c)$ *in* $H_T^{p+q+r}(G, A \otimes B \otimes C)$ *for* $a \in H_T^p(G,A)$, $b \in H_T^q(G,B)$, $c \in H_T^r(G,C)$;

(ii) $a \cup b = (-1)^{pq} \cdot b \cup a$ *under the isomorphism* $A \otimes B = B \otimes A$ *for* $a \in H_T^p(G,A)$, $b \in H_T^q(G,B)$;

(iii) $\operatorname{res}(a \cup b) = \operatorname{res} a \cup \operatorname{res} b \in H_T^{p+q}(H, A \otimes B)$ *for* $a \in H_T^p(G,A)$ *and* $b \in H_T^q(G,B)$;

(iv) (Projection formula) $\operatorname{cor}\big(a \cup \operatorname{res} b\big) = \operatorname{cor} a \cup b$ *for* $a \in H^p(H,A)$ *and* $b \in H^q(G,B)$.

Our last theorem shows that how the cup product enters in the periodicity of the cohomology of cyclic groups:

Theorem 2.4.4. *Let G be a cyclic group of order n and σ be a generator. Consider the 2-cocycle $f\colon G \times G \to \mathbb{Z}$ given by*

$$f(\sigma^i, \sigma^j) = \begin{cases} 1 & \text{if } i + j \geqslant n \\ 0 & \text{otherwise} \end{cases} \qquad \text{for} \quad 0 \leqslant i, j < n$$

Then for any G-module M the cup product $- \cup [f]$ gives an isomorphism

$$H_T^p(G,M) \underset{\approx}{\overset{\cup[f]}{\to}} H_T^{p+2}(G,M)$$

Proof. We have exact sequences of G-modules

$$0 \to I_G \to \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

$$0 \to \mathbb{Z} \xrightarrow{\mu} \mathbb{Z}[G] \xrightarrow{\sigma - 1} I_G \to 0$$

where ϵ is the augmentation map, μ denotes multiplication by $1 + \sigma + \cdots + \sigma^{n-1}$ and $\mathbb{Z}[G] \to I_G$ is given by multiplication by $\sigma - 1$. The connecting maps $\partial\colon H_T^p(G,\mathbb{Z}) \xrightarrow{\approx} H_T^{p+1}(G, I_G)$ and $\delta\colon H_T^{p+1}(G, I_G) \xrightarrow{\approx} H_T^{p+2}(G,\mathbb{Z})$ are isomorphisms and by explicit computations with cocycles we have that $[f] = \delta \circ \partial(\phi)$ where $\phi = 1 \bmod n \in H_T^0(G,\mathbb{Z}) = \mathbb{Z}/n$.
Since all terms of the above sequences are free $\mathbb{Z}$-modules they stay exact after tensoring with M. Therefore for any $\alpha \in H_T^p(G,M)$ we have that

$$\alpha \cup [f] = \alpha \cup \delta \circ \partial(\phi) = \delta \circ \partial(\alpha \cup \phi) = \delta \circ \partial(\alpha)$$

since $- \cup \phi$ is the identity, as can be easily checked by dimension shifting for example. Now since $\mathbb{Z}[G] \otimes M$ is induced the connecting maps $\partial\colon H_T^p(G,M) \xrightarrow{\approx} H_T^{p+1}(G, M \otimes I_G)$ and $\delta\colon H_T^{p+1}(G, M \otimes I_G) \xrightarrow{\approx} H_T^{p+2}(G,M)$ are isomorphisms, hence so is $\delta \circ \partial$ and we are done. $\square$

2.5 Tate-Nakayama theorem

Let G be a finite group. In this section, we prove a purely group theoretic result giving an isomorphism

$$G^{ab} \approx \frac{C^G}{N_G(C)}$$

where C is a G-module satisfying some conditions and $N_G\colon C \to C^G$ denotes the **norm map** of C.
We begin with a criterion for cohomological triviality for a G-module.

Theorem 2.5.1 (Twin number vanishing criterion)**.** *Let G be a finite group and M be a G-module. If there are two consecutive numbers i and $i+1$ for which*

$$H_T^i(H, M) = H_T^{i+1}(H, M) = 0 \qquad \textit{for all subgroups } H \leqslant G$$

then $H_T^r(G, M) = 0$ for all $r \in \mathbb{Z}$.

Proof. First observe that by dimension shifting it is enough to show the "weaker" conclusion $H_T^r(G, M) = 0$ for all $r \geqslant 1$. In fact, writing an exact sequence of G-modules

$$0 \to N \to P \to M \to 0$$

for some induced G-module P (which is thus also induced as an H-module for all $H \leqslant G$) we obtain an isomorphism $H_T^{j+1}(H, N) = H_T^j(H, M)$ for all $H \leqslant G$ and $j \in \mathbb{Z}$. Hence if we know that $H_T^r(H, M) = 0$ for all $r \geqslant 1$ then we know that $H_T^r(H, N) = 0$ for all $r \geqslant 2$, and applying the "weak twin number criterion" to N in place of M we conclude that $H_T^1(H, N) = 0$ as well, that is $H_T^0(H, M) = 0$, so that the conclusion of the weak criterion holds also for $r \geqslant 0$. Proceeding inductively in this manner, we extend the result to all $r \in \mathbb{Z}$.
A similar proof using dimension shifting (check!) also allows us to assume that $i = 1$ (or any other fixed number we deem convenient). Hence from now on we assume that $H^1(H, M) = H^2(H, M) = 0$ for all $H \leqslant G$ and prove that these conditions imply that $H^r(G, M) = 0$ for all $r \geqslant 1$.
Since $H^r(G, M)$ is a torsion abelian group, it is enough to show that its p-primary component vanishes for all prime numbers p. Let G_p be a p-Sylow subgroup of G. A restriction-corestriction argument shows that $\mathrm{res}\colon H^r(G, M) \to H^r(G_p, M)$ is injective on p-primary components, thus it is enough to show that $H^r(G_p, M) = 0$.
Hence we may assume that G is solvable and proceed by induction on the order of G. If G is cyclic, the theorem follows from the periodicity of cohomology. Now let $H \lhd G$ be a proper normal subgroup such that G/H is cyclic. By induction $H^r(H, M) = 0$ for all $r \geqslant 1$ and hence we have an exact inflation-restriction sequence

$$0 \to H^r(G/H, M^H) \to H^r(G, M) \to H^r(H, M) = 0$$

for all $r \geqslant 1$. Therefore $H^r(G/H, M^H) = H^r(G, M)$ for all $r \geqslant 1$ and in particular $H^1(G, M) = H^2(G, M) = 0$ implies that $H^1(G/H, M^H) = H^2(G/H, M^H) = 0$. But G/H is cyclic, so periodicity yields $H^r(G/H, M^H) = 0$ for all $r \geqslant 1$, and hence $H^r(G, M) = 0$ for all $r \geqslant 1$ too. $\square$

Now we can prove the main result of this section.

Theorem 2.5.2 (Tate-Nakayama)**.** *Let G be a finite group and let C be a G-module such that for all subgroups $H \leqslant G$*

(i) $H^1(H, C) = 0$

(ii) $H^2(H, C)$ *is cyclic of order* $|H|$

Let γ *be a generator of* $H^2(G, C)$*. Then for all* $r \in \mathbb{Z}$ *the cup product with* γ *gives an isomorphism*

$$\boxed{H_T^r(G, \mathbb{Z}) \underset{\approx}{\overset{\cup\gamma}{\to}} H_T^{r+2}(G, C)}$$

Observe that if $\gamma \in H^2(G, C)$ is a generator then $\mathrm{res}(\gamma) \in H^2(H, C)$ is also a generator since $\mathrm{cor} \circ \mathrm{res}(\gamma) = [G : H] \cdot \gamma$ has order $|H|$, hence $\mathrm{res}(\gamma)$ must have order $|H|$ as well in view of 2.

Proof. The key idea of the proof is to apply a "double dimension shifting" (wow!) and for that we construct a cohomologically trivial G-module $C(\gamma)$ fitting into an exact sequence

$$0 \to C \to C(\gamma) \to I_G \to 0 \tag{$*$}$$

Here $I_G = \langle \sigma - 1 \mid \sigma \in G \rangle$ is the kernel of the augmentation map $\mathbb{Z}[G] \to \mathbb{Z}$, so that we have an exact sequence of G-modules

$$0 \to I_G \to \mathbb{Z}[G] \to \mathbb{Z} \to 0 \tag{$**$}$$

Once $C(\gamma)$ is constructed, the proof of the theorem follows easily: from $(**)$, using the fact that $\mathbb{Z}[G]$ has trivial cohomology, we conclude that the connecting map $\delta_a \colon H_T^r(G, \mathbb{Z}) \approx H_T^{r+1}(G, I_G)$ is an isomorphism; similarly, from $(*)$, we have that the connecting map $\delta_c \colon H_T^{r+1}(G, I_G) \approx H_T^{r+2}(G, C)$ is also an isomorphism. The composition $\delta_c \circ \delta_a$ gives the desired isomorphism $H_T^r(G, \mathbb{Z}) \approx H_T^{r+2}(G, C)$, which equals to the cup product with γ, as we later show.

Let c be a 2-cocycle representing γ. Since we wish $H^2(G, C(\gamma))$ to vanish, the idea is to construct $C(\gamma)$ so that c becomes a coboundary in $C(\gamma)$. Take $C(\gamma)$ to be the direct sum of C with the free abelian group with basis x_σ, $\sigma \in G$, $\sigma \neq 1$:

$$C(\gamma) \overset{\text{def}}{=} C \oplus \bigoplus_{\substack{\sigma \in G \\ \sigma \neq 1}} \mathbb{Z} x_\sigma$$

We extend the G-action from C to $C(\gamma)$ in such a way that c becomes the coboundary of $\sigma \mapsto x_\sigma$:

$$\sigma \cdot x_\tau = x_{\sigma\tau} - x_\sigma + c(\sigma, \tau)$$

where we interpret "x_1" to be $c(1, 1)$. The 2-cocycle relation then guarantees that $1 \cdot x_\tau = x_\tau$ and $(\rho\sigma) \cdot x_\tau = \rho \cdot (\sigma \cdot x_\tau)$ hold for all $\rho, \sigma, \tau \in G$ (check!), turning $C(\gamma)$ into a G-module containing C as a G-submodule. Finally the map $C(\gamma) \to I_G$ in $(*)$ is given by $x_\sigma \mapsto \sigma - 1$ for $\sigma \neq 1$, and is identically zero on C. Another easy check shows that the latter map preserves the G-action.

In order to show that $C(\gamma)$ is cohomologically trivial we apply the twin number vanishing criterion: we need to verify that $H^1(H, C(\gamma)) = H^2(H, C(\gamma)) = 0$ for all subgroups $H \leqslant G$. Here the hypotheses 1 and 2 of the theorem come into play. First observe that from $(**)$ and the explicit description of the connecting map in terms of the standard resolution we obtain

(i) $H^1(H, I_G) = H_T^0(H, \mathbb{Z})$ is cyclic of order $|H|$ with generator given by the class $[f]$ of the 1-cocycle $f(\sigma) = \sigma - 1$, $\sigma \in H$.

(ii) $H^2(H, I_G) = H^1(H, \mathbb{Z}) = \mathrm{Hom}(H, \mathbb{Z}) = 0$

From $(*)$ we have an exact sequence

$$\begin{aligned} 0 = H^1(H, C) &\to H^1(H, C(\gamma)) \to H^1(H, I_G) \\ &\xrightarrow{\delta} H^2(H, C) \to H^2(H, C(\gamma)) \to H^2(H, I_G) = 0 \end{aligned}$$

Hence everything falls through if we can show that the connecting map δ is an isomorphism. At least we know that both $H^1(H, I_G)$ and $H^2(H, C)$ are cyclic groups of order $|H|$, so it is enough to show that δ is surjective. We show by explicit computation that $\delta([f]) = \mathrm{res}(\gamma)$ where $[f] \in H^1(H, I_G)$ is as in (i). First we lift f to the function $\tilde{f}\colon H \to C(\gamma)$ given by $\tilde{f}(\sigma) = x_\sigma$. Now $(d\tilde{f})(\sigma, \tau) = \sigma \cdot x_\tau - x_{\sigma\tau} + x_\sigma = c(\sigma, \tau)$ for all $\sigma, \tau \in G$, where d denotes the coboundary map, hence $\delta([f]) = [d\tilde{f}] = [c] = \mathrm{res}(\gamma)$, as required.
Finally, we verify that the composition $\delta_c \circ \delta_a$ of the two connecting maps $\delta_a\colon H_T^r(G, \mathbb{Z}) \xrightarrow{\approx} H_T^{r+1}(G, I_G)$ and $\delta_c\colon H_T^{r+1}(G, I_G) \xrightarrow{\approx} H_T^{r+2}(G, C)$ is indeed the cup product with γ. Denote by $\mu \in H_T^0(G, \mathbb{Z}) = \mathbb{Z}/|G|$ the generator 1 mod $|G|$ and by $\phi = [f] = \delta_a(\mu) \in H^1(G, I_G)$ where f is as in (i). By the compatibility of cup products with the connecting maps we obtain, for all $\alpha \in H_T^r(G, \mathbb{Z})$,

$$\begin{aligned} \alpha \cup \gamma = \alpha \cup \delta_c(\phi) &= (-1)^r \cdot \delta_c(\alpha \cup \phi) = (-1)^r \cdot \delta_c\big(\alpha \cup \delta_a(\mu)\big) \\ &= (-1)^r \cdot \delta_c\big((-1)^r \cdot \delta_a(\alpha \cup \mu)\big) = (-1)^r \cdot \delta_c\big((-1)^r \cdot \delta_a(\alpha)\big) = \delta_c \circ \delta_a(\alpha) \end{aligned}$$

since $- \cup \mu$ is the identity on $H_T^r(G, \mathbb{Z})$. □

Recall that $H_T^{-2}(G, \mathbb{Z}) = H_T^{-1}(G, I_G) = I_G/I_G^2$ and that $G^{ab} = I_G/I_G^2$ via the isomorphism $\sigma \cdot [G : G] \mapsto (\sigma - 1) \cdot I_G^2$. As a consequence we obtain

Corollary 2.5.3 ("Abstract" Reciprocity)**.** *With the notation and hypotheses of the Tate-Nakayama theorem, we have an isomorphism*

$$\boxed{\frac{C^G}{N_G(C)} = H_T^0(G, C) \approx H_T^{-2}(G, \mathbb{Z}) = G^{ab}}$$

We wish to apply the Tate-Nakayama's theorem to $G = \mathrm{Gal}(L/K)$ and $C = L^\times$ where $L \supset K$ is a finite Galois extension of local fields. Hence we need to verify that, for all subgroups $H \leqslant G$,

(i) $H^1(H, L^\times) = 0$;

(ii) $H^2(H, L^\times)$ is cyclic of order $|H|$.

The first condition is just Hilbert Satz 90, so we are left to compute $H^2(G, L^\times)$. This is not too difficult since we know the structure of $L^\times$ and since G is solvable so we will be able to reduce everything to the cyclic case. The starting point is to look at the cyclic factor G/G_0 corresponding to the maximal unramified extension of K in L. This is done in the next chapter.

Chapter 3

Local Class Field Theory

Local Class Field Theory is the study of finite **abelian extensions** $L \supseteq K$ of a local field K (i.e., finite Galois extensions $L \supseteq K$ with abelian Galois group $\mathrm{Gal}(L/K)$). For any such extension, the miraculous **local Artin reciprocity theorem** establishes a **canonical** isomorphism

$$\boxed{\theta_{L/K} \colon \frac{K^\times}{N_{L/K} L^\times} \xrightarrow{\approx} \mathrm{Gal}(L/K)}$$

Essentially the above isomorphism tells us that all information about such extensions is already contained, in an unexpectedly simple form, in the multiplicative group $K^\times$, whose structure is already known.

The local reciprocity theorem is beyond doubt one of the greatest achievements of contemporary Mathematics. Its (ravishingly beautiful!) proof will occupy us for most of this chapter.

Notation and General Remarks

Throughout this chapter, we adopt the following notations and conventions. For any field K we denote by

$$K_{sep} \stackrel{\text{def}}{=} \text{separable closure of } K$$
$$G_K \stackrel{\text{def}}{=} \mathrm{Gal}(K_{sep}/K) = \text{absolute Galois group of } K$$

Recall that G_K is a topological group: a topological basis consists of the left translates of the subgroups G_L where L runs over all finite extensions of K. Galois theory establishes a 1-1 correspondence between **closed** subgroups of G_K and fields between K and K_{sep}.

The group G_K is an example of a **profinite group** (i.e. a projective limit of finite groups): we have, both algebraically and topologically,

$$G_K \simeq \varprojlim_{L \supseteq K} \mathrm{Gal}(L/K)$$

where L runs over all finite Galois extensions of K and each $\mathrm{Gal}(L/K)$ is given the discrete topology.

Given any profinite group G, we denote by

$$\begin{aligned}
[G:G] &\overset{\text{def}}{=} \text{closure (in the profinite topology) of the commutator subgroup of } G \\
G^{ab} &\overset{\text{def}}{=} \frac{G}{[G:G]} = \text{maximal abelian quotient of } G
\end{aligned}$$

By the Galois correspondence we then have

$$\begin{aligned}
K^{ab} &\overset{\text{def}}{=} K_{sep}^{[G_K:G_K]} = \text{maximal abelian extension of } K \\
&\qquad\qquad = \text{compositum of all finite abelian extensions of } K \text{ inside } K_{sep} \\
G_K^{ab} &= \mathrm{Gal}(K^{ab}/K)
\end{aligned}$$

Finally, if K is a local field with residue field k we write (see section I.5)

$$\begin{aligned}
K_{nr} &\overset{\text{def}}{=} \text{maximal unramified extension of } K \text{ in } K_{sep} \\
&\quad = \text{compositum of all finite unramified extensions of } K \text{ inside } K_{sep} \\
G_K^{nr} &\overset{\text{def}}{=} \mathrm{Gal}(K_{nr}/K) \approx G_k = \widehat{\mathbb{Z}} \\
\mathrm{Frob}_K &\overset{\text{def}}{=} \text{Frobenius automorphism of } K_{nr} \supseteq K
\end{aligned}$$

where

$$\widehat{\mathbb{Z}} \overset{\text{def}}{=} \varprojlim_{n\in\mathbb{N}} \mathbb{Z}/(n) = \prod_p \mathbb{Z}_p$$

and p runs over all prime integers. The last isomorphism follows from the Chinese Remainder Theorem (check!). The Frobenius map Frob_K is a topological generator of G_K^{nr}, corresponding to the element $1 \in \widehat{\mathbb{Z}}$ under the above isomorphism.

3.1 Main results

In this section we state the main results. We begin with

Theorem 3.1.1 (Local Artin Reciprocity)**.** *Let K be a local field. There exists a unique injective group morphism, called* **local Artin map***,*

$$\theta_K : K^\times \hookrightarrow G_K^{ab}$$

that is "almost an isomorphism" in the following sense: for any finite abelian extension $L \supseteq K$, the composition

$$\theta_{L/K} : K^\times \overset{\theta_K}{\to} G_K^{ab} \overset{can.}{\twoheadrightarrow} \mathrm{Gal}(L/K)$$

of θ_K with the canonical projection satisfies:

(i) $\theta_{L/K}$ is surjective with kernel given by the **norm group** $N_{L/K}L^\times$. *Thus we have an induced isomorphism (which we still denote by $\theta_{L/K}$)*

$$\boxed{\theta_{L/K}\colon \frac{K^\times}{N_{L/K}(L^\times)} \xrightarrow{\approx} \mathrm{Gal}(L/K)}$$

(ii) if $L \supseteq K$ is unramified with Frobenius automorphism $\mathrm{Frob}_{L/K} \in \mathrm{Gal}(L/K)$, and $v\colon K^\times \to \mathbb{Z}$ denotes the normalised valuation of K, then

$$\boxed{\theta_{L/K}(a) = \mathrm{Frob}_{L/K}^{v(a)}}$$

for all $a \in K^\times$.

The first property is a "compatibility" one in that it says that the "big" Artin map θ_K is compatible with its "finite layers" in the sense that the diagram

$$\begin{tikzcd}
K^\times \arrow[r, hook, "\theta_K"] \arrow[d, two heads, "\text{can.}"] & G_K^{ab} \arrow[d, two heads, "\text{can.}"] \\
\dfrac{K^\times}{N_{L/K}(L^\times)} \arrow[r, "\theta_{L/K}", "\approx"'] & \mathrm{Gal}(L/K)
\end{tikzcd}$$

commutes. This implies that for all finite extensions $M \supseteq L \supseteq K$ with $M \supseteq K$ (and thus $L \supseteq K$) abelian we have a commutative diagram

$$\begin{tikzcd}
\dfrac{K^\times}{N_{M/K}(M^\times)} \arrow[r, "\theta_{M/K}", "\approx"'] \arrow[d, two heads, "\text{can.}"] & \mathrm{Gal}(M/K) \arrow[d, two heads, "\text{can.}"] \\
\dfrac{K^\times}{N_{L/K}(L^\times)} \arrow[r, "\theta_{L/K}", "\approx"'] & \mathrm{Gal}(L/K)
\end{tikzcd}$$

Conversely, to give θ_K is equivalent to give maps $\theta_{L/K}$, one for each finite abelian extension $L \supseteq K$, compatible in the above sense.

The second property is a "normalisation" property in that it fixes the value of the local Artin map for unramified extensions. Let $L \supseteq K$ be an unramified extension of degree n and let π be a uniformiser of K. From (i) and (ii) we conclude that $\pi^{n\mathbb{Z}} \cdot U_K = \ker \theta_{L/K} = N_{L/K}L^\times$ and therefore (see the explicit formula for the valuation of L in theorem 1.3.4) that the norm map $N_{L/K}\colon U_L \twoheadrightarrow U_K$ is surjective on units for unramified extensions (c.f. corollary 3.2.5).

The "functorial" properties of the reciprocity map immediately imply:

Theorem 3.1.2. *Let K be a local field and let L and L' be finite abelian extensions of K. Then*

(i) $L' \supseteq L \iff N_{L'/K}L'^{\times} \subseteq N_{L/K}L^{\times}$;

(ii) $N_{(L'\cap L)/K}(L' \cap L)^{\times} = N_{L'/K}L'^{\times} \cdot N_{L/K}L^{\times}$;

(iii) $N_{(L'\cdot L)/K}(L' \cdot L)^{\times} = N_{L'/K}L'^{\times} \cap N_{L/K}L^{\times}$.

Proof. We prove (iii) as an example and leave the other two as exercises. We have a commutative diagram

$$\begin{tikzcd} K^{\times} \arrow[r, "\theta_{(L'\cdot L)/K}"] \arrow[d, hook] & \mathrm{Gal}\big((L'\cdot L)/K\big) \arrow[d, hook, "\text{can.}"] \\ K^{\times} \arrow[r, "\theta_{L'/K}\times\theta_{L/K}"] & \mathrm{Gal}(L'/K) \times \mathrm{Gal}(L/K) \end{tikzcd}$$

where the right vertical arrow is injective. Hence

$$\begin{aligned} N_{(L'\cdot L)/K}(L' \cdot L)^{\times} &= \ker \theta_{(L'\cdot L)/K} = \ker(\theta_{L'/K} \times \theta_{L/K}) \\ &= \ker \theta_{L'/K} \cap \ker \theta_{L/K} = N_{L'/K}L'^{\times} \cap N_{L/K}L^{\times} \end{aligned}$$

□

Observe that (i) of the last theorem shows that there is a one-to-one containment reversing correspondence between finite abelian extensions of K and **norm groups** of $K^{\times}$, that is, subgroups of $K^{\times}$ which are of the form $N_{L/K}L^{\times}$ for some finite abelian extension $L \supseteq K$. The **local existence theorem** tells us which subgroups of $K^{\times}$ are norm groups.

Theorem 3.1.3 (Local Existence)**.** *Let K be a local field. A subgroup of finite index of $K^{\times}$ is a norm group if and only if it is open. Hence there is a 1-1 containment reversing correspondence*

$$\begin{aligned} \{\textit{finite abelian extensions of } K\} &\leftrightarrow \{\textit{open subgroups of } K^{\times} \textit{ of finite index}\} \\ L &\mapsto N_{L/K}L^{\times} \end{aligned}$$

Remark 3.1.4. *When* $\operatorname{char} K = 0$*, a subgroup $T \subseteq K^{\times}$ of finite index is automatically open: since T has finite index, $T \supseteq (K^{\times})^n$ for some n, and $(K^{\times})^n \supseteq U_K^{(m)}$ for m sufficiently large (exercise!). Hence if* $\operatorname{char} K = 0$ *there is a perfect 1-1 correspondence between subgroups of $K^{\times}$ of finite index and finite abelian extensions of K!*

In the first chapter, we gave a complete description of the unit group $K^{\times}$. Hence with the reciprocity and existence theorems we obtain a complete classification of **all** abelian extensions of a local field K! The proofs of these two deep theorems will be given later. In this section we give some applications to impress you with the power of these results.

Example 3.1.5. *Consider the Galois extension $M = \mathbb{Q}_3(\sqrt{2}, \sqrt{3})$ of $\mathbb{Q}_3$ with Galois group $\mathrm{Gal}(M/\mathbb{Q}_3) \cong \mathbb{Z}/2 \times \mathbb{Z}/2$, generated by automorphisms σ and τ given by*

$$\begin{cases} \sigma(\sqrt{2}) = -\sqrt{2} \\ \sigma(\sqrt{3}) = \sqrt{3} \end{cases} \qquad \begin{cases} \tau(\sqrt{2}) = \sqrt{2} \\ \tau(\sqrt{3}) = -\sqrt{3} \end{cases}$$

The lattice of subfields is

$$\begin{tikzcd}
 & M = \mathbb{Q}_3(\sqrt{2}, \sqrt{3}) \arrow[dl, dash, "\tau"'] \arrow[d, dash, "\sigma"] \arrow[dr, dash, "\sigma\tau"] & \\
L_0 = \mathbb{Q}_3(\sqrt{2}) \arrow[dr, dash, "\sigma|_{L_0}"'] & L_1 = \mathbb{Q}_3(\sqrt{3}) \arrow[d, dash, "\tau|_{L_1}"] & L_2 = \mathbb{Q}_3(\sqrt{6}) \arrow[dl, dash, "\sigma|_{L_2} = \tau|_{L_2}"] \\
 & K = \mathbb{Q}_3 &
\end{tikzcd}$$

Now we now identify the corresponding norm subgroups in (see section 2 of chapter I)

$$\mathbb{Q}_3^\times = 3^{\mathbb{Z}} \times \{\pm 1\} \times U_{\mathbb{Q}_3}^{(1)}$$

Since $[M : K] = 4$ and $[L_i : K] = 2$ for $i = 0, 1, 2$, the corresponding subgroups of $\mathbb{Q}_3^\times$ have indices dividing 4, thus all of them contain $(\mathbb{Q}_3^\times)^4$, and by the correspondence theorem they are in bijection with subgroups of indices 2 and 4 of the finite group

$$\mathbb{Q}_3^\times / (\mathbb{Q}_3^\times)^4 \cong \mathbb{Z}/4\mathbb{Z} \times \{\pm 1\}$$

(notice that $U_{\mathbb{Q}_3}^{(1)} \cong \mathbb{Z}_3$ by 1.4.1, thus $(U_{\mathbb{Q}_3}^{(1)})^4 = U_{\mathbb{Q}_3}^{(1)}$). Next,

- *since $L_0 \supseteq \mathbb{Q}_3$ is unramified we know that $N_{L_0/\mathbb{Q}_3}(L_0^\times) = 3^{2\mathbb{Z}} \times U_{\mathbb{Q}_3}$, corresponding to the index 2 subgroup $\{\overline{0}, \overline{2}\} \times \{\pm 1\} \subset \mathbb{Z}/4\mathbb{Z} \times \{\pm 1\}$.*
- *since $-3 \in N_{L_1/\mathbb{Q}_3}(L_1^\times)$ and $-3 = 3^1 \times (-1)$ generates the index 2 subgroup*

  $$\langle(\overline{1}, -1)\rangle = \{(\overline{0}, 1);\ (\overline{1}, -1);\ (\overline{2}, 1);\ (\overline{3}, -1)\} \subset \mathbb{Z}/4\mathbb{Z} \times \{\pm 1\} \cong \mathbb{Q}_3^\times / (\mathbb{Q}_3^\times)^4$$

  we conclude that $N_{L_1/\mathbb{Q}_3}(L_1^\times) = (-3)^{\mathbb{Z}} \times U_{\mathbb{Q}_3}^{(1)}$. Similarly, since $-6 \in N_{L_2/\mathbb{Q}_3}(L_2^\times)$ and $-2 \in U_{\mathbb{Q}_3}^{(1)}$, we have that $N_{L_2/\mathbb{Q}_3}(L_2^\times) = 3^{\mathbb{Z}} \times U_{\mathbb{Q}_3}^{(1)}$.
- *finally, M is the compositum of L_0 and L_2, therefore*

  $$\begin{aligned} N_{M/\mathbb{Q}_3}(M^\times) &= N_{L_0/\mathbb{Q}_3}(L_0^\times) \cap N_{L_2/\mathbb{Q}_3}(L_2^\times) = ((-3)^{\mathbb{Z}} \times U_{\mathbb{Q}_3}^{(1)}) \cap (3^{\mathbb{Z}} \times U_{\mathbb{Q}_3}^{(1)}) \\ &= 3^{2\mathbb{Z}} \times U_{\mathbb{Q}_3}^{(1)}) \end{aligned}$$

Putting everything together, we obtain the following lattice of subgroups of $\mathbb{Q}_3^\times$, drawn upside down:

$$\begin{tikzcd}
 & N_{M/\mathbb{Q}_3}(M^\times) = 3^{2\mathbb{Z}} \times U_{\mathbb{Q}_3}^{(1)} \arrow[dl, hook] \arrow[d, hook] \arrow[dr, hook] & \\
N_{L_0/\mathbb{Q}_3}(L_0^\times) = 3^{2\mathbb{Z}} \times \{\pm 1\} \times U_{\mathbb{Q}_3}^{(1)} \arrow[dr, hook] & N_{L_1/\mathbb{Q}_3}(L_1^\times) = (-3)^{\mathbb{Z}} \times U_{\mathbb{Q}_3}^{(1)} \arrow[d, hook] & N_{L_2/\mathbb{Q}_3}(L_2^\times) = 3^{\mathbb{Z}} \times U_{\mathbb{Q}_3}^{(1)} \arrow[dl, hook] \\
 & \mathbb{Q}_3^\times = 3^{\mathbb{Z}} \times \{\pm 1\} \times U_{\mathbb{Q}_3}^{(1)} &
\end{tikzcd}$$

Finally, we have that

$$\begin{cases} \theta_{L_0/\mathbb{Q}_3}\big(-1 \cdot N_{L_0/\mathbb{Q}_3}(L_0^\times)\big) & = 1 \\ \theta_{L_1/\mathbb{Q}_3}\big(-3 \cdot N_{L_1/\mathbb{Q}_3}(L_1^\times)\big) & = 1 \\ \theta_{L_2/\mathbb{Q}_3}\big(3 \cdot N_{L_2/\mathbb{Q}_3}(L_0^\times)\big) & = 1 \end{cases} \implies \begin{cases} \theta_{M/\mathbb{Q}_3}\big(-1 \cdot N_{M/\mathbb{Q}_3}(M^\times)\big) & = \tau \\ \theta_{M/\mathbb{Q}_3}\big(-3 \cdot N_{M/\mathbb{Q}_3}(M^\times)\big) & = \sigma \\ \theta_{M/\mathbb{Q}_3}\big(3 \cdot N_{M/\mathbb{Q}_3}(M^\times)\big) & = \sigma\tau \end{cases}$$

which completely determines $\theta_{M/\mathbb{Q}_3}$.

3.1.1 Some applications

For the next, we need the following explicit computation of norm groups of cyclotomic extensions.

Lemma 3.1.6. *Let n be a positive integer and consider the cyclotomic extension $L = \mathbb{Q}_p(\zeta_{p^n})$ of $\mathbb{Q}_p$ where ζ_{p^n} denotes a primitive p^n-th root of unity. Then the extension $L \supseteq \mathbb{Q}_p$ is totally ramified of degree $[L : \mathbb{Q}_p] = \phi(p^n) = (p-1) \cdot p^{n-1}$ (ϕ denotes the Euler function) and $1 - \zeta_{p^n}$ is a uniformiser of L. The corresponding norm group is*

$$N_{L/\mathbb{Q}_p}(L^\times) = p^{\mathbb{Z}} \cdot U_{\mathbb{Q}_p}^{(n)}$$

Hence the local Artin map gives a canonical isomorphism $U_{\mathbb{Q}_p}/U_{\mathbb{Q}_p}^{(n)} \approx \mathrm{Gal}(L/\mathbb{Q}_p)$ (which are canonically isomorphic to $(\mathbb{Z}/p^n)^\times$).

Proof. The polynomial in $\mathbb{Q}_p[x]$

$$f(x) \stackrel{\text{def}}{=} \frac{x^{p^n} - 1}{x^{p^{n-1}} - 1} = x^{(p-1)p^{n-1}} + x^{(p-2)p^{n-1}} + \cdots + x^{p^{n-1}} + 1$$

is irreducible by the usual argument combining Gauß' lemma and Eisenstein's criterion applied to $f(x+1)$. Hence $f(x)$ is the minimal polynomial of ζ_{p^n} over $\mathbb{Q}_p$ and thus $[L : \mathbb{Q}_p] = \phi(p^n) = (p-1) \cdot p^{n-1}$. Also

$$N_{L/K}(1 - \zeta_{p^n}) = \prod_{\substack{0 < i < p^n \\ p \nmid i}} (1 - \zeta_{p^n}^i) = f(1) = p$$

and from the explicit formula of theorem 1.3.4 we conclude that L is totally ramified over $\mathbb{Q}_p$ with uniformiser $1 - \zeta_{p^n}$ and residue field $\mathbb{F}_p$.

The last expression also shows that all powers of p belong to $N_{L/\mathbb{Q}_p}(L^\times)$. By computations of section I.4, we have that $\mathbb{Q}_p^\times = p^{\mathbb{Z}} \times \mu_{p-1} \times U_{\mathbb{Q}_p}^{(1)}$ and $L^\times \cong (\zeta_{p^n}-1)^{\mathbb{Z}} \times \mu_{p-1} \times U_L^{(1)}$, where $\mu_{p-1} \subseteq \mathbb{Q}_p^\times$ is the group of $(p-1)$-th roots of unity. Thus to show that $N_{L/\mathbb{Q}_p}(L^\times) = p^{\mathbb{Z}} \cdot U_{\mathbb{Q}_p}^{(n)}$ it suffices to show that $N_{L/\mathbb{Q}_p} U_L^{(1)} \supseteq U_{\mathbb{Q}_p}^{(n)}$. In fact, in that case $N_{L/\mathbb{Q}_p}(L^\times)$ will have index at most $|\mu_{p-1}| \cdot [U_{\mathbb{Q}_p}^{(1)} : U_{\mathbb{Q}_p}^{(n)}] = (p-1) \cdot p^{n-1}$ in $\mathbb{Q}_p^\times$, but from the local Artin reciprocity theorem we know that this index is precisely $(p-1) \cdot p^{n-1} = [L : \mathbb{Q}_p]$.

By 1.4.1, the exp and log functions give an isomorphism $U_{\mathbb{Q}_p}^{(i)} \cong (p^i)$ for $i \geqslant 1$ (respectively $i \geqslant 2$) when $p \neq 2$ (respectively $p = 2$). Hence, when $p \neq 2$ the map $x \mapsto x^{(p-1)\cdot p^{n-1}}$ gives an isomorphism $U_{\mathbb{Q}_p}^{(1)} \cong U_{\mathbb{Q}_p}^{(n)}$ since $x \mapsto (p-1) \cdot p^{n-1} \cdot x$ gives an isomorphism $(p) \cong (p^n)$, proving that $N_{L/\mathbb{Q}_p} U_L^{(1)} \supseteq \left(U_{\mathbb{Q}_p}^{(1)}\right)^{[L:\mathbb{Q}_p]} = U_{\mathbb{Q}_p}^{(n)}$ in this case. When $p = 2$ and $n \geqslant 2$ (for $n = 1$, $L = \mathbb{Q}_2$), we have that $x \mapsto x^{2^{n-1}}$ gives an isomorphism $U_{\mathbb{Q}_2}^{(2)} \cong U_{\mathbb{Q}_p}^{(n+1)}$ and only get that $N_{L/\mathbb{Q}_2} U_L^{(1)} \supseteq U_{\mathbb{Q}_2}^{(n+1)}$. However an explicit computation shows that $5^{2^{n-2}} = N_{L/\mathbb{Q}_p}(2+i)$, where $i = (\zeta_{2^n})^{2^{n-2}}$ is a primitive 4-th root of 1, and since $U_{\mathbb{Q}_2}^{(2)} = U_{\mathbb{Q}_2}^{(3)} \cup 5 \cdot U_{\mathbb{Q}_2}^{(3)}$ applying $x \mapsto x^{2^{n-2}}$ gives $U_{\mathbb{Q}_2}^{(n)} = U_{\mathbb{Q}_2}^{(n+1)} \cup 5^{2^{n-2}} \cdot U_{\mathbb{Q}_2}^{(n+1)}$, proving that $N_{L/\mathbb{Q}_2} U_L^{(1)} \supseteq U_{\mathbb{Q}_2}^{(n)}$ in this case as well. □

Remark 3.1.7. *It can be shown (using formal groups à la Lubin-Tate, see for instance Neukirch's book or Serre's article in Cassels-Fröhlich) that the isomorphism $U_{\mathbb{Q}_p}/U_{\mathbb{Q}_p}^{(n)} \approx \mathrm{Gal}(L/\mathbb{Q}_p)$ takes $u \bmod U_{\mathbb{Q}_p}^{(n)}$ to the automorphism given by $\zeta_{p^n} \mapsto \zeta_{p^n}^{u^{-1}}$.*

We are now ready to show the celebrated

Theorem 3.1.8 (Local Kronecker-Weber)**.** *Every finite abelian extension of $\mathbb{Q}_p$ is contained in a cyclotomic extension.*

Proof. Let $L \supseteq \mathbb{Q}_p$ be a finite abelian extension. Since $N_{L/\mathbb{Q}_p} L^\times$ has finite index in $\mathbb{Q}_p^\times$, there exists m such that $p^m \in N_{L/\mathbb{Q}_p} L^\times$, and since $N_{L/\mathbb{Q}_p} L^\times$ is open, there exists n such that $N_{L/\mathbb{Q}_p} L^\times \supseteq U_{\mathbb{Q}_p}^{(n)}$. Let $N = p^n \cdot (p^m - 1)$. We claim that $N_{\mathbb{Q}_p(\zeta_N)/\mathbb{Q}_p} \mathbb{Q}_p(\zeta_N)^\times = p^{m\mathbb{Z}} \times U_{\mathbb{Q}_p}^{(n)} \subseteq N_{L/\mathbb{Q}_p} L^\times$, which in turn implies that $\mathbb{Q}_p(\zeta_N) \supseteq L$, finishing the proof of the theorem.

To prove the claim, observe that $\mathbb{Q}_p(\zeta_N)$ is the compositum of $M_0 = \mathbb{Q}_p(\zeta_{p^m-1})$ and $M_1 = \mathbb{Q}_p(\zeta_{p^n})$, hence the norm group of $\mathbb{Q}_p(\zeta_N)$ is the intersection of the norm groups of M_0 and M_1. By the lemma, we already know that $N_{M_1/\mathbb{Q}_p} M_1^\times = p^{\mathbb{Z}} \times U_{\mathbb{Q}_p}^{(n)}$. On the other hand, we show below that M_0 is unramified over $\mathbb{Q}_p$ of degree m and hence has norm group $p^{m\mathbb{Z}} \times U_{\mathbb{Q}_p}$. Putting everything together, we conclude that $\mathbb{Q}_p(\zeta_N)$ has norm group $p^{m\mathbb{Z}} \times U_{\mathbb{Q}_p}^{(n)}$ as claimed.

Finally, to show that M_0 is the degree m unramified extension of $\mathbb{Q}_p$, let $f(x) \in \mathbb{Z}_p[x]$ be the minimal polynomial of ζ_{p^n-1} over $\mathbb{Q}_p$ and let $\bar{f}(x) \in \mathbb{F}_p[x]$ denote the image of $f(x)$ in $\mathbb{F}_p[x]$. Since $x^{p^m-1} - 1$ is separable over $\mathbb{F}_p[x]$, by Hensel's lemma the irreducible factors of $x^{p^m-1} - 1$ over $\mathbb{Q}_p$ and over $\mathbb{F}_p$ are in 1-1 correspondence, thus $\bar{f}(x)$ is irreducible over $\mathbb{F}_p[x]$. Hence $[M_0 : \mathbb{Q}_p] = \deg f(x) = \deg \bar{f}(x) \leqslant f_{M_0/\mathbb{Q}_p}$, but since we always have $[M_0 : \mathbb{Q}_p] \geqslant f_{M_0/\mathbb{Q}_p}$, equality holds (compare with 1.5.4). On the other hand M_0 and $\mathbb{F}_{p^m}$ are the splitting fields of $x^{p^m-1} - 1$ over $\mathbb{Q}_p$ and $\mathbb{F}_p$, respectively. Hence m is the largest of the degrees of the irreducible factors of $x^{p^m-1} - 1$ over $\mathbb{Q}_p$ and $\mathbb{F}_p$. This shows that $\deg \bar{f}(x) \leqslant m$ and also $[M_0 : \mathbb{Q}_p] \geqslant m$, hence both are equal to m. □

Example 3.1.9. *We have that $\mathbb{Q}_3(\sqrt{2}, \sqrt{3})$ has norm group $3^{2\mathbb{Z}} \times U_{\mathbb{Q}_3}^{(1)}$, so by the above proof we have that $\mathbb{Q}_3(\sqrt{2}, \sqrt{3}) \subseteq \mathbb{Q}_3(\zeta_{24})$. Explicitly, writing $i = \zeta_{24}^6$, a primitive 4-th root of 1, we have that*

$$\zeta_8 = \frac{\sqrt{2} + i\sqrt{2}}{2} \qquad \text{and} \qquad \zeta_3 = \frac{-1 + i\sqrt{3}}{2}$$

Hence $\sqrt{2} = 2\zeta_{24}^3 \cdot (1 + \zeta_{24}^6)^{-1}$ and $\sqrt{3} = (2\zeta_{24}^8 + 1) \cdot \zeta_{24}^{-6}$.

As an illustration of how the study of local fields is relevant in addressing global questions, we show how to derive the global Kronecker-Weber from the local one.

Theorem 3.1.10 (Kronecker-Weber)**.** *Every finite abelian extension of $\mathbb{Q}$ is contained in a cyclotomic extension.*

Proof. Let L be a finite abelian extension of $\mathbb{Q}$. The idea is to find n such that $M \overset{\text{def}}{=} L(\zeta_n)$ has exactly the same ramification data as $\mathbb{Q}(\zeta_n)$, that is, for each prime $p \in \mathbb{Z}$ the ramification degrees of the prime ideals of M and $\mathbb{Q}(\zeta_n)$ lying above p are the equal. We claim that this forces $M = \mathbb{Q}(\zeta_n)$ and therefore $L \subset \mathbb{Q}(\zeta_n)$.

In fact, denote by $I_p \subset \mathrm{Gal}(M/\mathbb{Q})$ the inertia group of a prime ideal of M lying above p (they are all equal since M is abelian over $\mathbb{Q}$). Consider the subgroup I of $\mathrm{Gal}(M/\mathbb{Q})$ generated by all the I_p. The fixed field M^I is unramified over $\mathbb{Q}$, hence by Minkowski's theorem (see for instance Neukirch's book) we have that $M^I = \mathbb{Q}$ and hence $I = \mathrm{Gal}(M/\mathbb{Q})$. On the other hand, the ramification degree of a prime ideal of $\mathbb{Q}(\zeta_n)$ lying above p is the same as the ramification degree of $\mathbb{Q}_p(\zeta_n)$ over $\mathbb{Q}_p$. Factoring $n = \prod_p p^{e_p}$, we have that $\mathbb{Q}_p(\zeta_n)$ is the compositum of the totally ramified extension $\mathbb{Q}_p(\zeta_{p^{e_p}}) \supset \mathbb{Q}_p$ and the unramified extension $\mathbb{Q}_p(\zeta_{n/p^{e_p}}) \supset \mathbb{Q}_p$ (see the proof of the local Kronecker-Weber), hence this ramification degree is $\phi(p^{e_p})$ (see 3.1.6). Therefore if M and $\mathbb{Q}(\zeta_n)$ have the same ramification data then $|I_p| = \phi(p^{e_p})$ for all p and

$$[M : \mathbb{Q}] = |I| \leqslant \prod_p |I_p| = \prod_p \phi(p^{e_p}) = \phi(n) = [\mathbb{Q}(\zeta_n) : \mathbb{Q}]$$

But since $M \supset \mathbb{Q}(\zeta_n)$, the two fields must be equal, as claimed.

Now we construct n with the above property. For each prime $p \in \mathbb{Z}$, choose a prime ideal $\mathfrak{m}_p$ of L lying above p and denote by L_p the completion of L with respect to this prime (all completions are isomorphic since L is Galois over $\mathbb{Q}$). By the local Kronecker-Weber, there exists an integer $n_p = p^{e_p} \cdot m_p$ with $p \nmid m_p$ such that $L_p \subset \mathbb{Q}_p(\zeta_{n_p})$. Set $n = \prod_{p \in S} p^{e_p}$ where S denotes the (finite) set of primes that ramify in L. Denoting by M_p the completion of M with respect to a prime lying above $\mathfrak{m}_p$, we have to check that the ramification degree of $M_p \supset \mathbb{Q}_p$ equals $\phi(p^{e_p})$, the ramification degree of $\mathbb{Q}_p(\zeta_n) \supset \mathbb{Q}_p$. But since $M_p = L_p(\zeta_n)$ and since $L_p \subset \mathbb{Q}_p(\zeta_{n_p})$ we have that $M_p(\zeta_{m_p}) = \mathbb{Q}_p(\zeta_n)(\zeta_{m_p})$, which has ramification degree $\phi(p^{e_p})$ over $\mathbb{Q}_p$. Since $\mathbb{Q}_p(\zeta_{m_p}) \supset \mathbb{Q}_p$ is unramified we have that $M_p(\zeta_{m_p}) \supset M$ is also unramified by corollary 1.5.2 and thus $M \supset \mathbb{Q}_p$ also has ramification degree $\phi(p^{e_p})$, as was to be shown. $\square$

3.1.2 Outline of the proofs

We give a general overview of the proof of the local reciprocity. The first part of the proof, the Tate-Nakayama theorem, is a purely group theoretical result that describes the maximal abelian quotient G^{ab} of a finite group G in terms of a $\mathbb{Z}[G]$-module C satisfying some conditions. The second part of the proof is to show that for a finite Galois extension of local fields $L \supseteq K$ with $G = \mathrm{Gal}(L/K)$ we may take $C = L^\times$, and for that we will make use of our knowledge of the structure of $L^\times$. Once the reciprocity theorem is proved, the proof of the existence theorem is then not too difficult.

Lemma 3.1.11. *Let G be a profinite group. For $r \in \mathbb{Z}$, denote by*

$$\delta \colon H_T^{r-1}(G, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\approx} H_T^r(G, \mathbb{Z})$$

the coboundary map in the long exact sequence associated to the short exact sequence (with trivial G-action)

$$0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0 \tag{3.1}$$

Then

(i) For all r, δ is an isomorphism.

(ii) For $r = 1$, let $\chi \in H^1(G, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}_c(G, \mathbb{Q}/\mathbb{Z})$ be a continuous morphism of order n, so that $G/\ker\chi \cong \frac{1}{n}\mathbb{Z}/\mathbb{Z} \subseteq \mathbb{Q}/\mathbb{Z}$. Let $\sigma \in G$ be such that $\chi(\sigma) = \frac{1}{n} \bmod \mathbb{Z}$, and $\overline{\sigma} \in G/\ker\chi$ denote its image in the cyclic group $G/\ker\chi$; observe that $\overline{\sigma}$ is then a generator. We have

$$\delta\chi = \inf([f]) \in H^2(G, \mathbb{Z})$$

where $\inf\colon H^2(G/\ker\chi, \mathbb{Z}) \to H^2(G, \mathbb{Z})$ denotes the inflation map, and $[f]$ is the class of the 2-cocycle

$$f(\overline{\sigma}^i, \overline{\sigma}^j) = \begin{cases} 0 & \text{if } i + j < n \\ 1 & \text{otherwise} \end{cases} \qquad (i, j \in \{0, 1, \ldots, n-1\})$$

Proof. Item (i) follows simply by $H^r_T(G, \mathbb{Q}) = 0$ for all r. The latter in turn follows from $H^r_T(G, \mathbb{Q})$ being torsion (it is killed by $|G|$) and the fact that multiplication by any non-zero $n \in \mathbb{Z}$ gives an automorphism of $\mathbb{Q}$, and thus also of $H^r_T(G, \mathbb{Q})$.
Item (ii) follows from an explicit computation:

(a) First, we pick a lift $\widetilde{\chi}\colon G \to \mathbb{Q}$ of $\chi\colon G \to \mathbb{Q}/\mathbb{Z}$ by choosing representatives of $\mathbb{Q}/\mathbb{Z}$ in the rational unit inverval $[0, 1) \cap \mathbb{Q}$. Be careful: although $\chi\colon G \to \mathbb{Q}/\mathbb{Z}$ is a group morphism, $\widetilde{\chi}\colon G \to \mathbb{Q}$ is not! However, given $\alpha, \beta \in G$ with $\widetilde{\chi}(\alpha) = \frac{i}{n}$ and $\widetilde{\chi}(\beta) = \frac{j}{n}$ with $0 \leqslant i, j < n$, we have

$$\widetilde{\chi}(\alpha\beta) = \begin{cases} \frac{i+j}{n} & \text{if } i + j < n \\ \frac{i+j-n}{n} & \text{if } i + j \geqslant n \end{cases} \qquad \text{as } \chi(\alpha\beta) = \frac{i+j}{n} \bmod \mathbb{Z}$$

(b) Next, computing the coboundary map $\delta\widetilde{\chi}$ we get the 2-cocycle f: for $\alpha, \beta \in G$ as above,

$$\begin{aligned} (\delta\widetilde{\chi})(\alpha, \beta) &= \alpha\big(\widetilde{\chi}(\beta)\big) - \widetilde{\chi}(\alpha\beta) + \widetilde{\chi}(\alpha) \\ &= \begin{cases} \frac{j}{n} - \frac{i+j}{n} + \frac{i}{n} = 0 & \text{if } i + j < n \\ \frac{j}{n} - \frac{i+j-n}{n} + \frac{i}{n} = 1 & \text{otherwise} \end{cases} \qquad \text{in } \mathbb{Z} \end{aligned}$$

□

Lemma 3.1.12. *Let $L \supseteq K$ be a finite Galois field extension. Denote by*

$$\mathrm{Br}(L/K) \overset{\text{def}}{=} \ker\big(\mathrm{Br}(K) \overset{\text{res}}{\to} \mathrm{Br}(L)\big)$$

the **relative Brauer group** *with respect to $L \supseteq K$ (i.e., the subgroup of $\mathrm{Br}(K)$ consisting of classes $\beta \in \mathrm{Br}(K)$ that are killed by $L \supseteq K$).*

(a) *One has* $\mathrm{Br}(L/K) = H^2(\mathrm{Gal}(L/K), L^\times)$. *Moreover if* $M \supseteq L \supseteq K$ *is a tower of finite extensions with both* $L \supseteq K$ *and* $M \supseteq K$ *Galois then the inflation map*

$$\inf\colon \mathrm{Br}(L/K) = H^2(\mathrm{Gal}(L/K), L^\times) \hookrightarrow \mathrm{Br}(M/K) = H^2(\mathrm{Gal}(M/K), M^\times)$$

is injective, and one has

$$\mathrm{Br}(K) = \varinjlim_{\substack{L\supseteq K \text{ Galois} \\ [L:K]<+\infty}} \mathrm{Br}(L/K) = \bigcup_{\substack{L\supseteq K \text{ Galois} \\ [L:K]<+\infty}} \mathrm{Br}(L/K)$$

where L *runs over all finite extensions of* K *inside* K_{sep}*, with transition maps are given by inflation.*

(b) *(Kummer) Let* $n \in \mathbb{N}$ *be prime to* $\mathrm{char}(K)$. *Consider the short exact sequence of* G_K*-modules (the so-called* **Kummer sequence***)*

$$1 \to \mu_n \to K_{sep}^\times \xrightarrow{n} K_{sep}^\times \to 1$$

Then the boundary maps $\delta_i\colon H^i(G_K, \mu_n) \to H^{i+1}(G_K, K_{sep}^\times)$ *in the associated long exact sequence induce natural isomorphims*

$$\overline{\delta}_0\colon K^\times/(K^\times)^n \xrightarrow{\approx} H^1(G_K, \mu_n) \qquad \text{and} \qquad \overline{\delta}_1\colon H^2(G_K, \mu_n) \xrightarrow{\approx} \mathrm{Br}(K)_n$$

where $\mathrm{Br}(K)_n$ *denotes the* n*-torsion subgroup of* $\mathrm{Br}(K)$.

Proof. Note that the second equality follows from the inflation-restriction sequence together with $H^1(\mathrm{Gal}(K_{nr}/K), K_{nr}^\times) = 0$ (Hilbert 90).

$$0 \to H^2(\mathrm{Gal}(K_{nr}/K), K_{nr}^\times) \xrightarrow{\inf} H^2(G_K, K_{sep}^\times) \xrightarrow{\mathrm{res}} H^2(G_{K_{nr}}, K_{sep}^\times)^{\mathrm{Gal}(K_{nr}/K)}$$

□

Lemma 3.1.13. *Suppose* $\mathrm{char}(K) = 0$. *Then* $\delta\pi \cup \circ$ *equals* $\pi \cup \delta\chi$

Proof. Let $\alpha \in K_{sep}$ be such that $\alpha^n = \pi$, and ζ_n be a primitive n-th root of unity. By example 2.2.2, the cup products are the classes of (ϕ is the Frobenius and $\rho(\alpha) = \zeta_n\alpha$)

$$\begin{aligned}
(\delta\pi \cup \chi)(\rho^a\phi^b, \rho^c\phi^d) &= \left(\frac{\rho^a\phi^b(\alpha)}{\alpha}\right)^{\chi(\rho^c\phi^d)} = (\zeta_n^a)^d = \zeta_n^{ad} \\
(\pi \cup \delta\chi)(\rho^a\phi^b, \rho^c\phi^d) &= \pi^{\lfloor \frac{b+d}{n} \rfloor}
\end{aligned}$$

Their quotient is the coboundary of $f(\sigma) \overset{\mathrm{def}}{=} \alpha^{\chi(\sigma)}$:

$$\begin{aligned}
(df)(\rho^a\phi^b, \rho^c\phi^d) &= \frac{\rho^a\phi^b\big(f(\rho^c\phi^d)\big) \cdot f(\rho^a\phi^b)}{f(\rho^a\phi^b\rho^c\phi^d)} = \frac{\rho^a\big(f(\phi^d)\big) \cdot f(\phi^b)}{f(\phi^{b+d})} \\
&= \frac{\rho^a\big(\alpha^d\big) \cdot \alpha^b}{\alpha^{b+d \bmod n}} = \frac{(\zeta_n^a\alpha)^d \cdot \alpha^b}{\alpha^{b+d \bmod n}} = \zeta_n^{ad}\alpha^{b+d-(b+d \bmod n)} \\
&= \zeta_n^{ad}\pi^{\lfloor \frac{b+d}{n} \rfloor}
\end{aligned}$$

□

3.2 Unramified Brauer group

Let K a local field with residue field k. Denote by K_{nr} its maximal unramified extension, a Galois extension with $\mathrm{Gal}(K_{nr}/K) = G_k$ (a cyclic group generated by the Frobenius automorphism Frob_K). In this section, our goal is to compute the **unramified Brauer group** of K

$$\mathrm{Br}(K_{nr}/K) = H^2(\mathrm{Gal}(K_{nr}/K), K_{nr}^\times)$$

(see lemma 3.1.12), i.e. the subgroup of $\mathrm{Br}(K)$ consisting of classes split by K_{nr}. The two main results in this section are:

Theorem 3.2.1 (Invariant map)**.** *In the above notation, for short write*

$$G_K^{nr} \overset{\text{def}}{=} \mathrm{Gal}(K_{nr}/K) = G_k = \widehat{\mathbb{Z}}.$$

(i) There is a canonical isomorphism (the so-called **invariant map***)*

$$\boxed{\mathrm{inv}_K\colon\ \mathrm{Br}(K_{nr}/K) = H^2(G_K^{nr}, K_{nr}^\times) \xrightarrow{\approx} \mathbb{Q}/\mathbb{Z}}$$

(ii) Let $\pi_K \in K$ be a uniformiser and let $\chi_n^{nr} \in \mathrm{Hom}_c(G_K^{nr}, \mathbb{Q}/\mathbb{Z})$ be the character of order n defined by $\chi_n^{nr}(\mathrm{Frob}_K) = 1 \bmod n$. Let $\delta\colon\ \mathrm{Hom}_c(G_K^{nr}, \mathbb{Q}/\mathbb{Z}) = H^1(G_K^{nr}, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\approx} H^2(G_K^{nr}, \mathbb{Z})$ be the isomorphism of lemma 3.1.11(i). Consider the composition (where $\cup$ is the cup product)

$$\begin{aligned}\phi\colon K^\times \otimes_{\mathbb{Z}} \mathrm{Hom}_c(G_K^{nr}, \mathbb{Q}/\mathbb{Z}) &\underset{\approx}{\xrightarrow{id\otimes\delta}} H^0(G_K^{nr}, K_{nr}^\times) \otimes_{\mathbb{Z}} H^2(G_K^{nr}, \mathbb{Z}) \\ &\xrightarrow{\cup} H^2(G_K^{nr}, K_{nr}^\times) = \mathrm{Br}(K_{nr}/K)\end{aligned}$$

Then

$$\boxed{\mathrm{inv}_K\, \phi(\pi_K \otimes \chi_n^{nr}) = \frac{1}{n} \bmod \mathbb{Z}}$$

Remark 3.2.2. *(i) Simply put, theorem (i) says that we can think of $\mathrm{Br}(K_{nr}/K)$ as a "rational circle" ($\mathbb{R}/\mathbb{Z}$ is topologically a circle, and $\mathbb{Q}/\mathbb{Z}$ is its torsion subgroup).*

(ii) It can be shown that the element $\phi(\pi_K \otimes \chi_n^{nr}) \in \mathrm{Br}(K_{nr}/K)$ whose invariant is $\frac{1}{n} \bmod \mathbb{Z}$ is represented by the following central simple algebra A over K: denote by $L_n \supseteq K$ the unique degree n unramified extension, then A is given by

$$A = \bigoplus_{0 \leqslant i < n} L_n \cdot e^i \quad \text{where } e^n = \pi_K \quad \text{and} \quad x \cdot a = \mathrm{Frob}_K(a) \cdot x \ (a \in L_n)$$

Theorem 3.2.3 (Functorial property of the invariant map)**.** *Let $K' \supseteq K$ be an arbitrary (possibly ramified) finite extension of local fields.*

(i) There are commutative diagrams

$$\begin{tikzcd} H^2(G_{K'}^{nr}, K'^{\times}_{nr}) \arrow[r, "\mathrm{inv}_{K'}", "\approx"'] & \mathbb{Q}/\mathbb{Z} \\ H^2(G_K^{nr}, K^{\times}_{nr}) \arrow[u, "\mathrm{res}"] \arrow[r, "\mathrm{inv}_K", "\approx"'] & \mathbb{Q}/\mathbb{Z} \arrow[u, "{[K':K]}"'] \end{tikzcd} \qquad \begin{tikzcd} H^2(G_{K'}^{nr}, K'^{\times}_{nr}) \arrow[r, "\mathrm{inv}_{K'}", "\approx"'] \arrow[d, "\mathrm{cor}"] & \mathbb{Q}/\mathbb{Z} \arrow[d, equal] \\ H^2(G_K^{nr}, K^{\times}_{nr}) \arrow[r, "\mathrm{inv}_K", "\approx"'] & \mathbb{Q}/\mathbb{Z} \end{tikzcd}$$

where res, cor *denote the restriction and corestriction maps, and in the first diagram the right vertical arrow is multiplication by the degree* $[K' : K]$.

(ii) Moreover, if $K' \supseteq K$ *is Galois with* $G = \mathrm{Gal}(K'/K)$ *then*

$$|\operatorname{Br}(K'/K) \cap \operatorname{Br}(K_{nr}/K)| = |H^2(G, K'^{\times}) \cap H^2(G_K^{nr}, K^{\times}_{nr})| = |G|.$$

Thus $\operatorname{Br}(K'/K)$ *contains a cyclic group of order* $|G|$.

In the next section, we show that, actually, $\operatorname{Br}(K'/K) \subseteq \operatorname{Br}(K_{nr}/K)$, i.e., each element of $\operatorname{Br}(K'/K)$ is split by an unramified extension. Since $\operatorname{Br}(K)$ is the union of all $\operatorname{Br}(K'/K)$ for $K' \supseteq K$ ranging over all finite Galois extensions of K, this will show that $\operatorname{Br}(K) = \operatorname{Br}(K_{nr}/K)$ and thus inv_K is defined on the whole Brauer group $\operatorname{Br}(K)$.

Proof of theorem 3.2.3. As usual, we reduce to finite extensions, and them patch everything together: let $L_d \supseteq K$ be the unique unramified extension of degree $d = [L_d : K] \geqslant 1$ contained in K_{nr}, and write $G_d = \mathrm{Gal}(L/K) = \langle \mathrm{Frob}_{L/K} \rangle$ (a cyclic group of order d generated by the Frobenius automorphism $\mathrm{Frob}_{L/K}$). We are going to show that there are canonical isomorphisms

$$\mathrm{inv}_d \colon H^2(G_d, L_d^{\times}) \xrightarrow{\approx} \frac{\frac{1}{d}\mathbb{Z}}{\mathbb{Z}} = \mathbb{Z}/d\mathbb{Z} \qquad (d \in \mathbb{N}_{>0}) \tag{3.2}$$

fitting together via inflation maps: if $d_1 \mid d_2$ (so that $L_{d_1} \subseteq L_{d_2}$) there is a commutative diagram

$$\begin{tikzcd} H^2(\mathrm{Gal}(L_{d_2}/K), L_{d_2}^{\times}) \arrow[r, "\mathrm{inv}_{d_2}", "\approx"'] & \frac{\frac{1}{d_2}\mathbb{Z}}{\mathbb{Z}} \\ H^2(\mathrm{Gal}(L_{d_1}/K), L_{d_1}^{\times}) \arrow[u, hook, "\mathrm{inf}"] \arrow[r, "\mathrm{inv}_{d_1}", "\approx"'] & \frac{\frac{1}{d_1}\mathbb{Z}}{\mathbb{Z}} \arrow[u, hook, "\mathrm{inclusion}"'] \end{tikzcd} \tag{3.3}$$

Taking limits (= unions), we get the desired isomorphism:

$$\mathrm{inv}_K \colon \operatorname{Br}(K_{nr}/K) = \varinjlim_d \operatorname{Br}(L_d/K) \xrightarrow{\approx} \varinjlim_d \frac{\frac{1}{d}\mathbb{Z}}{\mathbb{Z}} = \mathbb{Q}/\mathbb{Z}$$

To construct (3.2), the starting point is the valuation exact sequence for L_d ($v = v_{L_d}$ denotes the the normalised valuation of L_d):

$$\begin{tikzcd} 0 \arrow[r] & U_{L_d} \arrow[r] & L_d^\times \arrow[r, "v"] & \mathbb{Z} \arrow[r] \arrow[l, bend left] & 0 \end{tikzcd} \tag{3.4}$$

It splits by sending $1 \in \mathbb{Z}$ to a uniformiser $\pi_K \in K$ (beware that there are many choices for π_K, so this splitting $\mathbb{Z} \to L_d^\times$ is non-canonical, but this will fortunately not interfere in our proof). Note that as π_K is G_d-invariant, such a splitting is indeed a morphism of G_d-modules.
Thanks to the splitting, the long exact sequence associated to (3.4) breaks into short ones; in particular, we get

$$0 \to H^2(G_d, U_{L_d}) \to H^2(G_d, L_d^\times) \xrightarrow{v} H^2(G_d, \mathbb{Z}) \to 0 \tag{3.5}$$

Next, for each integer r,

(i) $H_T^r(G_d, U_{L_d}) = 0$ (see lemma 3.2.4 below) and

(ii) we have a natural isomorphism (see lemma 3.1.11(a))

$$\delta\colon H_T^{r-1}(G_d, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\approx} H_T^r(G_d, \mathbb{Z})$$

(iii) since $G_d = \langle \mathrm{Frob}_{L_d/K} \rangle$ is cyclic of order d with a **canonical** generator $\mathrm{Frob}_{L_d/K}$, evaluation at $\mathrm{Frob}_{L_d/K}$ gives a canonical identification

$$\begin{aligned} H^1(G_d, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}_c(G_d, \mathbb{Q}/\mathbb{Z}) &\xrightarrow{\approx} \frac{\frac{1}{d}\mathbb{Z}}{\mathbb{Z}} \\ \phi &\mapsto \phi(\mathrm{Frob}_{L_d/K}) \end{aligned}$$

Finally, (3.5) together with (i), (ii), (iii) (for $r = 2$) shows that the following composition is an isomorphism:

$$\mathrm{inv}_d\colon H^2(G_d, L_d^\times) \underset{\approx}{\xrightarrow{v}} H^2(G_d, \mathbb{Z}) \underset{\approx}{\xleftarrow{\delta}} \mathrm{Hom}_c(G_d, \mathbb{Q}/\mathbb{Z}) \underset{\approx}{\xrightarrow{\phi \mapsto \phi(\mathrm{Frob}_{L_d/K})}} \frac{\frac{1}{d}\mathbb{Z}}{\mathbb{Z}} \tag{3.6}$$

This is the desired isomorphism (3.2). Commutativity of (3.3) now follows from the following commutative diagram, whose rows are inv_{d_2} and inv_{d_1} by (3.6). Here we use the facts that $v_{L_{d_1}} = v_{L_{d_2}}|_{L_{d_1}}$ (by unramifiedness) and $\mathrm{Frob}_{L_{d_1}/K} = \mathrm{Frob}_{L_{d_2}/K}|_{L_{d_1}}$.

$$\begin{tikzcd} H^2(G_{d_2}, L_{d_2}^\times) \arrow[r, "v_{L_{d_2}}"] & H^2(G_{d_2}, \mathbb{Z}) & \mathrm{Hom}(G_{d_2}, \mathbb{Q}/\mathbb{Z}) \arrow[l, "\delta"'] \arrow[r] & \frac{\frac{1}{d_2}\mathbb{Z}}{\mathbb{Z}} \\ H^2(G_{d_1}, L_{d_1}^\times) \arrow[r, "v_{L_{d_1}}"] \arrow[u, "\mathrm{inf}"] & H^2(G_{d_1}, \mathbb{Z}) \arrow[u, "\mathrm{inf}"] & \mathrm{Hom}(G_{d_1}, \mathbb{Q}/\mathbb{Z}) \arrow[l, "\delta"'] \arrow[r] \arrow[u, "\mathrm{inf}"] & \frac{\frac{1}{d_1}\mathbb{Z}}{\mathbb{Z}} \arrow[u, "\mathrm{inclusion}"] \end{tikzcd}$$

Finally, to show (ii), as we $\phi(\pi_K \otimes \chi_n^{nr}) \in \mathrm{Br}(L_d/K)$, computing $\mathrm{inv}_n\, \phi(\pi_K \otimes \chi_n^{nr})$ using (3.6) we get $\square$

We still have to show

Lemma 3.2.4. *Let $L \supseteq K$ be a finite unramified extension with $G = \mathrm{Gal}(L/K)$. Then $H^r_T(G, U_L) = 0$ for all r.*

Proof. The lemma is a consequence of the fact that U_L has a filtration by $U_L \supseteq U_L^{(1)} \supseteq U^{(2)} \supseteq \cdots$ with cohomologically trivial subquotients $U_L/U_L^{(1)}$ and $U_L^{(r)}/U_L^{(r+1)}$ ($r \geqslant 1$). Indeed, let $l \supseteq k$ be the residue field extensions, and consider the exact sequences of G-modules (via the isomorphism $G \approx \mathrm{Gal}(l/k)$)

$$0 \to U_L^{(1)} \to U_L \to l^\times \to 0 \qquad\qquad 0 \to U_L^{(r+1)} \to U_L^{(r)} \to l^+ \to 0$$

Now, for all r, one has

(i) $H^r_T(G, l^+) = 0$ since l^+ is an induced module by the normal basis theorem (see appendix).

(ii) $H^r_T(G, l^\times) = 0$. By the periodicity of cohomology of cyclic groups, it is enough to prove that for $r = 0$ and $r = 1$. The case $H^1_T(G, l^\times) = 0$ is just Hilbert 90, while the case $H^r_T(G, l^\times) = k^\times/N_{l/k}(l^\times) = 0$ follows from the surjectivity of the norm map $N_{l/k}\colon l^\times \twoheadrightarrow k^\times$: if $k = \mathbb{F}_q$ and b is a generator of the group $l^\times$ then $N_{l/k}(b) = b^{1+q+\cdots+q^{n-1}}$ has order $q-1$ and hence is a generator of $k^\times$.

Hence, from the long exact sequences associated to the two short ones above, we conclude that

$$H^r_T(G, U_L^{(i+1)}) = H^r_T(G, U_L^{(i)}) \qquad \text{for all } r \text{ and all } i \geqslant 0.$$

To show that $H^r_T(G, U_L) = 0$ we may assume $r > 0$, again by periodicity. Let $f\colon G^r \to U_L$ be an r-cocycle and denote by d the $(r-1)$-th coboundary map. Since $H^r(G, U_L^{(1)}) = H^r(G, U_L)$, f differs by a coboundary from an r-cocycle $f_1\colon G^r \to U_L^{(1)}$, i.e., there exists $g_0\colon G^{r-1} \to U_L$ such that $f_1 = f \cdot d(g_0)^{-1}$. Proceeding in this manner, we inductively construct functions $g_i\colon G^{r-1} \to U_L^{(i)}$ such that $f \cdot d(g_0 g_1 \ldots g_i)^{-1}$ is an r-cocycle with values in $U_L^{(i+1)}$. Then the product $g_0 g_1 \ldots g_i$ converges to a function $g\colon G^{r-1} \to U_L$ (multiplicative Calculus Student Psychedelic Dream!) and we have that $dg = f$, that is, the class $[f]$ is trivial in $H^r(G, U_L)$. □

The case $r = 0$ is of special interest:

Corollary 3.2.5 (Norm groups of unramified extensions)**.** *Let $L \supseteq K$ be the unramified extension of degree n and let $\pi \in K$ be a common uniformiser. Then the norm map $N_{L/K}\colon U_L \twoheadrightarrow U_K$ is surjective on units, and thus*

$$N_{L/K} L^\times = \pi^{n\mathbb{Z}} \cdot U_K$$

We finish this section with the

Proof of theorem 3.2.3. Let e and f be the ramification and inertia degrees of $K' \supseteq K$ and denote by v and v' the normalised valuations of K and K' respectively. Thus

$$v'|_K = e \cdot v \qquad \text{and} \qquad \mathrm{Frob}_{K'}|_{K_{nr}} = \mathrm{Frob}_K^f\,.$$

First note that $K'_{nr} = K_{nr} \cdot K'$ by corollary 1.5.2 and the unramified Highlander's Philosophy. Hence the restriction map $\mathrm{Gal}(K_{nr}/K) \hookrightarrow \mathrm{Gal}(K_{nr}/K)$ given by $\sigma \mapsto \sigma|K_{nr}$ is injective (with image $\mathrm{Gal}(K_{nr}/(K' \cap K_{nr}))$), and thus $\mathrm{res}\colon H^2(\mathrm{Gal}(K_{nr}/K), K_{nr}^\times) \to H^2(\mathrm{Gal}(K'_{nr}/K'), K'^\times_{nr})$ is well-defined.

$$\begin{tikzcd}
 & & K'_{nr} \arrow[r, equal] \arrow[dd, "\mathrm{Gal}(K'_{nr}/K')"] & K_{nr} \cdot K' \\
 & K_{nr} \arrow[ur, dash] \arrow[dd, dash] \arrow[ddd, bend right, "\mathrm{Gal}(K'_{nr}/K')"'] & & \\
 & & K' \arrow[dl, dash, "e"] & \\
 & K' \cap K_{nr} \arrow[d, dash, "f"] & & \\
 & K & &
\end{tikzcd}$$

Hence, from the definition (3.6) of the invariant map, we obtain a commutative diagram whose top and bottom rows are $\mathrm{inv}_{K'}$ and inv_K, respectively (here Hom_c denotes the group of continuous morphisms):

$$\begin{tikzcd}
H^2(G^{nr}_{K'}, K'^\times_{nr}) \arrow[r, "v'"] & H^2(G^{nr}_{K'}, \mathbb{Z}) & \mathrm{Hom}_c(G^{nr}_{K'}, \mathbb{Q}/\mathbb{Z}) \arrow[l, "\delta", "\approx"'] \arrow[r, "\phi \mapsto \phi(\mathrm{Frob}_{K'})"] & \mathbb{Q}/\mathbb{Z} \\
H^2(G^{nr}_K, K^\times_{nr}) \arrow[r, "v"] \arrow[u, "\mathrm{res}"] & H^2(G^{nr}_K, \mathbb{Z}) \arrow[u, "e\cdot\mathrm{res}"] & \mathrm{Hom}_c(G^{nr}_K, \mathbb{Q}/\mathbb{Z}) \arrow[l, "\delta", "\approx"'] \arrow[r, "\phi \mapsto \phi(\mathrm{Frob}_K)"] \arrow[u, "e\cdot\mathrm{res}"] & \mathbb{Q}/\mathbb{Z} \arrow[u, "ef"]
\end{tikzcd}$$

Commutativity of the leftmost square results from $v'|_K = e \cdot v$, while that of the rightmost one from $\mathrm{Frob}_{K'}|_{K_{nr}} = \mathrm{Frob}_K^f$. Since $ef = [K' : K]$, the first diagram in (i) is commutative. For the second one, consider

$$\begin{tikzcd}
H^2(G^{nr}_{K'}, K'^\times_{nr}) \arrow[r, "v'"] \arrow[d, "\mathrm{cor}"] & H^2(G^{nr}_{K'}, \mathbb{Z}) \arrow[d, "f\,\mathrm{cor}"] & \mathrm{Hom}_c(G^{nr}_{K'}, \mathbb{Q}/\mathbb{Z}) \arrow[l, "\delta", "\approx"'] \arrow[r, "\phi \mapsto \phi(\mathrm{Frob}_{K'})"] \arrow[d, "f\,\mathrm{cor}"] & \mathbb{Q}/\mathbb{Z} \arrow[d, equal] \\
H^2(G^{nr}_K, K^\times_{nr}) \arrow[r, "v"] & H^2(G^{nr}_K, \mathbb{Z}) & \mathrm{Hom}_c(G^{nr}_K, \mathbb{Q}/\mathbb{Z}) \arrow[l, "\delta", "\approx"'] \arrow[r, "\phi \mapsto \phi(\mathrm{Frob}_K)"] & \mathbb{Q}/\mathbb{Z}
\end{tikzcd}$$

The leftmost square is commutative thanks to the relation $v' \circ N_{K'/K} = f \cdot v$, while that of the rightmost one again from $\mathrm{Frob}_{K'}|_{K_{nr}} = \mathrm{Frob}_K^f$.
To show (ii), let $n = |G| = [K' : K]$. By the first diagram of (i) we have

$$\ker\big(\mathrm{Br}(K_{nr}/K) \xrightarrow{\mathrm{res}} \mathrm{Br}(K'_{nr}/K')\big) = \ker(\mathbb{Q}/\mathbb{Z} \xrightarrow{n} \mathbb{Q}/\mathbb{Z}) = \frac{\frac{1}{n}\mathbb{Z}}{\mathbb{Z}}$$

But the left side is exactly $\mathrm{Br}(K'/K) \cap \mathrm{Br}(K_{nr}/K)$: by the inflation-restriction sequence, the top row of the following commutative diagram is exact,

and thus so is the bottom row (the intersection with the unramified Brauer group):

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Br}(K'/K) = H^2(G, K'^\times) \arrow[r, "\mathrm{inf}"] & \mathrm{Br}(K) \arrow[r, "\mathrm{res}"] & \mathrm{Br}(K') \\
0 \arrow[r] & \mathrm{Br}(K'/K) \cap \mathrm{Br}(K_{nr}/K) \arrow[r] \arrow[u, hook] & \mathrm{Br}(K_{nr}/K) \arrow[r, "\mathrm{res}"] \arrow[u, hook, "\mathrm{inf}"] & \mathrm{Br}(K'_{nr}/K') \arrow[u, hook, "\mathrm{inf}"]
\end{tikzcd}$$

□

3.3 Brauer group of local fields

The main of result of this section is

Theorem 3.3.1. *Let $L \supseteq K$ be an arbitrary finite Galois extension of local fields with $G = \mathrm{Gal}(L/K)$. Then the group $\mathrm{Br}(L/K) = H^2(G, L^\times)$ is cyclic of order $|G|$.*

As an immediate corollary, we show that the unramified Brauer group of the previous section actually equals the whole Brauer group!

Corollary 3.3.2. *Let K be a local field. The inflation map*

$$\mathrm{inf}\colon \mathrm{Br}(K_{nr}/K) = H^2(G_K^{nr}, K_{nr}^\times) \xrightarrow{\approx} H^2(G_K, K_{sep}^\times) = \mathrm{Br}(K)$$

is an isomorphism. Composing the invariant map $\mathrm{inv}_K\colon H^2(G_K^{nr}, K_{nr}^\times) \xrightarrow{\approx} \mathbb{Q}/\mathbb{Z}$ of last section with the inverse of the above inflation, we obtain a canonical isomorphism (still denoted inv_K)

$$\boxed{\mathrm{inv}_K\colon \mathrm{Br}(K) = H^2(G_K, K_{sep}^\times) \xrightarrow{\approx} \mathbb{Q}/\mathbb{Z}}$$

Proof. All we must show is that the inclusion $\mathrm{inf}\colon \mathrm{Br}(K_{nr}/K) \hookrightarrow \mathrm{Br}(K)$ is an isomorphism, i.e., that every class of $\mathrm{Br}(K)$ is split by K_{nr}. Since $\mathrm{Br}(K) = \varinjlim_L \mathrm{Br}(L/K) = \bigcup_L \mathrm{Br}(L/K)$, where L runs over all finite Galois extensions of K (inside K_{sep}), we must show that each $\mathrm{Br}(L/K) \subseteq \mathrm{Br}(K_{nr}/K)$, i.e., that each class of $\mathrm{Br}(L/K)$ is split by an unramified extension. But we already know by theorem 3.2.3(ii) that $|\mathrm{Br}(L/K) \cap \mathrm{Br}(K_{nr}/K)| = |G|$, hence theorem 3.3.1 above implies that $\mathrm{Br}(L/K) \subseteq \mathrm{Br}(K_{nr}/K)$, and we are done. □

There are two classical proofs of 3.3.1. The first one is via a counting argument, while the second one is more direct, using the theory of central simple algebras over local fields to show that $\mathrm{Br}(K_{nr})$ so that every class of $\mathrm{Br}(K)$ is split by the maximal unramified extension of K.

3.3.1 Proof of theorem 3.3.1 by counting

Since $H^2(G, L^\times)$ contains a cyclic group of order $|G|$ by 3.2.3, it is enough to show that the order of $H^2(G, L^\times)$ divides $|G|$. Moreover, it is easy to reduce to the case where G is solvable[1] and then to the case where G is cyclic. Indeed, assume the theorem has been shown in the solvable case. In the general, it suffices to show that, for each prime number p, the p-primary component of $H^2(G, L^\times)$ divides the order of a p-Sylow subgroup G_p of G: $H^2(G, L^\times)(p) \mid |G_p|$. But by a restriction-corestriction argument there is an injection on the p-primary components

$$\mathrm{res}\colon H^2(G, L^\times)(p) \hookrightarrow H^2(G_p, L^\times)(p)$$

and we may work with the solvable group G_p instead of G. Thus the solvable case implies the general case.

Now assume G is solvable, and assume that the theorem has been shown in the cyclic case. Let $H \lhd G$ be a normal subgroup such that $H \neq \{1\}$ is cyclic non-trivial. By Hilbert 90, we have an exact inflation-restriction sequence

$$0 \to H^2(G/H, (L^H)^\times) \to H^2(G, L^\times) \to H^2(H, L^\times)$$

Hence $|H^2(G, L^\times)|$ divides $|H^2(H, L^\times)| \cdot |H^2(G/H, (L^H)^\times)|$, which in turn divides $|H| \cdot |G/H| = |G|$ by induction hypothesis, and showing that $H^2(G, L^\times) \mid |G|$. So from now on we may assume G **to be cyclic.**

Before finishing the proof, we need to make an interlude and introduce a very useful tool in the cohomology of cyclic groups, the so-called **Herbrand quotient**. It plays a role similar to the Euler characteristic in Topology and Algebraic Geometry. Just to put all this into context, recall that for a complex projective variety X the Euler characteristic of a coherent sheaf $\mathcal{F}$ on X is defined as the (finite) alternating sum

$$\chi(X, \mathcal{F}) = \sum_{n \geqslant 0} (-1)^i \dim_{\mathbb{C}} H^i(X, \mathcal{F})$$

The Euler characteristic is **additive** in exact sequences, namely given a short exact sequence of coherent sheaves

$$0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$$

one has $\chi(X, \mathcal{G}) = \chi(X, \mathcal{F}) + \chi(X, \mathcal{H})$. Besides, the Euler characteristic is much more well-behaved than individual cohomology groups $H^i(X, \mathcal{F})$ in isolation, for instance, for a flat family $\mathcal{F}_s$ of sheaves, the Euler characteristic $\chi(\mathcal{F}_s)$ is locally constanst with respect to s, while the dimensions $\dim_{\mathbb{C}} H^i(X, \mathcal{F}_s)$ may change a lot.

The Herbrand quotient has similar properties to the Euler characteristic, however it is written multiplicatively (as opposed to additively), and only degrees 0 and 1 intervene in its definition, thanks to the periodicity of the cohomology of cyclic groups.

Definition 3.3.3. *Let G be a cyclic group and M be a G-module whose Tate cohomology groups are all finite. We define its* **Herbrand quotient** *as*

$$\boxed{h(G, M) = \frac{|H^0_T(G, M)|}{|H^1_T(G, M)|}}$$

[1] actually G is always solvable (see 1.5.7), but since it is easy to give the above direct argument and avoid resorting to 1.5.7

Lemma 3.3.4 (Multiplicativity)**.** *Let G be a cyclic group and consider an exact sequence of G-modules*

$$0 \to M' \to M \to M'' \to 0$$

If two of the Herbrand quotients $h(G, M)$, $h(G, M')$, $h(G, M'')$ are defined (i.e. have finite Tate cohomology groups) then so is the third and

$$\boxed{h(G, M) = h(G, M') \cdot h(G, M'')}$$

Proof. From the periodicity of cohomology of cyclic groups, the long exact sequence associated to the above short one becomes an "exact hexagon"

$$\begin{tikzcd}
H_T^0(G, M') \arrow[r, "f^0"] & H_T^0(G, M) \arrow[r, "g^0"] & H_T^0(G, M'') \arrow[d, "\delta^0"] \\
H_T^1(G, M'') \arrow[u, "\delta^1"] & H_T^1(G, M) \arrow[l, "g^1"] & H_T^1(G, M') \arrow[l, "f^1"]
\end{tikzcd}$$

The first result follows directly from from this hexagon. The second result also follows from this hexagon by elementary counting, as one has

$$\begin{aligned}
|H_T^0(G, M')| &= |\ker f^0| \cdot |\ker g^0| & \qquad |H_T^1(G, M')| &= |\ker f^1| \cdot |\ker g^1| \\
|H_T^0(G, M)| &= |\ker g^0| \cdot |\ker \delta^0| & \qquad |H_T^1(G, M)| &= |\ker g^1| \cdot |\ker \delta^1| \\
|H_T^0(G, M'')| &= |\ker \delta^0| \cdot |\ker f^1| & \qquad |H_T^1(G, M'')| &= |\ker \delta^1| \cdot |\ker f^0|
\end{aligned}$$

□

Lemma 3.3.5 (Finite Index Invariance)**.** *Let G be a cyclic group, M be a G-module and M' be a G-submodule of finite index. Then $h(G, M')$ is defined if and only if $h(G, M)$ is defined, in which case $h(G, M') = h(G, M)$.*

Proof. By the last lemma, it suffices to show that if M is a finite group then $h(G, M) = 1$. Let σ be a generator of G. Since $|M| < \infty$ one has that

$$h(G, M) = \frac{|H_T^0(G, M)|}{|H_T^1(G, M)|} = \frac{|M^G|}{|N_G(M)|} \cdot \frac{|(\sigma - 1) \cdot M|}{|\ker N_G|}.$$

By the isomorphism theorem applied to N_G and $\sigma - 1$, we get

$$|\ker N_G| \cdot |N_G(M)| = |M| = |M^G| \cdot |(\sigma - 1) \cdot M|$$

(observe that $\ker(\sigma - 1) = M^G$). Hence $h(G, M) = 1$. □

Now we finish the proof of theorem 3.3.1. Let $L \supseteq K$ be a cyclic extension of local fields with Galois group $G = \mathrm{Gal}(L/K)$ of order n. We apply the above to the valuation exact sequence

$$0 \to U_L \to L^\times \to \mathbb{Z} \to 0$$

It is easy to compute $h(G, \mathbb{Z}) = n$, and the lemma below shows that $h(G, U_L) = 1$. Hence we get

$$h(G, L^\times) = h(G, U_L) \cdot h(G, \mathbb{Z}) = 1 \cdot n = n.$$

Since $H^1(G, L^\times) = 0$ by Hilbert 90, we conclude that $|H^2(G, L^\times)| = n$, as desired.

Lemma 3.3.6. *With the above notation and hypotheses, $h(G, U_L) = 1$.*

Proof. By the previous lemma, it is enough to show that U_L contains an induced G-submodule of finite index. First assume that $\operatorname{char} K = 0$ and let π be a uniformiser of K. By 1.4.1 we have an isomorphism of G-modules $U_L^{(i)} \cong \mathfrak{m}_L^i$ for $i \gg 0$ sufficiently large. On the other hand, for $j \gg 0$ sufficiently large $\mathfrak{m}_L^i \supseteq \pi^j O_L \cong O_L$. Since $U_L/U_L^{(i)}$ and $\mathfrak{m}_L^i/\pi^j O_L$ are finite, it is enough to show that O_L contains an induced module of finite index. Now let $\omega_1, \ldots, \omega_n$ be a normal basis of $L \supseteq K$; multiplying by a convenient power of π we may assume that $\omega_1, \ldots, \omega_n \in O_L$. Since O_L is a finite O_K-module (theorem 1.3.9), we have that the induced module $M = O_K\omega_1 + \cdots + O_K\omega_n$ has finite index in O_L.
Now we sketch a proof that works even if $\operatorname{char} K \neq 0$. Let M be as above. Multiplying the ω_i by a sufficiently large power of π we may assume that $M \cdot M \subseteq \pi M$. Then we may consider the submodule of finite index $V = 1 + M$ of U_L and the filtration given by $V^{(i)} = 1 + \pi^i M$ for $i \geqslant 0$. It is easy to show that we have an isomorphism of G-modules $V^{(i)}/V^{(i+1)} \cong M/\pi M$, which has trivial cohomology since the latter is induced. As in the proof of 3.2.4, this implies that V itself has trivial cohomology. □

3.3.2 Direct proof of theorem 3.3.1 using Division Algebras

We now sketch a direct proof based on the fact that $\mathrm{Br}(K)$ is in one-to-one correspondence with the set of isomorphism classes of finite dimensional K-division algebras. Recall that by 3.2.3 all that is left to show is that $\mathrm{Br}(K) \subseteq \mathrm{Br}(K_{nr}/K)$, i.e., that any class of a K-division algebra D is split by some unramified extension of K. We will show that there exists a maximal unramified subfield of D, which will do the job.

Theorem 3.3.7. *Let K be a local field, and let D be a division algebra over K of degree n (i.e. the centre of D is precisely K and $[D : K] = n^2$). Then D contains a maximal commutative subfield L which is unramified over K, and consequently $D \otimes_K L \cong M_n(L)$. Hence $\mathrm{Br}(K_{nr}/K) = \mathrm{Br}(K)$.*

Proof. We begin with a few general remarks on the complete valued division algebra D. The valuation $v = v_K$ of K extends uniquely to D. Indeed, for each $\alpha \in D$, there is a unique extension v_α of v_K to the (commutative) subfield $K(\alpha) \subseteq D$:

$$v_\alpha \colon K(\alpha) \to \frac{1}{[K(\alpha) : K]}\mathbb{Z} \cup \{\infty\} \qquad \text{where} \quad v_\alpha(x) = \frac{v_K(x)}{[K(\alpha) : K]}$$

By uniqueness, these extensions v_α fit together and define a unique discrete valuation v_D on D, with respect to which D is complete. Set

$$\mathcal{O}_D \stackrel{\text{def}}{=} \{d \in D \mid v_D(d) \geqslant 0\} \qquad \mathfrak{m}_D \stackrel{\text{def}}{=} \{d \in D \mid v_D(d) > 0\}$$

(the non-commutative ring of integers of D and its two-sided maximal ideal). Let e be the ramification index $\mathcal{O}_D$ over $\mathcal{O}_K$, and $f = [\mathcal{O}_D/\mathfrak{m}_D : k]$ be the inertia degree, where k denotes the residue field of $\mathcal{O}_K$. Then

$$n^2 = ef$$

We claim that $e = f = n$. It suffices to show that $e \leqslant n$ and $f \leqslant n$. This will follow from the fact that any maximal commutative subfield of D has degree n over K, in particular given any $\alpha \in D$ the commutative subfield $K(\alpha) \subseteq D$ satisfies $[K(\alpha) : K] \leqslant n$.

- If $\beta \in D$ is such that $v_D(\beta) = e^{-1}$ then $e \leqslant [K(\beta) : K] \leqslant n$.
- As k is a finite field, $\mathrm{Br}(k) = 0$, and thus the residue division algebra $\mathcal{O}_D/\mathfrak{m}_D$ is commutative (Wedderburn's theorem). Since k is perfect, there exists $\alpha \in D$ whose residue class $\overline{\alpha}$ generates $\mathcal{O}_D/\mathfrak{m}_D$ over k. Then

$$f = [k(\overline{\alpha}) : k] \leqslant [K(\alpha) : K] \leqslant n. \tag{$*$}$$

Thus $e = f = n$, as claimed, and we must have equality in $(*)$ for $\overline{\alpha}$ such that $\mathcal{O}_D/\mathfrak{m}_D = k(\overline{\alpha})$. This shows that $K(\alpha)$ is a maximal subfield of D which is unramified over K. $\square$

3.4 Local reciprocity

3.4.1 Hilbert symbol

We now construct a $\mathbb{Z}$-bilinear pairing

$$\langle -, - \rangle \colon K^\times \otimes \mathrm{Hom}_c(G_K^{ab}, \mathbb{Q}/\mathbb{Z}) \to \mathbb{Q}/\mathbb{Z}$$

This is just the cup product

$$\begin{aligned} H^0(G_K, K_{sep}^\times) \otimes H^1(G_K, \mathbb{Q}/\mathbb{Z}) &\xrightarrow{\approx} H^0(G_K, K_{sep}^\times) \otimes H^2(G_K, \mathbb{Z}) \\ &\xrightarrow{\cup} H^2(G_K, \mathbb{Q}/\mathbb{Z}) = \mathrm{Br}(K) \end{aligned}$$

The relation of this pairing and the reciprocity map $\theta_K \colon K^\times \hookrightarrow G_K^{ab}$ is given by

$$\boxed{\chi\big(\theta_{L/K}(a)\big) = \langle a, \chi \rangle}$$

For any $a \in K^\times$ and any character $\chi \in H^1(G, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z})$ one has
where $\bar{a}$ is the image of a in $H_T^0(G, L^\times) = K^\times / N_{L/K} L^\times$.

Using this description of the reciprocity map let us first show that for an unramified extension of local fields $L \supseteq K$ one has $\theta_{L/K}(a) = \mathrm{Frob}_{L/K}^{v(a)}$, where v denotes the normalised valuation of K and $\mathrm{Frob}_{L/K}$ is the Frobenius automorphism of $L \supseteq K$. In fact, let $n = [L : K]$ and $G = \mathrm{Gal}(L/K)$ and consider the character $\chi \colon G \to \mathbb{Q}/\mathbb{Z}$ given by $\chi(\mathrm{Frob}_{L/K}) = \frac{1}{n} \bmod \mathbb{Z}$. Then $\bar{a} \cup \delta\chi$ is represented by the 2-cocycle $c \colon G \times G \to L^\times$ given by

$$c(\mathrm{Frob}_{L/K}^i, \mathrm{Frob}_{L/K}^j) = \begin{cases} a & \text{if } i + j \geqslant 0 \\ 0 & \text{otherwise} \end{cases} \qquad 0 \leqslant i, j < n$$

Now since the invariant map is given by the composition

$$H^2(G, L^\times) \underset{\approx}{\xrightarrow{v}} H^2(G, \mathbb{Z}) \underset{\approx}{\xleftarrow{\delta}} H^1(G, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) \approx \mathbb{Q}/\mathbb{Z}$$

and $v([c]) = \delta[f]$ where $f\colon G \to \mathbb{Q}/\mathbb{Z}$ is the 1-cocycle given by $f(\mathrm{Frob}_{L/K}) = \frac{v(a)}{n} \bmod \mathbb{Z}$, we conclude that

$$\chi\big(\theta_{L/K}(a)\big) = \mathrm{inv}_K(\bar{a} \cup \delta\chi) = \mathrm{inv}_K([c]) = \frac{v(a)}{n} \bmod \mathbb{Z} = \chi(\mathrm{Frob}_{L/K}^{v(a)})$$

showing that $\theta_{L/K}(a) = \mathrm{Frob}_{L/K}^{v(a)}$ since χ is injective.
We can also show the compatibility of the various maps $\theta_{L/K}$ in the sense that given finite extensions $M \supseteq L \supseteq K$ with M and L abelian over K we have a commutative diagram

$$\begin{tikzcd} K^\times \arrow[r, "\theta_{M/K}"] \arrow[dr, "\theta_{L/K}"'] & \mathrm{Gal}(M/K) \arrow[d, two heads, "\mathrm{can.}"] \\ & \mathrm{Gal}(L/K) \end{tikzcd}$$

This will prove that the maps $\theta_{L/K}$ fit together into a single map $\theta_K\colon K^\times \to G_K^{ab}$, which will then satisfy all the properties required in 3.1.1.
Let $a \in K^\times$, let $\chi \in \mathrm{Hom}(\mathrm{Gal}(L/K), \mathbb{Q}/\mathbb{Z})$ and denote by $\chi' = \inf(\chi) \in \mathrm{Hom}(\mathrm{Gal}(M/K), \mathbb{Q}/\mathbb{Z})$. Then $\inf(\bar{a} \cup \delta\chi) = \bar{a} \cup \delta\chi'$ and thus

$$\chi(\theta_{L/K}(a)) = \mathrm{inv}_K(\bar{a} \cup \delta\chi) = \mathrm{inv}_K(\bar{a} \cup \delta\chi') = \chi'(\theta_{M/K}(a)) = \chi(\theta_{M/K}(a)|_L)$$

Since this holds for all $\chi \in \mathrm{Hom}(\mathrm{Gal}(L/K), \mathbb{Q}/\mathbb{Z})$, we conclude that $\theta_{L/K}(a) = \theta_{M/K}(a)|_L$, as was to be shown.

Lemma 3.4.1. *Let $n > 0$. The pairing $\langle -, - \rangle$ is non-degenerate on n-torsion, i.e.*

(a) if $a \in K^\times$ is such that $\langle a, \chi \rangle = 0$ for all $\chi \in \mathrm{Hom}_c(G_K, \frac{1}{n}\mathbb{Z}/\mathbb{Z})$ then $a \in (K^\times)^n$.

(b) if $\chi \in \mathrm{Hom}_c(G_K, \frac{1}{n}\mathbb{Z}/\mathbb{Z})$ is such that $\langle a, \chi \rangle = 0$ for all $a \in K^\times$ then $\chi = 0$.

It is enough to show this for $n = p$ a prime number. Indeed, in (b) if $\chi \neq 0$ then there is a non-zero multiple $m\chi \in \mathrm{Hom}_c(G_K, \frac{1}{p}\mathbb{Z}/\mathbb{Z})$ of χ for some prime p dividing n. Since $\langle a, m\chi \rangle = m\langle a, \chi \rangle = 0$ for all $a \in K^\times$, if the lemma holds in the prime case we would get $m\chi = 0$, a contradiction. For (a), decomposing into primary components, we may assume that $n = p^r$ is a power of a prime p.

3.4.2 Reciprocity via Tate-Nakayama

Finally, we are ready to make

Definition 3.4.2. *For any finite abelian extension $L \supseteq K$ of local fields, we define the local reciprocity map $\theta_{L/K}\colon K^\times \to \mathrm{Gal}(L/K)$ as the composition of the natural projection map $K^\times \twoheadrightarrow K^\times/N_{L/K}L^\times$ with the inverse of the Tate-Nakayama isomorphism*

$$\mathrm{Gal}(L/K) \xrightarrow{\approx} \frac{K^\times}{N_{L/K}L^\times}$$

given by the cup product with the unique element $\gamma \in H^2(\mathrm{Gal}(L/K), L^\times)$ such that $\mathrm{inv}_{L/K}(\gamma) = \frac{1}{[L:K]} \bmod \mathbb{Z}$. Such element γ is called a **fundamental class**.

To prove that the above isomorphism satisfies the two properties of 3.1.1, we describe $\theta_{L/K}$ in terms of characters. Let $L \supseteq K$ be as above with $G = \mathrm{Gal}(L/K)$. Recall that we have two exact sequences of G-modules

$$0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0 \tag{$*$}$$

$$0 \to I_G \to \mathbb{Z}[G] \to \mathbb{Z} \to 0 \tag{$**$}$$

where $\mathbb{Q}$ and $\mathbb{Z}[G]$ have trivial cohomology, hence the connecting maps

$$\delta\colon H^1(G, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\approx} H^2(G, \mathbb{Z})$$
$$\delta_a\colon H_T^{-2}(G, \mathbb{Z}) \xrightarrow{\approx} H_T^{-1}(G, I_G)$$

are isomorphisms. Besides we have a natural isomorphism $G^{ab} = H_T^{-1}(G, I_G)$ given by

$$G^{ab} \xrightarrow{\approx} H_T^{-1}(G, I_G) = \frac{I_G}{I_G^2}$$
$$\sigma \cdot [G:G] \mapsto (\sigma - 1) \cdot I_G^2$$

Theorem 3.4.3 (Local Reciprocity Revisited)**.** *With the above notation, for any $a \in K^\times$ and any character $\chi \in H^1(G, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z})$ one has*

$$\boxed{\chi\big(\theta_{L/K}(a)\big) = \mathrm{inv}_K(\bar{a} \cup \delta\chi)}$$

where $\bar{a}$ is the image of a in $H_T^0(G, L^\times) = K^\times/N_{L/K}L^\times$. In other words, the cup product gives a perfect pairing

$$\begin{tikzcd}
H_T^0(G, L^\times) \otimes H_T^2(G, \mathbb{Z}) \arrow[r, "\cup"] & H_T^2(G, L^\times) \arrow[r, "\mathrm{inf}"] \arrow[d, "\approx"] & H_T^2(G_K, K_{sep}^\times) \arrow[d, "\mathrm{inv}_K", "\approx"'] \\
\frac{K^\times}{N_{L/K}L^\times} \otimes \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) \arrow[r] \arrow[u, "id\otimes\delta", "\approx"'] & \frac{\frac{1}{[G]}\mathbb{Z}}{\mathbb{Z}} \arrow[r] & \mathbb{Q}/\mathbb{Z}
\end{tikzcd}$$

Proof. Let $\gamma \in H^2(G, L^\times)$ be the fundamental class and $n = |G|$. Given $a \in K^\times$ and $\chi \in H^1(G, \mathbb{Q}/\mathbb{Z})$, write $\sigma \overset{\text{def}}{=} \theta_{L/K}(a) \in G$ and $\chi(\sigma) = \frac{i}{n} \bmod \mathbb{Z}$ with $0 \leqslant i < n$. Since $\mathrm{inv}_K(\gamma) = \frac{1}{n} \bmod \mathbb{Z}$, in order to show that $\chi(\sigma) = \mathrm{inv}_K(\bar{a} \cup \delta\chi)$ we have to show that $\bar{a} \cup \delta\chi = i \cdot \gamma$ in $H^2(G, L^\times)$. However by the very definition of $\theta_{L/K}$ we have that $\bar{\sigma} \cup \gamma = \bar{a}$, where $\bar{\sigma} \in H_T^{-2}(G, \mathbb{Z})$ denotes the image of σ under the isomorphism $H_T^{-2}(G, \mathbb{Z}) = G^{ab} = G$. Hence

$$\bar{a} \cup \delta\chi = \bar{\sigma} \cup \gamma \cup \delta\chi = \bar{\sigma} \cup \delta\chi \cup \gamma$$

and we are left to show that $\bar{\sigma} \cup \delta\chi = i \bmod n$ in $H_T^0(G, \mathbb{Z})$. Note that $\bar{\sigma} \cup \delta\chi = \delta(\bar{\sigma} \cup \chi)$ and by the lemma below $\bar{\sigma} \cup \chi \in H_T^{-1}(G, \mathbb{Q}/\mathbb{Z})$ is represented by $\chi(\sigma) = \frac{i}{n} \bmod \mathbb{Z} \in \mathbb{Q}/\mathbb{Z}$. On the other hand, the connecting map $\delta\colon H_T^{-1}(G, \mathbb{Q}/\mathbb{Z}) \overset{\approx}{\to} H_T^0(G, \mathbb{Z})$ is induced by the norm map (see appendix), which in our case is just multiplication by n. Hence $\delta(\bar{\sigma} \cup \chi) = i \bmod n$, as required.
Finally, the above identity implies that the pairing

$$H_T^0(G, L^\times) \otimes H_T^1(G, \mathbb{Q}/\mathbb{Z}) \underset{\approx}{\overset{\mathrm{id}\otimes\delta}{\to}} H_T^0(G, L^\times) \otimes H_T^2(G, \mathbb{Z}) \overset{\cup}{\to} H_T^2(G, L^\times)$$

is perfect. In fact, if $\chi \in H_T^1(G, \mathbb{Q}/\mathbb{Z})$ is such that $\bar{a} \cup \delta\chi = 0 \iff \chi(\theta_{L/K}(a)) = 0$ for all $a \in K^\times$ then $\chi = 0$ since $\theta_{L/K}\colon K^\times \twoheadrightarrow \mathrm{Gal}(L/K)$ is surjective. On the other hand, if $a \in K^\times$ is such that $\bar{a} \cup \delta\chi = 0 \iff \chi(\theta_{L/K}(a)) = 0$ for all $\chi \in H_T^1(G, \mathbb{Q}/\mathbb{Z})$ then clearly $\theta_{L/K}(a) = 0 \iff a \in \ker \theta_{L/K} = N_{L/K} L^\times$ and thus $\bar{a} = 0$, showing that the left kernel of this pairing is also trivial. □

Lemma 3.4.4. *Let G be a finite group and A and B be G-modules. Let $f\colon G \to B$ be a 1-cocycle.*

(i) for $a \in \ker(A \overset{N_G}{\to} A)$, the cup product of $[a] \in H_T^{-1}(G, A)$ and $[f] \in H_T^1(G, B)$ is given by

$$[a] \cup [f] = -\Big[\sum_{\sigma \in G} \sigma(a) \otimes f(\sigma)\Big] \in H_T^0(G, A \otimes B)$$

Here brackets denote the corresponding cohomology classes.

(ii) for $\sigma \in G$ denote by $\bar{\sigma} \in H_T^{-2}(G, \mathbb{Z})$ the image of σ under the isomorphism $G^{ab} \approx H_T^{-2}(G, \mathbb{Z})$. Then

$$\bar{\sigma} \cup [f] = [f(\sigma)] \in H_T^{-1}(G, B)$$

Proof. We use dimension shifting. Write an exact sequence of G-modules

$$0 \to B \to B' \to B'' \to 0$$

with B' induced, and such that it is **split** as a sequence of abelian groups, so that

$$0 \to A \otimes B \to A \otimes B' \to A \otimes B'' \to 0$$

is still exact (check the appendix for more details). Then the connecting map $\delta\colon H_T^0(G, B'') \overset{\approx}{\to} H_T^1(G, B)$ is an isomorphism and thus we may write $[f] = \delta[b'']$ for some $b'' \in B''^G$ and

$$[a] \cup [f] = [a] \cup \delta[b''] = -\delta[a \otimes b'']$$

On the other hand, the connecting map $\delta\colon H_T^{-1}(G, A\otimes B'') \to H_T^0(G, A\otimes B)$ is induced by the norm. There exists a pre-image $b' \in B'$ of b'' such that $f(\sigma) = \sigma(b') - b' \in B$ for all $\sigma \in G$. Since $a\otimes b'$ is a pre-image of $a\otimes b''$ under $A\otimes B' \to A\otimes B''$, we conclude that $\delta[a\otimes b'']$ is represented by

$$\begin{aligned} N_G(a\otimes b') &= \sum_{\sigma\in G} \sigma a\otimes \sigma(b') = \sum_{\sigma\in G} \sigma a\otimes f(\sigma) + \sum_{\sigma\in G} \sigma a\otimes b' \\ &= \sum_{\sigma\in G} \sigma a\otimes f(\sigma) + N_G(a)\otimes b' = \sum_{\sigma\in G} \sigma a\otimes f(\sigma) \end{aligned}$$

To show 2, observe that tensoring the augmentation sequence $(**)$ with B we obtain an exact sequence

$$0 \to I_G\otimes B \to \mathbb{Z}[G]\otimes B \to \mathbb{Z}\otimes B = B \to 0$$

since $\mathbb{Z}$ is a free $\mathbb{Z}$-module and hence $\mathrm{Tor}_1^{\mathbb{Z}}(B,\mathbb{Z}) = 0$. Since $\mathbb{Z}[G]\otimes B$ is induced (see appendix), we have that the connecting map $\delta_a\colon H_T^{-1}(G,B) \xrightarrow{\approx} H_T^0(G, I_G\otimes B)$ is an isomorphism, hence it suffices to show that $\delta_a(\bar\sigma\cup[f]) = \delta_a[f(\sigma)]$. Since $\delta_a(\bar\sigma\cup[f]) = \delta_a(\bar\sigma)\cup[f]$ and $\delta_a(\bar\sigma) = [\sigma-1] \in H_T^{-1}(G, I_G)$, applying 1 we obtain

$$\begin{aligned} \delta_a(\bar\sigma\cup[f]) = [\sigma-1]\cup[f] &= -\Big[\sum_{\tau\in G} \tau(\sigma-1)\otimes f(\tau)\Big] \\ &= \Big[\sum_{\tau\in G} \tau\otimes f(\tau) - \sum_{\tau\in G} \tau\sigma\otimes f(\tau)\Big] = \Big[\sum_{\tau\in G} \tau\sigma\otimes f(\tau\sigma) - \sum_{\tau\in G} \tau\sigma\otimes f(\tau)\Big] \\ &= \Big[\sum_{\tau\in G} \tau\sigma\otimes \tau f(\sigma)\Big] \in H_T^0(G, I_G\otimes B) \end{aligned}$$

On the other hand, since $\delta_a\colon H_T^{-1}(G,B) \xrightarrow{\approx} H_T^0(G, I_G\otimes B)$ is induced by the norm and $1\otimes f(\sigma) \in \mathbb{Z}[G]\otimes B$ is a pre-image of $f(\sigma) \in B = \mathbb{Z}\otimes B$ we have that

$$\delta_a[f(\sigma)] = \big[N_G\big(1\otimes f(\sigma)\big)\big] = \Big[\sum_{\tau\in G} \tau\otimes\tau f(\sigma)\Big] \in H_T^0(G, I_G\otimes B)$$

Hence the two classes $\delta_a(\bar\sigma\cup[f])$ and $\delta_a[f(\sigma)]$ are equal in $H_T^0(G, I_G\otimes B)$ since

$$\begin{aligned} \sum_{\tau\in G} \tau\sigma\otimes\tau f(\sigma) - \sum_{\tau\in G} \tau\otimes\tau f(\sigma) &= \sum_{\tau\in G} \tau(\sigma-1)\otimes\tau f(\sigma) \\ &= N_G\big((\sigma-1)\otimes f(\sigma)\big) \in N_G(I_G\otimes B) \end{aligned}$$

□

Using the description of the reciprocity map given in 3.4.3, it is easy to show that for an unramified extension of local fields $L \supseteq K$ one has $\theta_{L/K}(a) = \mathrm{Frob}_{L/K}^{v(a)}$, where v denotes the normalised valuation of K and $\mathrm{Frob}_{L/K}$ is the Frobenius automorphism of $L \supseteq K$. In fact, let $n = [L:K]$

and $G = \mathrm{Gal}(L/K)$ and consider the character $\chi\colon G \to \mathbb{Q}/\mathbb{Z}$ given by $\chi(\mathrm{Frob}_{L/K}) = \frac{1}{n} \bmod \mathbb{Z}$. Then $\bar{a} \cup \delta\chi$ is represented by the 2-cocycle $c\colon G \times G \to L^\times$ given by

$$c(\mathrm{Frob}_{L/K}^i, \mathrm{Frob}_{L/K}^j) = \begin{cases} a & \text{if } i + j \geqslant 0 \\ 0 & \text{otherwise} \end{cases} \qquad 0 \leqslant i, j < n$$

Now since the invariant map is given by the composition

$$H^2(G, L^\times) \underset{\approx}{\xrightarrow{v}} H^2(G, \mathbb{Z}) \underset{\approx}{\xleftarrow{\delta}} H^1(G, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) \approx \mathbb{Q}/\mathbb{Z}$$

and $v([c]) = \delta[f]$ where $f\colon G \to \mathbb{Q}/\mathbb{Z}$ is the 1-cocycle given by $f(\mathrm{Frob}_{L/K}) = \frac{v(a)}{n} \bmod \mathbb{Z}$, we conclude that

$$\chi\big(\theta_{L/K}(a)\big) = \mathrm{inv}_K(\bar{a} \cup \delta\chi) = \mathrm{inv}_K([c]) = \frac{v(a)}{n} \bmod \mathbb{Z} = \chi(\mathrm{Frob}_{L/K}^{v(a)})$$

showing that $\theta_{L/K}(a) = \mathrm{Frob}_{L/K}^{v(a)}$ since χ is injective.
We can also show the compatibility of the various maps $\theta_{L/K}$ in the sense that given finite extensions $M \supseteq L \supseteq K$ with M and L abelian over K we have a commutative diagram

$$\begin{tikzcd} K^\times \arrow[r, "\theta_{M/K}"] \arrow[dr, "\theta_{L/K}"'] & \mathrm{Gal}(M/K) \arrow[d, two heads, "\mathrm{can.}"] \\ & \mathrm{Gal}(L/K) \end{tikzcd}$$

This will prove that the maps $\theta_{L/K}$ fit together into a single map $\theta_K\colon K^\times \to G_K^{ab}$, which will then satisfy all the properties required in 3.1.1.
Let $a \in K^\times$, let $\chi \in \mathrm{Hom}(\mathrm{Gal}(L/K), \mathbb{Q}/\mathbb{Z})$ and denote by $\chi' = \inf(\chi) \in \mathrm{Hom}(\mathrm{Gal}(M/K), \mathbb{Q}/\mathbb{Z})$. Then $\inf(\bar{a} \cup \delta\chi) = \bar{a} \cup \delta\chi'$ and thus

$$\chi(\theta_{L/K}(a)) = \mathrm{inv}_K(\bar{a} \cup \delta\chi) = \mathrm{inv}_K(\bar{a} \cup \delta\chi') = \chi'(\theta_{M/K}(a)) = \chi(\theta_{M/K}(a)|_L)$$

Since this holds for all $\chi \in \mathrm{Hom}(\mathrm{Gal}(L/K), \mathbb{Q}/\mathbb{Z})$, we conclude that $\theta_{L/K}(a) = \theta_{M/K}(a)|_L$, as was to be shown.
With the above, we finish the proof of the local reciprocity theorem. Hurray!

3.5 Hilbert Symbol and Proof of Existence Theorem

In this section, we prove the existence theorem, and for that we shall need a few results from Kummer theory and its relation with local reciprocity. This is accomplished in the first subsection.

3.5.1 Hilbert symbol

For any field K and any positive integer n prime to $\operatorname{char} K$, denote by $\mu_n \subset K^\times_{sep}$ the group of n-th roots of 1 of K.

Definition 3.5.1. *Let K be a local field such that $K \supset \mu_n$. For $a, b \in K^\times$ we define the* **Hilbert symbol** (a, b) *as*

$$(a,b) \overset{\text{def}}{=} \frac{\theta_K(b)(\alpha)}{\alpha} \in \mu_n \qquad \text{where} \quad \alpha^n = a, \quad \alpha \in K^\times_{sep}$$

Observe that since $K \supset \mu_n$ this definition is independent of the choice of α.

Lemma 3.5.2. *Let K be a local field such that $K \supset \mu_n$. The Hilbert symbol $(-,-)\colon K^\times \times K^\times \to \mu_n$ has the following properties:*

(i) Bilinearity: for all $a, a_1, a_2, b, b_1, b_2 \in K^\times$,

$$(a_1 \cdot a_2, b) = (a_1, b) \cdot (a_2, b) \quad \textit{and} \quad (a, b_1 \cdot b_2) = (a, b_1) \cdot (a, b_2)$$

(ii) Steinberg relations:

$$(a, 1-a) = 1 \textit{ for all } a \neq 0, 1 \quad \textit{and} \quad (a, -a) = 1 \textit{ for all } a \in K^\times$$

(iii) Skew-symmetry: $(a, b) = (b, a)^{-1}$ for all $a, b \in K^\times$

(iv) $(a, b) = 1$ if and only if $b \in N_{L/K} L^\times$ where $L = K(\sqrt[n]{a})$

(v) The Hilbert symbol gives a perfect pairing $K^\times/(K^\times)^n \times K^\times/(K^\times)^n \to \mu_n$: if $a \in K^\times$ is such that $(a, b) = 1$ for all $b \in K^\times$ then $a \in (K^\times)^n$, and similarly for the other entry.

Proof. Write $\alpha = \sqrt[n]{a}$. The Hilbert symbol is clearly linear in the first entry, and also in the second since $\theta_K(b_2)(\alpha)$ is an n-th root of a and thus

$$(a, b_1 \cdot b_2) = \frac{\theta_K(b_1 b_2)(\alpha)}{\alpha} = \frac{\theta_K(b_1)\big(\theta_K(b_2)(\alpha)\big)}{\theta_K(b_2)(\alpha)} \cdot \frac{\theta_K(b_2)(\alpha)}{\alpha} = (a, b_1) \cdot (a, b_2)$$

Also, 4 is a direct consequence of the definition of the Hilbert symbol and the reciprocity **??**. Now we show that 4 $\implies$ 2. Since (a, b) is a $[L : K]$-th root of 1 (by Kummer theory for instance), we may wlog assume that $[L : K] = n$. Let ζ be a primitive n-th root of 1. Then

$$x^n - a = \prod_{0 \leqslant i \leqslant n} (x - \zeta^i \alpha) \implies \begin{cases} 1 - a = N_{L/K}(1 - \alpha) \\ -a = N_{L/K}(-\alpha) \end{cases}$$

and hence $(a, 1-a) = 1$ and $(a, -a) = 1$ by 4 (actually, $(a, -a) = 1$ can be shown to be a formal consequence of bilinearity and the identity $(a, 1-a) = 1$, as the reader may verify).

Now we show that 3 follows formally from 1 and 2. In fact:

$$(ab, -ab) = 1 \iff (a, -a) \cdot (a, b) \cdot (b, a) \cdot (b, -b) = 1 \iff (a, b) \cdot (b, a) = 1$$

Finally, suppose that $(a, b) = 1$ for all $b \in K^\times$, and let $L = K(\alpha)$ as above. Since $\theta_{L/K}\colon K^\times \twoheadrightarrow \operatorname{Gal}(L/K)$ is surjective (see **??**) we conclude that $\alpha \in K$, i.e., $a \in (K^\times)^n$, as was to be shown. By the skew-symmetry 3, the same holds for the other entry, and we are done. $\square$

Properties 3–5 immediately yield the following interesting corollary, which will be used below in the proof of the Existence Theorem.

Corollary 3.5.3. *Let n be a positive integer and let K be a local field containing a primitive n-th root of 1. If an element $a \in K^\times$ belongs to $N_{L/K}L^\times$ for every cyclic extension $L \supset K$ of degree n then $a \in (K^\times)^n$.*

The Hilbert symbol can be defined more intrinsically in terms of the so-called **Galois symbol**. Recall that for any field K and any positive integer n prime to $\operatorname{char} K$ (we do not assume $K \supset \mu_n$), we have an exact sequence of G_K-modules (the *Kummer sequence*)

$$1 \to \mu_n \to K_{sep}^\times \xrightarrow{n} K_{sep}^\times \to 1$$

where the last map is given by $x \mapsto x^n$. By Hilbert 90 the connecting map $\partial\colon K^\times \to H^1(G_K, \mu_n)$ induces an isomorphism $K^\times/(K^\times)^n = H^1(G_K, \mu_n)$, and we also have that $H^2(G_K, \mu_n)$ is the n-torsion of $H^2(G_K, K_{sep}^\times)$. Explicitly, ∂a is given by the 1-cocycle $\sigma \mapsto \sigma(\alpha)/\alpha \in \mu_n$ where $\alpha = \sqrt[n]{a}$ is any n-th root of a in K_{sep}.
The Galois symbol is defined to be the pairing $\{-,-\}\colon K^\times \times K^\times \to H^2(G_K, \mu_n \otimes \mu_n)$ given by

$$K^\times \otimes K^\times \xrightarrow{\partial\otimes\partial} H^1(G_K, \mu_n) \otimes H^1(G_K, \mu_n) \xrightarrow{\cup} H^2(G_K, \mu_n \otimes \mu_n) \tag{$\dagger$}$$

It has the following properties:

(i) Bilinearity: $\{a_1 \cdot a_2, b\} = \{a_1, b\} \cdot \{a_2, b\}$ and $\{a, b_1 \cdot b_2\} = \{a, b_1\} \cdot \{a, b_2\}$

(ii) Steinberg relations: $\{a, 1-a\} = 1$ for all $a \neq 0, 1$ and $\{a, -a\} = 1$ for all $a \in K^\times$

(iii) Skew-symmetry: $\{a, b\} = \{b, a\}^{-1}$ for all $a, b \in K^\times$

(iv) If $\mu_n \subset K$ then $\{a, b\} = 1 \iff a \in N_{L/K}L^\times$ where $L = K(\sqrt[n]{b})$.

Property 1 is clear, and 2–3 are formal consequences of $\{a, 1-a\} = 1$ (one can also prove 3 using the skew-symmetry of the cup product). To prove the last Steinberg relation, one factors $x^n - a = \prod_i f_i(x)$ into irreducible polynomials $f_i(x) \in K[x]$ and set $K_i = K(\alpha_i)$ where $\alpha_i \in K_{sep}$ is any root of $f_i(x)$. Then

$$\begin{aligned}\{1-a, a\} &= \prod_i \{N_{K_i/K}(1-\alpha_i), a\} = \prod_i \operatorname{cor}_{K_i/K}\{1-\alpha_i, a\} \\ &= \prod_i \operatorname{cor}_{K_i/K}\{1-\alpha_i, \alpha_i\}^n = 1\end{aligned}$$

where $\operatorname{cor}_{K_i/K}\colon H^2(G_{K_i}, \mu_n \otimes \mu_n) \to H^2(G_K, \mu_n \otimes \mu_n)$ is the corestriction map.

Finally, to show 4 let $\zeta \in K$ be a primitive n-th root of unity. The choice of ζ defines a *non-canonical* isomorphism of G_K-modules $\phi_\zeta \colon \mu_n \to \mathbb{Z}/n$ given by $\phi_\zeta(\zeta) = 1 \bmod n$ and hence we obtain a commutative diagram

$$\begin{tikzcd}
H^1(G_K, \mu_n) \otimes H^1(G_K, \mu_n) \arrow[r, "\cup"] \arrow[d, "\cong", "\mathrm{id}\otimes\phi_\zeta"'] & H^2(G_K, \mu_n \otimes \mu_n) \arrow[d, "\mathrm{id}\otimes\phi_\zeta"', "\cong"] \\
H^1(G_K, \mu_n) \otimes H^1(G_K, \mathbb{Z}/n) \arrow[r, "\cup"] & H^2(G_K, \mu_n)
\end{tikzcd}$$

where the vertical arrows are isomorphisms. Then the symbol $\{a, b\}$ corresponds to the element $\partial a \cup \chi_b \in H^2(G_K, \mu_n)$ where $\chi_b \in H^1(G_K, \mathbb{Z}/n)$ is the character given by $\chi_b(\sigma) = \phi_\zeta\big(\partial b(\sigma)\big) = \phi_\zeta\big(\sigma(\sqrt[n]{b})/\sqrt[n]{b}\big)$. Hence we have to show that $\partial a \cup \chi_b = 0 \iff a \in N_{L/K}L$ where $L = K_{sep}^{\ker \chi_b} = K(\sqrt[n]{b})$ denotes the cyclic extension of K defined by χ_b.

We make the identifications $H^1(G_K, \mathbb{Z}/n) = {}_nH^1(G_K, \mathbb{Q}/\mathbb{Z})$ and ${}_nH^2(G_K, K_{sep}^\times) = H^2(G_K, \mu_n)$ where the subscript denotes the n-torsion part of the corresponding groups. Now an explicit computation shows that, for all $a \in K^\times$ and $\chi \in H^1(G_K, \mathbb{Z}/n) = {}_nH^1(G_K, \mathbb{Q}/\mathbb{Z})$,

$$a \cup \delta\chi = -\partial a \cup \chi \in {}_nH^2(G_K, K_{sep}^\times) = H^2(G_K, \mu_n)$$

where δ is as in 3.4.3. In fact, let $G = G_K/\ker\chi$ and σ be a generator of the cyclic group G. Let $n = |G|$ and $0 \leqslant t < n$ be such that $\chi(\sigma) = t \bmod n$. We have that $a \cup \delta\chi$ and $-\partial a \cup \chi$ are represented by the inflations of the 2-cocycles $f \colon G \times G \to L^\times$ and $g \colon G \times G \to \mu_n$ given by

$$f(\sigma^i, \sigma^j) = \begin{cases} a^t & \text{if } i + j \geqslant n \\ 1 & \text{otherwise} \end{cases} \qquad \text{and} \qquad g(\sigma^i, \sigma^j) = \left(\frac{\sigma^i \sqrt[n]{a}}{\sqrt[n]{a}}\right)^{-j\cdot t}$$

for $0 \leqslant i, j < n$, and it is easy to check that they differ by the coboundary of the function $h \colon G \to L^\times$ given by

$$h(\sigma^i) = (\sqrt[n]{a})^{i\cdot t} \qquad \text{for } 0 \leqslant i < n$$

We apply the above to $\chi = \chi_b$ and $G = \mathrm{Gal}(L/K)$. Since the cup product $- \cup \delta\chi_b$ gives an isomorphism $H_T^0(G, L^\times) \cong H_T^2(G, L^\times) \subset H^2(G_K, K_{sep}^\times)$ (see appendix) we have that

$$\partial a \cup \chi_b = 0 \iff a \cup \delta\chi_b = 0 \iff a \in N_{L/K}L^\times$$

as was to be shown.

What all this has to do with the Hilbert symbol? We now specialise the above discussion to the case when K is a local field containing μ_n. Since $H^2(G_K, \mu_n)$ is the n-torsion of $H^2(G_K, K_{sep}^\times) \overset{\mathrm{inv}_K}{=} \mathbb{Q}/\mathbb{Z}$, we have that $H^2(G_K, \mu_n) = \frac{1}{n}\mathbb{Z}/\mathbb{Z}$ and thus we obtain a **canonical** isomorphism

$$H^2(G_K, \mu_n \otimes \mu_n) \underset{\approx}{\overset{\cup}{\leftarrow}} H^2(G_K, \mu_n) \otimes H^0(G_K, \mu_n) \underset{\approx}{\overset{\mathrm{inv}_K \otimes \mathrm{id}}{\to}} \frac{\frac{1}{n}\mathbb{Z}}{\mathbb{Z}} \otimes \mu_n = \mu_n$$

which allows us to write the Galois symbol pairing (†) simply as a pairing $K^\times \otimes K^\times \to \mu_n$. We claim that this pairing is precisely the Hilbert symbol. In fact, let $\zeta \in K$ be a primitive n-th root of 1; given $a, b \in K^\times$, the pairing we have just defined takes $a \otimes b$ to

$$\zeta^{n\cdot \mathrm{inv}_K(\partial a \cup \chi_b)} = \zeta^{-n\cdot \mathrm{inv}_K(a \cup \delta\chi_b)} = \zeta^{-\chi_b(\theta_K(a))}$$

by the reciprocity 3.4.3. Here χ_b is defined via ϕ_ζ using the **same** chosen root of unity ζ, so that the dependency on this choice "cancels." Now if $\beta = \sqrt[n]{b}$ by definition of χ_b we have that

$$\zeta^{-\chi_b(\theta_K(a))} = \left(\frac{\theta_K(a)(\beta)}{\beta}\right)^{-1} = (b,a)^{-1} = (a,b)$$

proving the claim. This shows that properties 1–4 of the lemma are consequences of the corresponding properties of the Galois symbol. The reciprocity is only needed to show the perfectness of the pairing.

Remark 3.5.4. *When K has positive characteristic p, one has an exact sequence of G_K-modules (the* **Artin-Schreier** *sequence)*

$$0 \to \mathbb{Z}/p \to K^+_{sep} \xrightarrow{\wp} K^+_{sep} \to 0$$

where $\wp(x) = x^p - x$. Then it is possible to define a pairing

$$K^+ \times K^\times \to \mathbb{Z}/p$$

taking $a \in K^+$ and $b \in K^\times$ to $[a,b) \overset{\text{def}}{=} \delta\psi_a \cup b \in H^2(G_K, K^\times_{sep})$ where the character $\psi_a \in H^1(G_K, \mathbb{Z}/p)$ is defined to be the image of a with respect to the connecting map of the Artin-Schreier sequence.
One shows that for a local field K

$$[a,b) = \theta_K(b)(\alpha) - \alpha \in \mathbb{Z}/p \qquad \text{where} \qquad \wp(\alpha) = a$$

and that this pairing satisfies

(i) Bilinearity: $[a_1 + a_2, b) = [a_1, b) + [a_2, b)$ and $[a, b_1 b_2) = [a, b_1) + [a, b_2)$

(ii) Steinberg relation: $[a, -a) = 0$ for all $a \neq 0$

(iii) $[a,b) = 0 \iff b \in N_{L/K}L^\times$ where $L = K(\alpha)$ with $\wp(\alpha) = a$

(iv) If $[a,b) = 0$ for all $b \in K^\times$ then $a \in \wp(K^+)$.

The proofs are mutatis mutandis the same as for the Hilbert symbol; we refer the reader to Serre's "Local Fields" for further details.

Lemma 3.5.5. *Let p be a prime, q be a power of p and $K = \mathbb{F}_q((t))$. If an element $b \in K^\times$ belongs to $N_{L/K}L^\times$ for every cyclic extension $L \supset K$ of degree p then $b \in (K^\times)^p$.*

Proof. We sketch the proof and refer the reader to Serre's book for details. For any $a \in K^+$ and $b \in K^\times$ one has *Schmid's formula*

$$[a,b) = T_{\mathbb{F}_q/\mathbb{F}_p} \operatorname{res}\left(a \cdot \frac{db}{b}\right)$$

where $db/b = (\sum_{n \geqslant n_0} n c_n t^{n-1}) \cdot (\sum_{n \geqslant n_0} c_n t^n)^{-1}$ is the *logarithmic derivative* of $b = \sum_{n \geqslant n_0} c_n t^n$, and res denotes the **residue**, that is, the coefficient of t^{-1}. Schmid's formula can be proved by an explicit computation using the above properties of the pairing $[-,-)$.
Now if $b \in N_{L/K}L^\times$ for every cyclic extension $L \supset K$ of degree p then $[a,b) = 0$ for all $a \in K^+$. If b were not a p-th power then $db/b \neq 0$, and we would be able to find a such that $\operatorname{res}\left(a \cdot \frac{db}{b}\right) = c$ for any given $c \in \mathbb{F}_q$. But then by Schmid's formula $[a,b) = T_{\mathbb{F}_q/\mathbb{F}_p}(c) = 0$ for all $c \in \mathbb{F}_q$, contradicting the fact that the trace is surjective. $\square$

3.5.2 Proof of the Existence Theorem

Now we are ready to prove the existence theorem. First we remark that in the definition of a norm group $N_{L/K}L^\times$ it is indifferent whether we require $L \supset K$ to be an abelian Galois extension or not:

Theorem 3.5.6 (Norm Limitation)**.** *Let $L \supset K$ be a finite separable extension of local fields and let E be the maximal abelian extension of K contained in L. Then $N_{E/K}E^\times = N_{L/K}L^\times$.*

Proof. Let M be a finite Galois extension of K that contains L, let $G = \mathrm{Gal}(M/K)$ and $H = \mathrm{Gal}(M/L)$ so that E is the fixed field of $H \cdot [G : G]$. Since the cup product commutes with corestriction, we have a commutative diagram (check!)

$$\begin{tikzcd}
\frac{L^\times}{N_{M/L}M^\times} \arrow[r, "\theta_{M/L}", "\approx"'] \arrow[d, "N_{L/K}"] & \frac{H}{[H:H]} \arrow[d, "\text{can.}"] \\
\frac{K^\times}{N_{M/K}M^\times} \arrow[r, "\theta_{M/K}", "\approx"'] & \frac{G}{[G:G]}
\end{tikzcd}$$

where the horizontal arrows are isomorphisms by the reciprocity theorem. Therefore $\theta_{M/K}$ gives an isomorphism between the cokernels of two vertical arrows and thus $[K^\times : N_{L/K}L^\times] = \big[G : H \cdot [G : G]\big] = |\,\mathrm{Gal}(E/K)|$. But we also have an isomorphism $\theta_{E/K} \colon K^\times/N_{E/K}E^\times \approx \mathrm{Gal}(E/K)$ and hence both $N_{E/K}E^\times$ and $N_{L/K}L^\times$ have the same index in $K^\times$. However $N_{E/K}E^\times \supset N_{L/K}L^\times$ since $E \subset L$ and therefore we must have $N_{E/K}E^\times = N_{L/K}L^\times$. □

Before plunging into the proof of the existence theorem let us make a couple of the preliminary observations. Let G be any topological group and $H \leqslant G$ be a subgroup. If H is open then it is also closed, since H is the complement of the union of its (open) left cosets different from H. The very same argument shows that if H is closed and of finite index then H is also open so that for subgroups of finite index "open" and "closed" are equivalent notions. Moreover in order to show that H is open it is enough to show that it contains an open subgroup T, for then H will be the union of the left translates of T. Finally observe that the norm map $N_{L/K} \colon L \to K$ is continuous for all finite extensions $L \supset K$ of local fields since it is given by a product of automorphisms of some finite Galois extension M of K, and these automorphisms are continuous since they preserve the valuation of M (Highlander's Philosophy!).

We first show the easy direction of the existence theorem:

Lemma 3.5.7. *Let $L \supset K$ be a finite extension of local fields. Then $N_{L/K}L^\times$ is an open (and closed) subgroup of finite index of $K^\times$. Moreover the kernel of $N_{L/K} \colon L^\times \to K^\times$ is compact.*

Proof. We already know that $N_{L/K}L^\times$ has finite index in $K^\times$ by the reciprocity **??**. Also, $U_K \cap N_{L/K}L^\times = N_{L/K}U_L$ (see the formula for the valuation of L in theorem 1.3.4), so that we have an injection $U_K/N_{L/K}U_L \hookrightarrow K^\times/N_{L/K}L^\times$ showing that $N_{L/K}U_L$ also has finite index in U_K. Since U_L is compact, $N_{L/K}U_L$ is compact and hence closed and open in U_K (since the latter group is Hausdorff). Since U_K is open in $K^\times$ we have that $N_{L/K}U_L$ is open in $K^\times$ as well. Therefore $N_{L/K}L^\times$ is open too as it contains the open subgroup $N_{L/K}U_L$. Moreover since $N_{L/K}$ is continuous we have that $\ker N_{L/K}$ is closed. And since it is contained in the compact subgroup U_L we conclude that $\ker N_{L/K}$ is compact as well. □

We now prove the converse. Let K is a local field and $T \subset K^\times$ be an open subgroup of finite index. First observe that if T contains a norm group $N_{L/K}L^\times$ for some finite abelian extension $L \supset K$ then T itself is a norm group: by the functorial properties of the reciprocity map we have that $T = N_{M/K}M^\times$ where $M = L^{\theta_K T}$. Also the intersection of two norm groups $N_{L_1/K}L_1^\times$ and $N_{L_2/K}L_2^\times$ is a norm group, namely the norm group of the compositum $L_1 \cdot L_2$. Hence the family of norm groups is **directed** (i.e. the intersection of any two members of this family contains a third member).

Definition 3.5.8. *Let K be a field. Then the* **universal norm group** *D_K of K is defined to be the intersection of all norm groups:*

$$D_K \overset{\text{def}}{=} \bigcap_{[K':K]<\infty} N_{K'/K}(K'^\times) \subset K^\times$$

The existence theorem is essentially the statement that for a local field K its universal norm group D_K is trivial. In fact, considering the images of the norm groups in the finite group $K^\times/T$ and using the fact that this family is directed, $D_K = 1$ implies that some member of this family must have trivial image, therefore T contains a norm group as was to be shown.

Hence all that is left to prove is

Lemma 3.5.9. *Let K be a local field. Then $D_K = 1$.*

Proof. First observe that if L is a finite extension of K then $D_K \subset N_{L/K}D_L$. To show this let $a \in D_K$. For each finite extension $L' \supset L$ we have that

$$F(L') \overset{\text{def}}{=} N_{L/K}^{-1}(a) \cap N_{L'/L}(L'^\times) \subset L^\times$$

is a non-empty compact set: by hypothesis there exists $c \in N_{L'/K}L'^\times$ such that $a = N_{L'/K}(c)$, hence $N_{L'/L}(c) \in F(L')$; moreover $N_{L'/L}(L'^\times)$ is closed and $N_{L/K}^{-1}(a)$ is compact by the previous lemma. Clearly $F(L' \cdot L'') \subset F(L') \cap F(L'')$, hence by compactness (finite intersection property) we have that $\bigcap_{L'} F(L') \neq \varnothing$, and any element in this intersection belongs to D_L and has norm a.

Now we prove that D_K is trivial. It is enough to show that D_K is divisible since this implies $D_K = \bigcap_{n\geqslant 1}(K^\times)^n = 1$. Let p be a prime number. Given $a \in D_K$ we have to show that $x^p = a$ has a solution with $x \in D_K$. Assume first that $p \neq \operatorname{char} K$ and let ζ be a primitive p-th root of 1. For each finite extension $L \supset K(\zeta)$ consider the finite set

$$E(L) \overset{\text{def}}{=} \{b \in K^\times \mid b^p = a \text{ with } b \in N_{L/K}L^\times\}$$

Clearly $E(L \cdot L') \subset E(L) \cap E(L')$. Moreover each $E(L)$ is non-empty: writing $a = N_{L/K}(c)$ with $c \in D_L$ (which is possible by the above) then **??** implies that $c = d^p$ for some $d \in L^\times$ and hence $N_{L/K}(d) \in E(L)$. By compactness, $\bigcap_L E(L) \neq \varnothing$ and any element x in this intersection is in D_K and satisfies $x^p = a$, as required. If $p = \operatorname{char} K$, the proof is similar, but using 3.5.5 instead. $\square$

Remark 3.5.10. *The lemma together with* **??** *shows that $\theta_K \colon K^\times \to G_K^{ab}$ is injective and has dense image. Therefore we have a commutative diagram*

$$\begin{tikzcd}
0 \arrow[r] & U_K \arrow[r] \arrow[d, "\theta_K", "\approx"'] & K^\times \arrow[r, "v"] \arrow[d, hook, "\theta_K"] & \mathbb{Z} \arrow[r] \arrow[d, hook, "\text{inclusion}"] & 0 \\
0 \arrow[r] & I_K^{ab} \arrow[r] & G_K^{ab} \arrow[r, "\text{can.}"] & G_K^{nr} = \hat{\mathbb{Z}} \arrow[r] & 0
\end{tikzcd}$$

where I_K^{ab} is the **inertia group** *of $K^{ab} \supset K$.*

3.6 Exercises

3.6.1. *Show that* $\theta_{\mathbb{Q}_5(\zeta_5)/\mathbb{Q}_5}(2)(\zeta_5) = \zeta_5^3$.

3.6.2. *Let* $M \supset L \supset K$ *be finite extensions of local fields with* $M \supset K$ *abelian. Show that the following diagrams commute:*

$$\begin{tikzcd}
\frac{L^\times}{N_{M/L}(M^\times)} \arrow[r, "\theta_{M/L}", "\approx"'] \arrow[d, "N_{L/K}"] & \mathrm{Gal}(M/L) \arrow[d, hook] \\
\frac{K^\times}{N_{M/K}(M^\times)} \arrow[r, "\theta_{M/K}", "\approx"'] & \mathrm{Gal}(M/K)
\end{tikzcd}
\qquad
\begin{tikzcd}
\frac{L^\times}{N_{M/L}(M^\times)} \arrow[r, "\theta_{M/L}", "\approx"'] & \mathrm{Gal}(M/L) \\
\frac{K^\times}{N_{M/K}(M^\times)} \arrow[r, "\theta_{M/K}", "\approx"'] \arrow[u] & \mathrm{Gal}(M/K) \arrow[u, "\mathrm{Ver}"]
\end{tikzcd}$$

Here the unlabelled maps are the canonical ones and Ver *denotes the* **transfer map** *(Verlagerung in German): given finite groups* $H \leqslant G$, $\mathrm{Ver}\colon G^{ab} \to H^{ab}$ *is defined via the restriction map* $\mathrm{res}\colon G^{ab} = H_T^{-2}(G, \mathbb{Z}) \to H^{ab} = H_T^{-2}(H, \mathbb{Z})$.

3.6.3. *Compute the reciprocity map for the cyclotomic extension* $\mathbb{Q}_3(\zeta_{18}) \supset \mathbb{Q}_3$ *and draw the lattices of subfields and corresponding norm groups.*

3.6.4. *Show that every finite abelian group is the Galois group of some extension of* $\mathbb{Q}$.

3.6.5 (Witt)**.** *Let* $(R, \mathfrak{m}, k)$ *be a complete discrete valuation ring with* $K = \mathrm{Frac}\, R$. *Let* $L \supseteq K$ *be a finite unramified extension with corresponding residue field extension* $l \supseteq k$. *Write* $G \overset{\text{def}}{=} \mathrm{Gal}(L/K) \approx \mathrm{Gal}(l/k)$ *for its Galois group. Show that for all integers* $i \geqslant 1$ *there is an exact sequence*

$$0 \to H^i(G, l^\times) \to H^i(G, L^\times) \xrightarrow{v_L} H^{i-1}(G, \mathbb{Q}/\mathbb{Z}) \to 0$$

3.6.6 (Witt)**.** *which is splits non-canonically, depending on the choice a uniformiser* $\pi_K \in K$:

$$\begin{aligned} H^i(G, l^\times) \oplus H^{i-1}(G, \mathbb{Q}/\mathbb{Z}) &\xrightarrow{\approx} H^i(G, L^\times) \\ (a, \chi) &\mapsto a \cup \delta(\chi) \end{aligned}$$

Here $\delta\colon H^{i-1}(G, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\approx} H^i(G, \mathbb{Z})$ *is the coboundary map in the long exact sequence associated to the short one*

$$0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0 \tag{3.7}$$

Chapter 4

Global Class Field Theory

By a **global field** we mean either a **function field** or a **number field**, that is, a finite extension of $\mathbb{F}_p(t)$ or a finite extension of $\mathbb{Q}$, respectively. The main result of this chapter is the following computation:

Theorem 4.0.1 (Brauer group of global fields)**.** *Let K be a global field. One has an exact sequence*

$$\boxed{0 \to \mathrm{Br}(K) \overset{\oplus_v \mathrm{res}_v}{\to} \bigoplus_v \mathrm{Br}(K_v) \overset{\oplus_v \mathrm{inv}_{K_v}\, i}{\to} \mathbb{Q}/\mathbb{Z} \to 0}$$

Here K_v denotes the completion of K with respect to the place v of K, $\mathrm{res}_v\colon \mathrm{Br}(K) \to \mathrm{Br}(K_v)$ and $\mathrm{inv}_{K_v}\colon \mathrm{Br}(K_v) \to \mathbb{Q}/\mathbb{Z}$ are the usual restriction and invariant maps, and the direct sum runs over all places of K, including the archimedean ones.

The proof of the above theorem is the main step of the proof of the global reciprocity law. In terms of central simple algebras, it describes the **local-global** or **Haße principle** for division algebras: a division algebra D over a global field K is completely determined by its restrictions $D \otimes_K K_v$ to the local fields K_v. In particular, D is trivial if and only if $D \otimes_K K_v$ is trivial for all v.

4.1 The geometric case

In the next section, we will present a proof that works over a general global field K. However the geometric case, that is, the case when K is a finite extension of $\mathbb{F}_p(t)$, and therefore the function field of a non-singular projective curve C, is much easier to deal with, and serves as inspiration for the general method. We thus give a sketch of the proof of this special case first, assuming a few facts from the theory of algebraic curves.

Here is a quick review of the facts we shall need. Given a non-singular projective curve C over a field k, we can view it as a collage of two Dedekind domains, namely the integral closures of $k[t]$ and $k[\frac{1}{t}]$ in its function field $K = k(C)$. For technical reasons, we assume that k is algebraically closed in K from now on. Each closed point $P \in C_0$ (not necessarily defined over k) corresponds to a maximal ideal $\mathfrak{m}_P$ of one of these two Dedekind domains, and hence P defines a discrete valuation $v_P\colon K \to \mathbb{Z} \cup \{\infty\}$: $v_P(h)$ equals the exponent of $\mathfrak{m}_P$ in the factorisation of the principal fractional ideal (h). In geometric terms, $v_P(h)$ measures the order of the zero or pole of the function h at the point P.

A **divisor** in C is just an element of the free abelian group generated by the closed points of C:

$$\mathrm{Div}(C) = \bigoplus_{P \in C_0} \mathbb{Z}P$$

Roughly, $\mathrm{Div}(C)$ corresponds to the multiplicative group of non-zero fractional ideals of a Dedekind domain, written additively. The **degree** $\deg D$ of a divisor $D = \sum_P n_P P \in \mathrm{Div}(C)$, $n_P \in \mathbb{Z}$, is defined as a weighted sum of the finitely many non-zero coefficients n_P of D: $\deg D = \sum_P n_P f_P \in \mathbb{Z}$, where $f_P = [k(P) : k]$ is the residue degree of P, that is, the residue degree of the discrete valuation v_P over its restriction to $k[t]$ or $k[\frac{1}{t}]$.
Each element $h \in K^\times$ defines a **principal divisor**

$$\mathrm{div}(h) \overset{\mathrm{def}}{=} \sum_{P \in C_0} v_P(h) P$$

since $v_P(h) = 0$ for all but finitely many P. A principal divisor corresponds to a principal fractional ideal in a Dedekind domain. An explicit computation shows that $\deg \mathrm{div}(h) = 0$ for any principal divisor $\mathrm{div}(h)$ (do it first for $\mathbb{P}^1_k$ and then use the formula $\sum_i e_i f_i = n$ for extension of primes in Dedekind domains). It is also easy to check that $\mathrm{div}(h) = 0 \iff h \in k^\times$ (only constant functions have no zeros or poles).
The **Picard group** $\mathrm{Pic}(C)$ of C is defined as the quotient of $\mathrm{Div}(C)$ by the subgroup of principal divisors. It corresponds to a globalised form of the usual class group of a Dedekind domain. We have thus an exact sequence

$$0 \to k^\times \to K^\times \overset{\mathrm{div}}{\to} \mathrm{Div}(C) \to \mathrm{Pic}(C) \to 0$$

Since the degree of a principal divisor is zero, the degree of an element of $\mathrm{Pic}(C)$ is well-defined. The kernel of the degree map is denoted by $\mathrm{Pic}^0(C)$. If C has k-rational points (i.e. points with residue degree 1, or points "defined over k"), the degree map is surjective and we have an exact sequence

$$0 \to \mathrm{Pic}^0(C) \to \mathrm{Pic}(C) \overset{\deg}{\to} \mathbb{Z} \to 0$$

End of the review. From now on, let C be a non-singular projective curve defined over a finite field k, and let $K = k(C)$ be its function field, which is a global field. Let us assume for simplicity that C has a k-rational point (even though the proof can be harnessed to eliminate this hypothesis). The computation of $\mathrm{Br}(K)$ is based on the following theorem:

Theorem 4.1.1. *Let C be a non-singular projective curve defined over a finite field k. Write $\overline{C} = C \times_{\mathrm{Spec}\, k} \mathrm{Spec}\, k_{\mathrm{sep}}$ for the curve over k_{sep} obtained by base change from C, and $\overline{K} = K \cdot k_{\mathrm{sep}}$ for the function field of $\overline{C}$. Then*

(i) $\mathrm{Pic}^0(\overline{C})$ *is a torsion abelian group;*

(ii) (Lang) $H^i(G_k, \mathrm{Pic}^0(\overline{C})) = 0$ *for* $i > 0$*;*

(iii) (Tsen) $\mathrm{Br}(\overline{K}) = 0$.

Proof. Omitted. The proof of (1) follows from the fact that there is a natural bijection $\mathrm{Pic}^0(\overline{C}) = J_C(k_{\mathrm{sep}})$ where J_C is the Jacobian variety of C (see Milne's notes "Abelian varieties", for instance), and $J_C(k_{\mathrm{sep}})$ is the union of the finite groups $J_C(l)$ where l runs over all finite extensions of k. On the other hand, (2) holds more generally for any connected algebraic group instead of Pic^0 by Lang's article *Algebraic groups over finite fields*, Amer. J. Math. **78** (1956), 555–563. For a proof of Tsen's theorem, see Shatz's book for instance. □

From Lang's theorem and the long exact sequence associated to the short one

$$0 \to \mathrm{Pic}^0(\overline{C}) \to \mathrm{Pic}(\overline{C}) \xrightarrow{\deg} \mathbb{Z} \to 0$$

we obtain an isomorphism $\deg\colon H^i(G_k, \mathrm{Pic}(\overline{C})) \xrightarrow{\approx} H^i(G_k, \mathbb{Z})$ for $i > 0$ (actually, these groups are 0 for $i > 2$ since $G_k \cong \hat{\mathbb{Z}}$ has cohomological dimension 1). Finally, from the exact sequence of G_k-modules

$$0 \to \overline{K}^\times / k_{\mathrm{sep}}^\times \to \mathrm{Div}(\overline{C}) \to \mathrm{Pic}(\overline{C}) \to 0$$

and the above computations, we obtain an exact sequence

$$\begin{aligned} & H^1(G_k, \mathrm{Div}(\overline{C})) \xrightarrow{\deg} H^1(G_k, \mathbb{Z}) \\ \to\ & H^2(G_k, \overline{K}^\times / k_{\mathrm{sep}}^\times) \to H^2(G_k, \mathrm{Div}(\overline{C})) \to H^2(G_k, \mathbb{Z}) \\ \to\ & H^3(G_k, K^\times / k_{\mathrm{sep}}^\times) = 0 (*) \end{aligned}$$

Here the last term vanishes since G_k has cohomological dimension 1. Since we are assuming that C has a k-rational point, we have that the map deg is surjective. On the other hand, we already know from the computations of last chapter that $H^2(G_k, \mathbb{Z}) = \mathbb{Q}/\mathbb{Z}$. We also have that

$$H^2(G_k, \mathrm{Div}(\overline{C})) = H^2(G_k, \bigoplus_{P \in C_0} \mathrm{Ind}\, \mathbb{Z}P) = \bigoplus_{P \in C_0} H^2(G_k, \mathbb{Z})$$

and each factor $H^2(G_k, \mathbb{Z}) = \mathbb{Q}/\mathbb{Z}$ can be identified with $\mathrm{Br}(K_P)$, the Brauer group of the completion of K with respect to the valuation v_P. Finally, using once more the fact that G_k has cohomological dimension 1, from the exact sequence

$$0 \to k_{\mathrm{sep}}^\times \to \overline{K}^\times \to \overline{K}^\times / k_{\mathrm{sep}}^\times \to 0$$

we obtain an isomorphism

$$H^2(G_k, \overline{K}^\times) = H^2(G_k, \overline{K}^\times / k_{\mathrm{sep}}^\times)$$

But the first group is $\mathrm{Br}(\overline{K}/K)$, which equals $\mathrm{Br}(K)$ by Tsen's theorem.
Putting everything together, we obtain from $(*)$ the exact sequence

$$0 \to \mathrm{Br}(K) \to \bigoplus_{P \in C_0} \mathrm{Br}(K_P) \to \mathbb{Q}/\mathbb{Z} \to 0$$

Following the isomorphisms, it is easy to check that the second map is just the restriction map while the third is the sum of the invariant maps.

4.2 The general case

Now we turn to the general case. Let K be an arbitrary global field. One of the key steps in the computation of $\mathrm{Br}(K)$ of the geometric case was Lang's theorem, which shows that $H^1(G_k, \mathrm{Pic}^0(\overline{C})) = H^1(G_k, J_C(k_{\mathrm{sep}})) = 0$. For number fields, we do not have an "arithmetic Jacobian" at our disposal. Instead, we shall define another object, the **idèle class group**, which will play an analogous role to the Jacobian (actually $J_C \times \mathbb{Z}$) in the geometric case.

In what follows, K will denote an arbitrary global field, and for any place v of K (archimedean or not), we write K_v for the completion of K with respect to v. For K a number field, an archimedean place v corresponds to an embedding $\sigma\colon K \hookrightarrow K^{\mathrm{alg}}$, which defines an absolute value

$$\|a\|_v \overset{\mathrm{def}}{=} \begin{cases} |\sigma(a)| & \text{if } \sigma \text{ is a real embedding} \\ |\sigma(a)|^2 & \text{if } \sigma \text{ is a complex embedding} \end{cases} \qquad a \in K$$

and K_v is isomorphic to $\mathbb{R}$ and $\mathbb{C}$, respectively. On the other hand, when v is non-archimedean, that is, $v\colon K \twoheadrightarrow \mathbb{Z} \cup \{\infty\}$ is a (surjective) valuation, we define its **normalised absolute value** by

$$\|a\|_v \overset{\mathrm{def}}{=} N(v)^{-v(a)} \qquad a \in K$$

where $N(v)$ is the cardinality of the residue field of v (which is a finite field). This definition is made so that the **product formula** holds:

$$\prod_v \|a\|_v = 1 \qquad a \in K^\times$$

Here v runs over all places of v, both archimedean or not. For non-archimedean places we also write

$$\begin{aligned} \mathcal{O}_v &= \{x \in K_v \mid v(x) \geqslant 0\} \\ U_v &= \mathcal{O}_v^\times = \{x \in K_v \mid v(x) = 0\} \end{aligned}$$

for its ring of integers and corresponding unit group, respectively.

Now let $L \supset K$ be a finite Galois extension of global fields with $G = \mathrm{Gal}(L/K)$. Recall that G acts transitively on the set of places $w_1, \ldots, w_g$ of L that divide v, via $\sigma(w_i) = w_i \circ \sigma^{-1}$. For $w \mid v$, denote by

$$G_w = \{\sigma \in G \mid \sigma w = w\}$$

the **decomposition group** of w. Since L is dense in L_w, each $\sigma \in G$ extends uniquely to a continuous isomorphism $\sigma\colon L_w \xrightarrow{\approx} L_{\sigma w}$. In particular, if $\sigma \in G_w$, σ extends to an automorphism of L_w, and this gives a natural map $G_w \xrightarrow{\approx} \mathrm{Gal}(L_w/K_v)$, which is an isomorphism.

It will be convenient to write L^v for the completion of L with respect to any place w dividing v. They will all be isomorphic to each other by the above, and the corresponding decomposition groups will be conjugate in G. In particular, if G is abelian, the decomposition groups will be all equal and we will be able to identify $\mathrm{Gal}(L^v/K_v)$ with a subgroup of $G = \mathrm{Gal}(L/K)$. In particular, if $L^v \supset K_v$ is unramified, we will be able to view the **Frobenius automorphism** Φ_v of L^v as an element of G.

4.2.1 Idèles

We begin with a

Definition 4.2.1. *The* **idèle group** $\mathbb{I}_K$ *of* K *is the multiplicative subgroup of* $\prod_v K_v^\times$ *given by*

$$\mathbb{I}_K = \{(a_v) \in \prod_v K_v^\times \mid a_v \in U_v \text{ for all but finitely many } v\}$$

where v *runs over all places (both archimedean and non-archimedean) of* K*. We turn* $\mathbb{I}_K$ *into a topological group as follows: as* S *runs over all finite sets of places of* K *containing all archimedean ones, the subsets of the form*

$$\prod_{v \in S} W_v \times \prod_{v \notin S} U_v \subset \mathbb{I}_K, \qquad \text{where } W_v \text{ is an open subset of } K_v^\times,$$

define a basis of neighbourhoods of 1.

The idèle group plays the role of the divisor group $\mathrm{Div}(C)$ of the geometric case. Note however that it is actually **larger** when $K = k(C)$ is the function field of a curve C, since in that case one has a surjective map

$$\begin{aligned} \mathbb{I}_K &\twoheadrightarrow \mathrm{Div}(C) \\ (a_P)_{P \in C_0} &\mapsto \sum_{P \in C_0} v_P(a_P) P \end{aligned}$$

with a nontrivial kernel. Yet $\mathbb{I}_K$ is not too large: it can be shown that $\mathbb{I}_K$ is a locally compact topological group, and we will manage to compute its cohomology without too much effort.
The embeddings $K \hookrightarrow K_v$ allow us to view $K^\times$ as a subgroup of $\mathbb{I}_K$ via the diagonal map $a \mapsto (\ldots, a, a, a, \ldots)$. An element of the image of $K^\times \hookrightarrow \mathbb{I}_K$ is called a **principal idèle**; principal idèles will play the role of principal divisors of the geometric case.

Definition 4.2.2. *The* **idèle class group** C_K *of* K *is defined to be the quotient*

$$C_K = \frac{\mathbb{I}_K}{K^\times}$$

The idèle class group will play the role of the Jacobian (actually $J_C \times \mathbb{Z}$) of the geometric case. It is not a compact group nor a countable union of compact ones (as one would expect from a generalisation of a projective group variety), but it is "almost compact," which will do for our purposes.

Example 4.2.3. *The group* $\mathbb{I}_\mathbb{Q}$ *consists of the tuples* $(a_\infty, a_2, a_3, a_5, \ldots) \in \mathbb{R}^\times \times \mathbb{Q}_2^\times \times \mathbb{Q}_3^\times \times \mathbb{Q}_5^\times \times \cdots$ *with* $a_p \in U_p$ *for all but finitely many primes* p*. Given an idèle* $\alpha = (a_\infty, a_2, a_3, \ldots) \in \mathbb{I}_\mathbb{Q}$*, let*

$$r = (\operatorname{sign} a_\infty) \prod_p p^{v_p(a_p)} \in \mathbb{Q}^\times$$

where $\operatorname{sign} a_\infty = \pm 1$ *depending on whether* a_∞ *is positive or negative. This is is well-defined since* $v_p(a_p) = 0$ *for all but finitely many primes* p*. We have* $\alpha r^{-1} \in \mathbb{R}_{>0} \times U_2 \times U_3 \times \cdots$*. On the other hand, if* $r \in \mathbb{Q}^\times \cap (\mathbb{R}_{>0} \times U_2 \times U_3 \times \cdots)$*, then* $r = 1$*. Hence*

$$C_\mathbb{Q} = \mathbb{I}_\mathbb{Q}/\mathbb{Q}^\times = \mathbb{R}_{>0} \times U_2 \times U_3 \times \cdots$$

Each U_p *is a compact group, but* $\mathbb{R}_{>0}$ *is not (the archimedean place is the "culprit" for the non-compactness of* $C_\mathbb{Q}$*).*

Now let $L \supset K$ be a finite Galois extension of number fields with $G = \mathrm{Gal}(L/K)$. There is an injection $\mathbb{I}_K \hookrightarrow \mathbb{I}_L$ which maps the v-th component a_v to $(a_v, \ldots, a_v) \in L_{w_1}^\times \times \cdots \times L_{w_g}^\times$ where the w_i denote the places of L dividing v. Hence we can view $\mathbb{I}_K$ as a subgroup of $\mathbb{I}_L$. Moreover, the G-action on the w_i induces a G-action on $\mathbb{I}_L$, defined by $\sigma(a_w) = (b_w)$, where $b_{\sigma w} = \sigma a_w$ for $\sigma \in G$ and $(a_w) \in \mathbb{I}_L$. Pictorially,

$$\begin{array}{ll} (a_1, \ldots, a_g) & \in L_{w_1}^\times \times \cdots \times L_{w_g}^\times \\ \Big\downarrow \sigma & \\ (\sigma a_1, \ldots, \sigma a_g) & \in L_{\sigma w_1}^\times \times \cdots \times L_{\sigma w_g}^\times \end{array}$$

Hence $\mathbb{I}_L$ is a G-module, with "induced action" on each component over v. This fact will be quite useful later since it will allow us to apply Shapiro's lemma during the cohomology computations.
If $(a_w) \in \mathbb{I}_L$ is fixed by all $\sigma \in G$, taking $\sigma \in G_w = \mathrm{Gal}(L_w/K_v)$ we have that $a_w \in K_v^\times$ for each $w \mid v$, and hence $\mathbb{I}_L^G = \mathbb{I}_K$. Therefore we may define a **norm map**

$$\begin{aligned} N_{L/K} \colon \mathbb{I}_L &\to \mathbb{I}_K \\ \alpha &\mapsto \prod_{\sigma \in G} \sigma(\alpha) \end{aligned}$$

We can also define a norm map between idèle class groups thanks to

Lemma 4.2.4 (Norm compatibility)**.** *The diagram*

$$\begin{tikzcd} L^\times \arrow[r, hook] \arrow[d, "N_{L/K}"'] & \mathbb{I}_L \arrow[d, "N_{L/K}"] \\ K^\times \arrow[r, hook] & \mathbb{I}_K \end{tikzcd}$$

commutes. Hence $N_{L/K} \colon C_L \to C_K$ *is well-defined.*

Proof. Let $w \overset{\text{def}}{=} w_1, w_2, \ldots, w_g$ be the places dividing a fixed place v of K, and let $\sigma_i \in G \overset{\text{def}}{=} \mathrm{Gal}(L/K)$ be such that $\sigma_i w_i = w$. Since $\{\tau \in G \mid \tau w_i = w\} = G_w \sigma_i$, we have that the w-th component of the norm of $b \in L^\times$, viewed as an element of $\mathbb{I}_L$, is just

$$\prod_{1 \leqslant i \leqslant g} \prod_{\tau \in G_w \sigma_i} \tau(b) = \prod_{\tau \in G} \tau(b) = N_{L/K}(b)$$

□

Definition 4.2.5. *For any global field K and any finite set S of places of K containing all archimedean ones, define the following subgroup of $\mathbb{I}_K$:*

$$\mathbb{I}_K^S \overset{\text{def}}{=} \{(a_v) \in \mathbb{I}_K \mid a_v \in U_v \text{ for } v \notin S\} = \prod_{v \in S} K_v^\times \times \prod_{v \notin S} U_v$$

The next lemma is just an idelic version of the finiteness of the class number.

Lemma 4.2.6 (Idelic Class Group Finiteness)**.** *Let K be a global field. There exists a finite set S of places of K such that*

$$\mathbb{I}_K = K^\times \mathbb{I}_K^S$$

Hence $C_K = \mathbb{I}_K^S / (\mathbb{I}_K^S \cap K^\times)$.

Proof. Take S to be the set of all archimedean places together with enough non-archimedean ones to generate the whole class group of K. Then, given any $(a_v) \in \mathbb{I}_K$, consider the corresponding fractional ideal $\mathfrak{a} = \prod_v \mathfrak{p}_v^{v(a_v)}$, where now v runs over all non-archimedean places of K, and $\mathfrak{p}_v$ denotes the prime ideal of $\mathcal{O}_K$ corresponding to v. By the choice of S, there exists $f \in K^\times$ and $n_v \in \mathbb{Z}$ such that

$$(f) \cdot \mathfrak{a} = \prod_{\substack{v \in S \\ v \text{ non-arch.}}} \mathfrak{p}_v^{n_v}$$

But that amounts to saying that $f \cdot a_v \in U_v$ for $v \notin S$, that is, $f \cdot (a_v) \in \mathbb{I}_K^S$. □

4.2.2 Global reciprocity

In this section, we want to give a description of the abelian extensions of a global field in terms of the idèle class group.

Definition 4.2.7. *Let $L \supset K$ be a finite Galois extension of global fields with abelian $G = \mathrm{Gal}(L/K)$. The* global Artin reciprocity map *$\theta_{L/K}$ is defined as the product of the local reciprocity maps:*

$$\begin{aligned} \theta_{L/K} \colon \mathbb{I}_K &\to G \\ (a_v) &\mapsto \prod_v \theta_{L^v/K_v}(a_v) \end{aligned}$$

Here, if v is archimedean and $L^v \supset K_v$ is isomorphic to $\mathbb{C} \supset \mathbb{R}$, we define $\theta_{L^v/K_v}(a_v)$ to be the conjugation map if $a_v < 0$ and the identity map otherwise, so that we have a trivial reciprocity isomorphism (the "sign map")

$$\theta_{\mathbb{C}/\mathbb{R}} \colon \frac{\mathbb{R}^\times}{N_{\mathbb{C}/\mathbb{R}}\mathbb{C}^\times} = \frac{\mathbb{R}^\times}{\mathbb{R}_{>0}} \xrightarrow{\approx} \mathrm{Gal}(\mathbb{C}/\mathbb{R})$$

Also, observe that for all but finitely many v, the extension $L^v \supset K_v$ is unramified and $a_v \in U_v$, hence $\theta_{L^v/K_v}(a_v) = 1$. Hence the above product does make sense.

We know from the local reciprocity theorem that $N_{L/K}\mathbb{I}_L$ is contained in the kernel of $\theta_{L/K}$. We actually want to show that the principal idèles are also killed by $\theta_{L/K}$ and that

Theorem 4.2.8 (Global Artin Reciprocity)**.** *Let $L \supset K$ be a finite Galois extension of global fields with abelian $G = \mathrm{Gal}(L/K)$. The global Artin map induces an isomorphism*

$$\boxed{\theta_{L/K} \colon \frac{C_K}{N_{L/K}C_L} \xrightarrow{\approx} G}$$

The relation between the Brauer group and the Artin reciprocity map is given by the following

Lemma 4.2.9. *Let $L \supset K$ be a finite abelian Galois extension of global fields with $G = \mathrm{Gal}(L/K)$. Consider the following statements:*

(i) the sequence

$$0 \to \mathrm{Br}(L/K) \to \bigoplus_v \mathrm{Br}(L^v/K_v) \xrightarrow{\sum_v \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z} \to 0$$

is a complex.

(ii) $K^\times \subset \ker \theta_{L/K}$

Then $1 \implies 2$ unconditionally, and $2 \implies 1$ if G is cyclic.

Proof. Let $\chi \in H^1(G, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z})$ and denote by $\delta\chi \in H^2(G, \mathbb{Z})$ the image of χ under the isomorphism given by the connecting map associated to the short exact sequence

$$0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$$

Consider the following diagram:

$$\begin{tikzcd}
K^\times \arrow[r] \arrow[d, "\cup\delta\chi"] & \mathbb{I}_K \arrow[r, "\prod_v \theta_v"] \arrow[d, "\cup\delta\chi"] & \bigoplus_v G_v \arrow[r, "\prod"] \arrow[d, "\sum_v \chi"] & G \arrow[d, "\chi"] \\
\mathrm{Br}(L/K) \arrow[r] & \bigoplus_v \mathrm{Br}(L^v/K_v) \arrow[r, "\sum_v \mathrm{inv}_v"] & \mathbb{Q}/\mathbb{Z} \arrow[r, equal] & \mathbb{Q}/\mathbb{Z}
\end{tikzcd}$$

Cup product with $\delta\chi$ gives a morphism $K^\times/N_{L/K}L^\times = H^0_T(G, L^\times) \to H^2(G, L^\times) = \mathrm{Br}(L/K)$, which is an isomorphism if G is cyclic and χ generates $H^1(G, \mathbb{Q}/\mathbb{Z})$ (the "periodicity isomorphism", see the appendix). Similarly, still denoting by χ its restriction to $L^v \supset K_v$, cup product with $\delta\chi$ gives morphisms $K_v^\times \to \mathrm{Br}(L^v/K_v)$, which induce the second vertical map of the diagram: given $(a_v) \in \mathbb{I}_K$, for all but finitely many v, a_v is a unit and the extension $L^v \supset K_v$ is unramified, hence $a_v \in U_{K_v} = N_{L^v/K_v}U_{L^v}$ and $a_v \cup \delta\chi = 0$ (see **??** and **??**).

Now the whole diagram commutes: the left square, by the compatibility of the restriction maps with the cup product; the middle square, by the formula $\chi(\theta_v(a_v)) = \mathrm{inv}_v(a_v \cup \delta\chi)$ of the local reciprocity law (see 3.4.3); and finally, the right square, by the definition of group morphism.

Now assume (1) holds. To show that $\theta_{L/K}(a) = 1$ for $a \in K^\times$, we have to show that $\chi(\theta_{L/K}(a)) = 0$ for all $\chi \in H^1(G, \mathbb{Q}/\mathbb{Z})$. But from the commutativity of the diagram, $\chi(\theta_{L/K}(a)) = \sum_v \mathrm{inv}_v(a \cup \delta\chi)$, which vanishes since the bottom row is a complex by assumption. Conversely, assume that G is cyclic and (2) holds. Let χ be a generator of $H^1(G, \mathbb{Q}/\mathbb{Z})$. Then the leftmost vertical arrow is surjective, hence any element of $\mathrm{Br}(L/K)$ can be written as $a \cup \delta\chi$ for some $a \in K^\times$. But then again from the diagram we conclude that $\sum_v \mathrm{inv}_v(a \cup \delta\chi) = \chi(\theta_{L/K}(a)) = 0$, and hence the bottom row is a complex. $\square$

In other words, the computation of the Brauer group of a global field is essentially equivalent to proving Artin's reciprocity law! As a first step toward this common goal, let us prove

Theorem 4.2.10 (Reciprocity Law for Cyclotomic Fields)**.** *Let K be a number field and let $L \supset K$ be a subextension of the cyclotomic extension $K(\zeta_n) \supset K$, where ζ_n denotes a primitive n-th root of unity. Then statement (2) of the last lemma holds.*

Proof. First we show that it is enough to prove the theorem for $K = \mathbb{Q}$ and $L = \mathbb{Q}(\zeta_n)$. This follows from the functorial properties of the local reciprocity map, which imply that the following diagram commutes for any abelian extension $L \supset \mathbb{Q}$ (use the fact that the local reciprocity map is given by the cup product with the fundamental class and that the cup product commutes with corestriction):

$$\begin{tikzcd}
\mathbb{I}_K \arrow[r, "\theta_{KL/K}"] \arrow[d, "N_{K/\mathbb{Q}}"] & \mathrm{Gal}(KL/K) \arrow[d, hook] \\
\mathbb{I}_\mathbb{Q} \arrow[r, "\theta_{L/\mathbb{Q}}"] & \mathrm{Gal}(L/\mathbb{Q})
\end{tikzcd}$$

Since the right vertical arrow is injective, if we can show that $\mathbb{Q}^\times \subset \ker\theta_{L/\mathbb{Q}}$ then $\theta_{KL/K}(a)|_L = \theta_{K/\mathbb{Q}}(N_{L/\mathbb{Q}}(a)) = 1 \implies \theta_{KL/K}(a) = 1$ for all $a \in K^\times$. Let $\mathbb{Q}^{\mathrm{mc}} \overset{\mathrm{def}}{=} \bigcup_{n\geqslant 1}\mathbb{Q}(\zeta_n)$ be the maximal cyclotomic extension of $\mathbb{Q}$. We have that

$$\mathrm{Gal}(\mathbb{Q}^{\mathrm{mc}}/\mathbb{Q}) = \varprojlim_{n\in\mathbb{N}} \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) = \varprojlim_{n\in\mathbb{N}} (\mathbb{Z}/n)^\times = \hat{\mathbb{Z}}^\times$$

We now construct a continuous map $\psi\colon \mathbb{I}_\mathbb{Q} \to \mathrm{Gal}(\mathbb{Q}^{\mathrm{mc}}/\mathbb{Q})$, which we secretly know to coincide with the inverse limit $\theta_\mathbb{Q}\colon \mathbb{I}_\mathbb{Q} \to \mathrm{Gal}(\mathbb{Q}^{\mathrm{mc}}/\mathbb{Q})$ of the reciprocity maps $\theta_{\mathbb{Q}(\zeta_n)/\mathbb{Q}}\colon \mathbb{I}_\mathbb{Q} \to \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ (they fit together by the functorial properties of the local reciprocity maps). By construction, we will have that $\mathbb{Q}^\times \subset \ker\psi$, and then we will have to show that $\psi = \theta_\mathbb{Q}$.

The map $\psi\colon \mathbb{I}_{\mathbb{Q}} \to \mathrm{Gal}(\mathbb{Q}^{\mathrm{mc}}/\mathbb{Q})$ will be the inverse limit of $\psi_n\colon \mathbb{I}_{\mathbb{Q}} \to \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$, defined as follows. Let S be a finite set of places containing ∞ and the ramified ones (i.e., those dividing n). For $(a_v) \in \mathbb{I}_{\mathbb{Q}}$, we use the approximation theorem to find $f \in \mathbb{Q}^\times$ such that

$$\|f - a_v\|_v < \min\{\|na_v\|_v, \|a_v\|_v\} \implies \|1 - f^{-1}a_v\|_v < \|n\|_v \qquad \text{for each } v \in S$$

Set

$$\psi_n(a_v) \overset{\text{def}}{=} \prod_{v \notin S} \Phi_v^{v(f^{-1}a_v)} = \prod_{v \notin S} \theta_{\mathbb{Q}_v(\zeta_n)/\mathbb{Q}_v}(f^{-1}a_v)$$

where Φ_v denotes the Frobenius automorphism of $\mathbb{Q}_v(\zeta_n)$, namely the element of $\mathrm{Gal}(\mathbb{Q}_v(\zeta_n)/\mathbb{Q}_v) \subset \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ given by $\Phi_v(\zeta_n) = \zeta_n^p$, where p is the prime corresponding to the place v. In other words, $\psi_n(a_v)$ is the automorphism given by $\zeta_n \mapsto \zeta_n^r$, where $r = \prod_{v \notin S} \|f^{-1}a_v\|_v^{-1} \bmod n$.

The product formula

$$\prod_v \|a\|_v = 1 \qquad a \in \mathbb{Q}^\times$$

ensures that the definition of ψ_n does not depend on the choice of S or f. For instance, if $g \in \mathbb{Q}^\times$ is another element such that $\|g - a_v\|_v < \min\{\|na_v\|_v, \|a_v\|_v\}$ for all $v \in S$, then since $\|f\|_v = \|g\|_v = \|a_v\|_v$ for all $v \in S$ we have that

$$\prod_{v \notin S} \|f^{-1}a_v\|_v^{-1} \equiv \prod_{v \notin S} \|g^{-1}a_v\|_v^{-1} \pmod{n} \iff \prod_{v \in S} \|f^{-1}a_v\|_v \equiv \prod_{v \in S} \|g^{-1}a_v\|_v \pmod{n}$$

holds. It is also easy to check that $\psi_m(a_v) = \psi_n(a_v)|_{\mathbb{Q}(\zeta_m)}$ for $m \mid n$, so that we may define $\psi(a_v) = \varprojlim \psi_n(a_v)$, and another routine check shows that the resulting map $\psi\colon \mathbb{I}_{\mathbb{Q}} \to \mathrm{Gal}(\mathbb{Q}^{\mathrm{mc}}/\mathbb{Q})$ is continuous.

In particular, we get that $\mathbb{Q}^\times \subset \ker\psi$ for free since for a principal idèle $a \in \mathbb{Q}^\times$ we may choose $f = a$. In other words, ψ is defined so as to coincide with the product of the local reciprocity maps at the unramified places, and so that $\mathbb{Q}^\times \subset \ker\psi$: this allow us to replace an arbitrary idèle by an equivalent one which is "very close to 1" at the ramified places, so that the corresponding local reciprocity maps also vanish there. Besides, for any finite subextension $L \supset \mathbb{Q}$ of $\mathbb{Q}^{\mathrm{mc}} \supset \mathbb{Q}$ we also have that $N_{L/\mathbb{Q}}\,\mathbb{I}_L \subset \ker\psi|_L$ (which is no surprise if we believe ψ equals the reciprocity map $\theta_{\mathbb{Q}}$). In fact, for $\beta \in \mathbb{I}_L$, by the approximation theorem we may find $b \in L^\times$ such that $b\beta$ is arbitrarily close to 1 at the archimedean and ramified places of L. Since the norm map is continuous, $N_{L/\mathbb{Q}}(b\beta) = N_{L/\mathbb{Q}}(b) \cdot N_{L/\mathbb{Q}}(\beta)$ will also be arbitrarily close to 1 at the archimedean and ramified places of $\mathbb{Q}$, and since we already know that $N_{L/\mathbb{Q}}(b) \in \mathbb{Q}^\times$ belongs to $\ker\psi$, in order to prove that $N_{L/\mathbb{Q}}(\beta) \in \ker\psi|_L$ it suffices to show that $N_{L/\mathbb{Q}}(b\beta) \in \ker\psi|_L$. But now $\psi\big(N_{L/\mathbb{Q}}(b\beta)\big)|_L$ coincides with the product of the local maps at the unramified local extensions, and the result follows from the corresponding property of the local maps.

For each v, let $\iota_v\colon \mathbb{Q}_v^\times \to \mathbb{I}_K$ be v-th inclusion map given by $\iota_v(a) = (\ldots, 1, 1, a, 1, \ldots)$ which equals the identity map at the v-th component. Let v be a fixed place of $\mathbb{Q}$ and $\mathbb{Q}_v^{\mathrm{mc}} \overset{\text{def}}{=} \bigcup_{n \geqslant 1} \mathbb{Q}_v(\zeta_n)$ be the maximal cyclotomic extension of $\mathbb{Q}_v$. We now show that the composition $\psi_v \overset{\text{def}}{=} \psi \circ \iota_v$ induces a continuous map $\psi_v\colon \mathbb{Q}_v^\times \to \mathrm{Gal}(\mathbb{Q}_v^{\mathrm{mc}}/\mathbb{Q}_v)$ which coincides with the local reciprocity map θ_v. This will prove that $\psi = \theta_{\mathbb{Q}}$, finishing the proof of the theorem.

We now show that the image of ψ_v indeed lies in $\mathrm{Gal}(\mathbb{Q}_v^{\mathrm{mc}}/\mathbb{Q}_v) \subset \mathrm{Gal}(\mathbb{Q}^{\mathrm{mc}}/\mathbb{Q})$. We have to show that, for any finite subextension $L \supset \mathbb{Q}$ of $\mathbb{Q}^{\mathrm{mc}} \supset \mathbb{Q}$ with $G = \mathrm{Gal}(L/\mathbb{Q})$, $\psi_v(a)|_L$ belongs to the decomposition group $G_v = \mathrm{Gal}(L^v/\mathbb{Q}_v)$ of v for all $a \in \mathbb{Q}_v$. Let $M = L^{G_v}$; then $M^v = \mathbb{Q}_v$, and hence $\iota_v(a) \in N_{M/\mathbb{Q}}\mathbb{I}_M$. Thus $\psi_v(a)|_M$ is trivial, that is, $\psi_v(a)|_L \in G_v$, as required.

Next we show that $\psi_v = \theta_v$. First assume that $v \neq \infty$. Let $\mathbb{Q}_v^{\mathrm{nr}} \subset \mathbb{Q}_v^{\mathrm{mc}}$ be the maximal unramified extension of $\mathbb{Q}_v$, and for a uniformiser $\pi \in \mathbb{Q}_v^\times$ of v, denote by $\mathbb{Q}_v^\pi$ the fixed field of $\theta_v(\pi)$. Then $\mathbb{Q}_v^{\mathrm{mc}} = \mathbb{Q}_v^{\mathrm{nr}} \cdot \mathbb{Q}_v^\pi$. Since the uniformisers π generate $\mathbb{Q}_v^\times$, to prove that $\psi_v = \theta_v$ it suffices to show that $\psi_v(\pi)$ and $\theta_v(\pi)$ agree on $\mathbb{Q}_v^{\mathrm{nr}}$ and $\mathbb{Q}_v^\pi$. But both equal the Frobenius automorphism Φ_v on $\mathbb{Q}_v^{\mathrm{nr}}$, so we are left to show that $\psi_v(\pi)$ is trivial on $\mathbb{Q}_v^\pi$, or equivalently, on any finite subextension $M \supset \mathbb{Q}_v$ of $\mathbb{Q}_v^\pi \supset \mathbb{Q}_v$. Let $L \subset \mathbb{Q}^{\mathrm{mc}}$ be the fixed field of $\mathrm{Gal}(\mathbb{Q}_v^{\mathrm{mc}}/\mathbb{Q}_v) \subset \mathrm{Gal}(\mathbb{Q}^{\mathrm{mc}}/\mathbb{Q})$ so that $M = L^v$. We have that $\pi \in N_{L^v/\mathbb{Q}_v}(L^v)^\times$ since $\theta_v(\pi)$ is trivial on L^v, therefore $\iota_v(\pi) \in N_{L/\mathbb{Q}}\mathbb{I}_L$, showing that $\psi_v(\pi)$ is trivial on $M = L^v$ as well.

Now if $v = \infty$, $\theta_v(a)$ is the conjugation map if $a < 0$ and equals the identity otherwise. Since $\mathbb{R}_{>0} = N_{\mathbb{C}/\mathbb{R}}\mathbb{C}^\times$ is contained in $\ker \psi_v$ as well, $\psi_v(a)^2 = \mathrm{id}$ for all $a \in \mathbb{R}^\times$, and so $\psi_v(a)$ must be the conjugation map if it is not trivial. Hence we just have to check that ψ_v is not trivial, and for that we look at its restriction to $\mathbb{Q}(i)$, which ramifies only at 2 and ∞ over $\mathbb{Q}$. Therefore

$$1 = \psi(-7) = \psi_\infty(-7) \cdot \psi_2(-7) \cdot \psi_7(-7)$$

We have that -7 is a 2-adic norm so $\psi_2(-7) = 1$. We have thus to show that $\psi_7(-7)$ is the conjugation map. But $\psi_7(-7)$ is the Frobenius map $i \mapsto i^7 = -i$, so we are done. $\square$

4.2.3 The Brauer group of a global field

The computation of the Brauer group $\mathrm{Br}(K)$ of a global field and the proof of the reciprocity law is a consequence of the following computation, which we shall dub "class field theory axioms."

Theorem 4.2.11 (Class Field Theory Axioms)**.** *Let $L \supset K$ be a Galois extension of global fields with abelian $G = \mathrm{Gal}(L/K)$. Then*

(i) (Cohomology of Idèles) For all $i \geqslant 0$,

$$H_T^i(G, \mathbb{I}_L) = \bigoplus_v H_T^i(G_v, (L^v)^\times)$$

(ii) (Cohomology of C_L) Suppose further that G is cyclic of prime order p, and that $\mu_p \subset K^\times$. Then

$$|H_T^0(G, C_L)| = |G| \qquad \text{and} \qquad |H_T^1(G, C_L)| = 1$$

We postpone the proof of this theorem to the next section. We now derive some important consequences from it. Note that the vanishing of $H^1(G, C_L)$ is the analogue of Lang's theorem for a general global field.

Corollary 4.2.12. *Let $L \supset K$ be a Galois extension of global fields with $G = \mathrm{Gal}(L/K)$. Then*

(i) $H_T^0(G, C_L)$ is finite, and its order divides $|G|$;

(ii) $H^1(G, C_L) = 0$;

(iii) $H^2(G, C_L)$ is finite, and its order divides $|G|$.

Proof. Let us first show that $H^1(G, C_L)$ vanishes. First observe that the class field theory axiom $H^1(G, C_L) = 1$ holds for all cyclic extensions $L \supset K$ of prime degree p without the extra hypothesis $\mu_p \subset K^\times$. In fact, since the degree $[K(\zeta_p) : K]$ divides $p - 1$, it is prime to p. But $H^1(G, C_L)$ is p-torsion (it is killed by $|G|$), hence $\mathrm{res}\colon H^1(G, C_L) \hookrightarrow H^1(\mathrm{Gal}(L(\zeta_p)/K(\zeta_p)), C_{L(\zeta_p)})$ is injective by the restriction-corestriction argument (see appendix). By the class field theory axiom, $H^1(\mathrm{Gal}(L(\zeta_p)/K(\zeta_p)), C_{L(\zeta_p)}) = 1$, and therefore $H^1(G, C_L) = 1$ as well.

For the general case, since $H^1(G, C_L)$ is torsion, it is enough to show that its p-primary part vanishes for all primes p. If $S_p \subset G$ is a p-Sylow subgroup, then by the restriction-corestriction argument, we have that $\mathrm{res}\colon H^1(G, C_L) \hookrightarrow H^1(S_p, C_L)$ is injective on p-primary parts, hence it is enough to consider the case where $G = S_p$ is a p-group, hence solvable. Now we proceed by induction on $|G|$, the base case $|G| = p$ already solved. Let $H \lhd G$ be a non-trivial normal subgroup, and denote by $M = L^H$ the fixed field of L by H. By induction hypothesis, $H^1(H, C_L) = 1$, and hence from the short exact sequence

$$1 \to L^\times \to \mathbb{I}_L \to C_L \to 1$$

we have that

$$1 \to M^\times \to \mathbb{I}_M \to C_L^H \to H^1(H, C_L) = 1$$

is exact and thus $C_L^H = C_M$. By the inflation-restriction sequence (see appendix), we thus obtain an exact sequence

$$0 \to H^1(G/H, C_M) \to \inf H^1(G, C_L) \xrightarrow{\mathrm{res}} H^1(H, C_L)$$

We conclude that the middle term vanishes since the left and right terms vanish by induction hypothesis.

The proof of (1) proceeds along similar lines. Since $H_T^0(G, C_L)$ is torsion, it is enough to show that its p-primary part is finite and divides $|S_p|$ where $S_p \subset G$ denotes once more a p-Sylow subgroup of G. Restriction-corestriction allows us to assume that $G = S_p$, and shows that we may dispense with the roots of unity condition of the class field theory axiom $|H_T^0(G, C_L)| = |G|$ for G cyclic. The only difference is that induction on the order of the p-group G utilises the following observation instead of the inflation-restriction sequence: if $H \lhd G$ then since

$$[C_K : N_{L/K} C_L] = [C_K : N_{M/K} C_M] \cdot [N_{M/K} C_M : N_{L/K} C_L]$$

and $N_{M/K}\colon C_M/N_{L/M}C_L \twoheadrightarrow N_{M/K}C_M/N_{L/K}C_L$ is surjective, we obtain that $|H_T^0(G, C_L)|$ divides $|H_T^0(G/H, C_M)| \cdot |H_T^0(H, C_L)|$.

We also leave the details of the proof of (2) to the reader. We just observe that the base case $|G| = p$ of the induction now follows from $|H^2(G, C_L)| = |H_T^0(G, C_L)| = |G|$ by the periodicity of cohomology of cyclic groups (see appendix), and that the inflation-restriction sequence now reads

$$0 \to H^2(G/H, C_M) \xrightarrow{\inf} H^2(G, C_L) \xrightarrow{\mathrm{res}} H^2(H, C_L)$$

since we have already shown that $H^1(H, C_L) = 1$. □

Theorem 4.2.13. *Let $L \supsetneq K$ be a solvable extension of global fields with $G = \mathrm{Gal}(L/K)$. Then*

(i) there exist infinitely many prime ideals $\mathfrak{p}$ in $\mathcal{O}_K$ that do not split completely in $\mathcal{O}_L$;

(ii) if G is abelian, the Frobenius automorphisms $\Phi_v \in \mathrm{Gal}(L^v/K_v) \subset G$ generate G, as v runs over all valuations that do not belong to a fixed finite set S of places of K containing all archimedean and ramified ones.

Proof. Replacing $L \supset K$ by some cyclic subextension of prime order, we may assume that $|H^0_T(G, C_L)| \geqslant |G|$ by the class field theory axioms. Suppose that the set S of primes that do not split completely in L is finite. Given $\alpha = (a_v) \in \mathbb{I}_K$, using the approximation theorem (c.f. the proof of 4.2.6) we may find $f \in K^\times$ such that $f \cdot a_v \in N_{L^v/K_v}(L^v)^\times$ for all $v \in S$ (norm groups of local fields are open of finite index by local class field theory). But then $f \cdot \alpha \in N_{L/K}\mathbb{I}_L$ since $L^v = K_v$ for all $v \notin S$ by assumption. This proves that $K^\times \cdot N_{L/K}\mathbb{I}_L = \mathbb{I}_K$. But since $\mathbb{I}_K/(K^\times \cdot N_{L/K}\mathbb{I}_L) = C_K/N_{L/K}C_L = H^0_T(G, C_L)$, this cannot happen, proving (1).

To obtain (2), observe that the prime corresponding to a valuation v of K splits completely in L if and only if $\Phi_v = 1$. Hence, applying (1) to the fixed field M of all Φ_v, $v \notin S$, we conclude that $M = K$, that is, the Φ_v, $v \notin S$, generate G, as required. □

Remark 4.2.14. *Observe that the above proof only relies on the fact that $|H^0_T(G, C_L)| \geqslant |G|$. This will be used later in the proof of the so-called second inequality.*

Theorem 4.2.15 (Haße Principle for CSA)**.** *Let $L \supset K$ be a Galois extension of global fields with $G = \mathrm{Gal}(L/K)$. Then*

$$0 \to \mathrm{Br}(L/K) \to \bigoplus_v \mathrm{Br}(L^v/K_v) \to H^2(G, C_L) \to 0$$

is complex that is exact at the first two terms. If G is cyclic, it is also exact at the last term. In particular, taking the limit over all L, we have that

$$0 \to \mathrm{Br}(K) \to \bigoplus_v \mathrm{Br}(K_v)$$

is exact, that is, an element $\alpha \in \mathrm{Br}(K)$ is trivial if and only if it is locally trivial.

Proof. Let $G = \mathrm{Gal}(L/K)$. From the short exact sequence of G-modules

$$1 \to L^\times \to \mathbb{I}_L \to C_L \to 1$$

we obtain an exact sequence

$$0 = H^1(G, C_L) \to H^2(G, L^\times) \to H^2(G, \mathbb{I}_L) \to H^2(G, C_L) \to H^3(G, L^\times)$$

which can be rewritten as

$$0 \to \mathrm{Br}(L/K) \to \bigoplus_v \mathrm{Br}(L^v/K_v) \to H^2(G, C_L) \to H^3(G, L^\times)$$

using the class field axioms. Moreover, if G is cyclic, by Hilbert 90 and the periodicity of cohomology of G, $H^3(G, L^\times) = H^1(G, L^\times) = 0$ as well, which finishes the proof. □

Corollary 4.2.16 (Haße Norm Theorem)**.** *Let $L \supset K$ be a cyclic extension of global fields. Then an element of $K^\times$ is a norm if and only if it is a local norm, that is, $a \in K^\times$ belongs to $N_{L/K}L^\times$ if and only if $a \in N_{L^v/K_v}(L^v)^\times$ for all v.*

Proof. Just use the previous theorem together with the commutative diagram:

$$\begin{tikzcd}
\dfrac{K^\times}{N_{L/K}L^\times} = H^0_T(G, L^\times) \arrow[r, "\cup\delta\chi", "\approx"'] \arrow[d] & H^2(G, L^\times) = \mathrm{Br}(L/K) \arrow[d] \\
\dfrac{K_v^\times}{N_{L^v/K_v}(L^v)^\times} = H^0_T(G, (L^v)^\times) \arrow[r, "\cup\delta\chi", "\approx"'] & H^2(G, (L^v)^\times) = \mathrm{Br}(L^v/K_v)
\end{tikzcd}$$

Here $\chi \in H^1(G, \mathbb{Q}/\mathbb{Z})$ is a generator of this group, and the rows are isomorphisms by the periodicity of the cohomology of the cyclic group G. $\square$

Now we can show

Lemma 4.2.17. *Let K be a number field. Any $\alpha \in \mathrm{Br}(K)$ is split by a cyclic extension L of K contained in some cyclotomic extension $K(\zeta)$ of K.*

Proof. From the commutative diagram with exact top row

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Br}(L^v/K_v) \arrow[r] & \mathrm{Br}(K_v) \arrow[r, "\mathrm{res}"] \arrow[d, "\approx", "\mathrm{inv}"'] & \mathrm{Br}(L^v) \arrow[d, "\approx", "\mathrm{inv}"'] \\
& & \mathbb{Q}/\mathbb{Z} \arrow[r, "{[L^v:K_v]}"] & \mathbb{Q}/\mathbb{Z}
\end{tikzcd}$$

together with the Haße principle for CSA, we reduce the problem to proving the following claim: given an integer n and a finite set of places S of K, there is a cyclic extension $L \supset K$ contained in a cyclotomic extension of K such that n divides the local degrees $[L^v : K_v]$ for all $v \in S$.

Clearly, it suffices to prove the claim for $K = \mathbb{Q}$. Let p be a prime. Since $\mathrm{Gal}(\mathbb{Q}(\zeta_{p^r})/\mathbb{Q}) = (\mathbb{Z}/p^r)^\times$, $\mathbb{Q}(\zeta_{p^r}) \supset \mathbb{Q}$ contains a cyclic subextension $F(p^r) \supset \mathbb{Q}$ of degree p^{r-2}. Now if l is another prime, we have that $[\mathbb{Q}_l(\zeta_{p^r}) : \mathbb{Q}_l]$ equals $\phi(l^r) = l^{r-1}(l-1)$ if $p = l$ (totally ramified case), and equals the smallest positive integer f such that $p^r \mid l^f - 1$ if $p \neq l$ (totally unramified case). In either case, $[F(p^r)_l : \mathbb{Q}_l]$ is a power of p that tends to infinity as r grows.

Now let $n = p_1^{e_1} \dots p_t^{e_t}$ be the factorisation of n into powers of distinct primes p_i. Choose r large enough so that $p_i^{e_i} \mid [F(p_i^r)_l : \mathbb{Q}_l]$ for all $i = 1, 2, \dots, t$ and $l \in S$. We may take $L = F(p_1^r) \dots F(p_t^r)$, which is a cyclic extension of $\mathbb{Q}$ with the required properties. $\square$

Observe that the above theorem shows that $\mathrm{Br}(\mathbb{Q}^{\mathrm{mc}}) = 0$ for the maximal cyclotomic extension $\mathbb{Q}^{\mathrm{mc}}$ of $\mathbb{Q}$, much in the same way that Tsen's theorem asserts that the Brauer group of the function field of a curve over $\mathbb{F}_p^{\mathrm{sep}}$ is trivial. The field $\mathbb{Q}^{\mathrm{mc}}$ (more precisely, a cyclic quotient of it) plays the same role as the $\mathbb{F}_p^{\mathrm{sep}}$ in the geometric case.

We now can finish the proof of the Reciprocity Law:

Proof. (of the Reciprocity Law) Let $L \supset K$ be the cyclic extension contained in a cyclotomic extension of K, as the previous theorem. Then by the explicit computation of 4.2.10 we know that $K^\times \subset \ker \theta_{L/K}$. Hence, by 4.2.9, the sequence

$$0 \to \mathrm{Br}(L/K) \to \bigoplus_v \mathrm{Br}(L^v/K_v) \xrightarrow{\sum_v \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z} \qquad (*)$$

is a complex. But the previous theorem says that $\mathrm{Br}(K)$ is the union of all $\mathrm{Br}(L/K)$ as L runs over all cyclic extension contained in a cyclotomic extension of K, hence we obtain a complex

$$0 \to \mathrm{Br}(K) \to \bigoplus_v \mathrm{Br}(K_v) \xrightarrow{\sum_v \mathrm{inv}_v} \mathbb{Q}/\mathbb{Z} \to 0 \tag{$**$}$$

This, in turn, implies that $(*)$ is a complex for **all** L. Finally, we conclude that $K^\times \subset \ker \theta_{L/K}$ for all L by 4.2.9.

Now suppose that G is abelian, and observe that

$$\theta_{L/K} \colon \mathbb{I}_K \twoheadrightarrow G$$

is surjective since the Frobenius maps Φ_v generate G (4.2.13). But $\ker \theta_{L/K} \supset K^\times \cdot N_{L/K}\mathbb{I}_L$, and since the subgroup $K^\times \cdot N_{L/K}\mathbb{I}_L$ has index at most $|G|$ in $\mathbb{I}_K$ by 4.2.12, it must equal the kernel, and $\theta_{L/K}$ induces an isomorphism

$$\theta_{L/K} \colon \frac{\mathbb{I}_K}{K^\times \cdot N_{L/K}\mathbb{I}_L} = \frac{C_K}{N_{L/K}C_L} \xrightarrow{\approx} G$$

Finally, to show that $(**)$ is exact, it is enough to show that

$$0 \to \mathrm{Br}(L/K) \to \bigoplus_v \mathrm{Br}(L^v/K_v) \xrightarrow{\sum_v \mathrm{inv}_v} \frac{\frac{1}{[L:K]}\mathbb{Z}}{\mathbb{Z}} \to 0 \tag{$\dagger$}$$

is exact for the cyclic extension of the beginning of the proof. Observe that we already know that $(\dagger)$ is a complex and that it is exact at the first (Haße principle) and last term since the Frobenius maps Φ_v generate G, hence $\mathrm{inv}_v(\Phi_v \cup \delta\pi_v) = \frac{1}{f_v} \bmod \mathbb{Z}$ generate the last group: here $f_v = [L^v : K_v]$ is the order of Φ_v and π_v is a uniformiser of v.

Comparison of the exact sequence in 4.2.15 with the last sequence yields a surjective map $H^2(G, C_L) \twoheadrightarrow \frac{\frac{1}{[L:K]}\mathbb{Z}}{\mathbb{Z}}$, and we have to show that this map is an isomorphism. For that, just observe that $H^2(G, C_L)$ is cyclic of order $|G| = [L : K]$ since by the global Artin reciprocity we have isomorphisms

$$G \underset{\approx}{\overset{\theta_{L/K}}{H}}{}^0_T(G, C_L) \xrightarrow[\approx]{\cup\delta\chi} H^2(G, C_L)$$

where $\chi \in H^1(G, \mathbb{Q}/\mathbb{Z})$ is a generator and cup product with $\delta\chi \in H^2(G, \mathbb{Z})$ is the periodicity isomorphism. $\square$

4.3 Proof of the Class Field Theory Axioms

We are left to compute the cohomology groups of $\mathbb{I}_L$ and C_L. The first computation is fairly straightforward, but verifying that $|H^0_T(G, C_L)| = |G|$ and $|H^1_T(G, C_L)| = 1$ is quite delicate, and the proof is broken into two parts, traditionally called "first and second inequalities:" the first inequality is the computation of the Herbrand quotient $h(G, C_L) = |H^0_T(G, C_L)|/|H^1_T(G, C_L)| = |G|$, which already shows that $|H^0_T(G, C_L)| \geqslant |G|$, while the second inequality is the opposite one.

4.3.1 Cohomology of $\mathbb{I}_L$

Recall that for any set S of valuations of K, we write

$$\mathbb{I}_K^S = \prod_{v \in S} K_v^\times \times \prod_{v \notin S} U_v$$

Theorem 4.3.1 (Cohomology of Idèles)**.** *Let $L \supset K$ be a finite Galois extension of global fields with abelian $G = \mathrm{Gal}(L/K)$. Then*

$$H^p(G, \mathbb{I}_L) = \bigoplus_v H^p(G_v, (L^v)^\times)$$

In particular, $H^1(G, \mathbb{I}_L) = 1$ and $H^2(G, \mathbb{I}_L) = \bigoplus_v \frac{\frac{1}{n_v}\mathbb{Z}}{\mathbb{Z}}$ where $n_v = [L^v : K_v]$.

Proof. Let S be a finite set of primes containing all archimedean and ramified ones. Using the fact that cohomology commutes with products,

$$H^p(G, \mathbb{I}_L^S) = \prod_{\substack{w|v \\ v \in S}} H^p(G, L_w^\times) \times \prod_{\substack{w|v \\ v \notin S}} H^p(G, U_w)$$

By Shapiro's lemma,

$$H^p(G, \mathbb{I}_L^S) = \prod_{v \in S} H^p(G_v, (L^v)^\times) \times \prod_{v \notin S} H^p(G_v, U^v)$$

Since every $v \notin S$ is unramified, $H^p(G_v, U^v) = 0$, and therefore

$$H^p(G, \mathbb{I}_L^S) = \prod_{v \in S} H^p(G_v, (L^v)^\times)$$

Taking the injective limit over all finite sets S yields the result. □

From now on we concentrate on the number field case, which is the difficult one. We leave to the reader the task of adapting the proofs to the function field case.

4.3.2 First inequality: Herbrand quotient

Let $L \supset K$ be a cyclic extension of number fields with Galois group $G = \mathrm{Gal}(L/K)$. Our goal is to compute the Herbrand quotient $h(G, C_L)$.

Theorem 4.3.2 (First Inequality)**.** *Let $L \supset K$ be a Galois extension of number fields with cyclic $G = \mathrm{Gal}(L/K)$. Then*

$$h(G, C_L) = [L : K]$$

In particular, $|H_T^0(G, C_L)| \geqslant [L : K]$.

Proof. We invoke 4.2.6 and choose a finite set T of valuations of L which

(i) includes all archimedean and ramified primes;

(ii) $\mathbb{I}_L = L^\times \mathbb{I}_L^T$;

(iii) is G-stable, i.e., $w \in T \implies \sigma(w) \in T$ for all $\sigma \in G$.

and let S be the set of places of K dividing the places of T. We have

$$h(G, C_L) = h\Big(G, \frac{\mathbb{I}_L^T}{\mathbb{I}_L^T \cap L^\times}\Big) = \frac{h(G, \mathbb{I}_L^T)}{h(G, \mathbb{I}_L^T \cap L^\times)}$$

We already know that

$$h(G, \mathbb{I}_L^T) = \frac{|H_T^0(G, \mathbb{I}_L^T)|}{|H_T^1(G, \mathbb{I}_L^T)|} = \prod_{v \in S} \frac{|H_T^0(G_v, L^v)|}{|H_T^1(G_v, L^v)|} = \prod_{v \in S} n_v$$

where $n_v = [L^v : K_v]$. On the other hand, $L^T \overset{\text{def}}{=} \mathbb{I}_L^S \cap L^\times$ is just the group of T-units of L. Denote by

$$V \overset{\text{def}}{=} \operatorname{Hom}(T, \mathbb{R}) = \bigoplus_{w \in T} \mathbb{R} e_w$$

the $|T|$-dimensional real vector space equipped with G-action

$$(\sigma f)(w) = f(\sigma^{-1} w), \qquad f \in V \quad \text{and} \quad \sigma \in G, w \in T$$

Here e_w denotes the dual of the standard basis of $\mathbb{R}^{|T|}$, that is, $e_w(w) = 1$ and $e_w(w') = 0$ if $w' \neq w$. By Dirichlet's unity theorem, the map of G-modules

$$\begin{aligned} \log u \colon L^T &\to V \\ u &\mapsto \sum_{w \in T} \log \|u\|_w \cdot e_w \end{aligned}$$

has a finite kernel $\mu \subset L^\times$ and a G-lattice M^0 of rank $|T| - 1$ as image, contained in the G-invariant hyperplane $\{\sum_{w \in T} x_w e_w \mid \sum_{w \in T} x_w = 0\}$ by the product formula. We have thus exact sequences of G-modules

$$0 \to \mu \to L^T \overset{\log}{\to} M^0 \to 0$$

$$0 \to M^0 \to M \to \mathbb{Z} \to 0$$

where $M = M^0 \oplus \mathbb{Z}\mathbf{v}$ and $\mathbf{v} = (1, 1, \ldots, 1) \in V$. Hence

$$h(G, L^T) = \frac{h(G, M) h(G, \mathbb{Z})}{h(G, \mu)} = h(G, M) \cdot |G|$$

To compute $h(G, M)$, we may replace M with any G-lattice N such that $M \otimes_{\mathbb{Z}} \mathbb{Q} \cong N \otimes_{\mathbb{Z}} \mathbb{Q}$ (as G-modules) since Herbrand quotients of a finite G-modules are trivial. Take $N \overset{\text{def}}{=} \operatorname{Hom}(T, \mathbb{Z}) \subset V$. We have that $M \otimes_{\mathbb{Z}} \mathbb{R} \cong N \otimes_{\mathbb{Z}} \mathbb{R} \implies M \otimes_{\mathbb{Z}} \mathbb{Q} \cong N \otimes_{\mathbb{Z}} \mathbb{Q}$ and

$$h(G, M) = h(G, N) = h(G, \prod_{v \in S} \operatorname{Ind}_{G_v}^G \mathbb{Z}) = \prod_{v \in S} h(G_v, \mathbb{Z}) = \prod_{v \in S} n_v$$

Putting everything together, we obtain the desired result. □

4.3.3 Second inequality

Now we finish the proof of the class field theory axioms by showing

Theorem 4.3.3 (Second Inequality)**.** *Let $L \supset K$ be a Galois extension of number fields with cyclic $G = \operatorname{Gal}(L/K)$ of prime order n. Suppose that $\mu_n \subset K^\times$. Then*

$$|H_T^0(G, C_L)| \leqslant n$$

The key idea of the proof is to use Kummer theory in place of the reciprocity law, and a variant of Haße's norm theorem dealing with n-th powers instead of norms, given by the first lemma below.
By Kummer theory, $L = K(\sqrt[n]{a})$ for some $a \in K^\times$. The second inequality follows from

Theorem 4.3.4. *Let $L \supset K$ be as above, and let S be a sufficiently large finite set of primes of K such that*

- *all archimedean primes are in S;*
- *all primes that ramify in L are in S;*
- *all primes dividing a or n are in S;*
- $\mathbb{I}_K = K^\times \mathbb{I}_K^S$ *Suppose that there is a set T of $|S| - 1$ primes of K, disjoint from S, satisfying*

  (i) $L^v = K_v$ for all $v \in T$.

  (ii) (local power) $K^{S \cup T} \cap D(S, T) = (K^{S \cup T})^n$, where

$$D(S, T) = \prod_{v \in S} (K_v^\times)^n \times \prod_{v \in T} K_v^\times \times \prod_{v \notin S \cup T} U_v$$

  and $K^{S \cup T}$ denotes the group of $(S \cup T)$-units.

Then $|H_T^0(G, C_L)|$ divides n.

Proof. Since $D(S,T) \subset N_{L/K}\mathbb{I}_L$ (use **??**), it is enough to prove that $|\frac{\mathbb{I}_K}{K^\times D(S,T)}|$ divides n. Using the relation for subgroups A, B, C of an abelian group

$$[A : B] = [CA : CB][C \cap A : C \cap B]$$

and condition (ii), we have that

$$\left|\frac{\mathbb{I}_K}{K^\times D(S,T)}\right| = \left|\frac{K^\times \mathbb{I}_K^{S\cup T}}{K^\times D(S,T)}\right| = \frac{\left|\frac{\mathbb{I}_K^{S\cup T}}{D(S,T)}\right|}{\left|\frac{K^\times \cap \mathbb{I}_K^{S\cup T}}{K^\times \cap D(S,T)}\right|} = \frac{\prod_{v\in S}[K_v^\times : (K_v^\times)^n]}{[K^{S\cup T} : (K^{S\cup T})^n]}$$

But by Kummer theory and the structure theorem of the multiplicative group of a local field, we have that $[K_v^\times : (K_v^\times)^n] = n^2/\|n\|_v$ (which also works for archimedean v in view of the assumption $\mu_n \subset K^\times$). On the other hand, since $\mu_n \subset K^\times$, Dirichlet's unit theorem implies that $[K^{S\cup T} : (K^{S\cup T})^n] = n^{|S|+|T|} = n^{2|S|-1}$. Hence

$$\left|\frac{\mathbb{I}_K}{K^\times D(S,T)}\right| = \frac{\prod_{v\in S}\frac{n^2}{\|n\|_v}}{n^{2|S|-1}} = \frac{n^{2|S|}}{n^{2|S|-1}} = n$$

using the product formula. □

We now construct T. First we make the following reduction:

Lemma 4.3.5. *Let $L \supset K$ and S be as above. Suppose T is a set of places of K, disjoint from S, such that the map*

$$K^S \twoheadrightarrow \prod_{v\in T} U_v/U_v^n$$

is surjective. Then $K^{S\cup T} \cap D(S,T) = (K^{S\cup T})^n$.

Proof. Observe that $x \in K$ is an n-th power if and only if the field $M = K(\sqrt[n]{x})$ is equal to K, and by the first inequality (or rather 4.2.13) this happens if and only if

$$H_T^0(\mathrm{Gal}(M/K), C_M) = 1 \iff \mathbb{I}_K = K^\times N_{M/K}\mathbb{I}_M$$

Hence we have to show that if $x \in K^{S\cup T} \cap D(S,T)$ then $\mathbb{I}_K = K^\times N_{M/K}\mathbb{I}_M$, so we need to find many norms. Notice that

$$N_{M/K}\mathbb{I}_M \supset E \overset{\text{def}}{=} \prod_{v\in S} K_v^\times \times \prod_{v\in T} U_v^n \times \prod_{v\notin S\cup T} U_v$$

since $x \in D(S,T)$, hence $x \in (K_v^\times)^n$, and $x \in K^{S\cup T}$, hence M is unramified outside $S \cup T$ and norm maps are surjective by **??**. But now given $\alpha \in \mathbb{I}_K^S$ there exists $f \in K^S$ such that $f\alpha \in E \subset N_{M/K}\mathbb{I}_M$ by assumption, and therefore $\mathbb{I}_K = K^\times \mathbb{I}_K^S = K^\times N_{M/K}\mathbb{I}_M$, as required. □

The next lemma finishes the proof of the second inequality.

Lemma 4.3.6. *With the above notation and hypotheses, there is a set T of $|S| - 1$ primes in K such that*

(i) $L^v = K_v$ *for all* $v \in T$.

(ii) the map $K^S \twoheadrightarrow \prod_{v \in T} U_v/U_v^n$ *is surjective.*

Proof. Let $s = |S|$, and H be the kernel of the map in (2). We claim that it suffices to choose T so that $L = K(\sqrt[n]{H})$ (by the Kummer correspondence). In fact, (1) is clear, while in order to show (2), it suffices to show that K^S/H and $\prod_{v \in T} U_v/U_v^n$ have the same cardinality. But

$$|K^S/H| = \frac{[K^S : (K^S)^n]}{[H : (K^S)^n]} = \frac{n^s}{[L : K]} = n^{s-1}$$

by Dirichlet's unit theorem and Kummer theory, while

$$\prod_{v \in T} |U_v/U_v^n| = \prod_{v \in T} n/\|n\|_v = n^{s-1}$$

by the structure theorem of units of local fields and the product formula.
Now choose representatives $a_1 = a, a_2, \ldots, a_s \in K^S$ of a basis of the s-dimensional $\mathbb{Z}/n$-vector space $K^S/(K^S)^n$. Let

$$M = K(\sqrt[n]{K^S}) \qquad \text{and} \qquad M_i = K(\sqrt[n]{a_1}, \ldots, \sqrt[n]{a_{i-1}}, \sqrt[n]{a_{i+1}} \ldots, \sqrt[n]{a_s})$$

Then by Kummer theory we have that $L = \bigcap_{2 \leqslant i \leqslant s} M_i$, and each $M \supset M_i$ is a cyclic extension of degree n. Hence by 4.2.13 we may choose places w_i of M_i such that their restrictions v_i to K are all distinct and lie outside S, and such that w_i does not split completely in M. Hence M_i is the decomposition field of the unique place of M lying over w_i, and therefore $L \supset K$ is the maximal subextension of $M \supset K$ where all places v_i split completely. Let $T = \{v_2, \ldots, v_s\}$. Now $H = (L^\times)^n \cap K^S$ since, for $x \in K^S$, we have

$$\begin{aligned} x \in (L^\times)^n &\iff K(\sqrt[n]{x}) \subset L \\ &\iff K_{v_i}(\sqrt[n]{x}) = K_{v_i} \text{ for } i = 2, \ldots, s \qquad \text{(since each } v_i \text{ splits completely)} \\ &\iff x \in U_{v_i}^n \text{ for } i = 2, \ldots, s \\ &\iff x \in H \end{aligned}$$

and by Kummer theory we have that $L = K(\sqrt[n]{H})$, as required. $\square$

Chapter 5

Appendix

5.1 Integral Extensions

Definition 5.1.1. *Let $S \supset R$ be an extension of commutative rings. An element $s \in S$ is* **integral** *over R if it satisfies a monic polynomial with coefficients in R:*

$$s^n + r_{n-1}s^{n-1} + \cdots + r_0 = 0, \qquad r_i \in R$$

The extension $S \supset R$ is **integral** *if every element of S is integral over R.*

Integral extensions are to rings what algebraic extensions are to fields. The next lemma explicits this relation, showing how to "clear out" the denominators.

Lemma 5.1.2. *Let R be a domain. Then for any element α in the algebraic closure of the field* $\operatorname{Frac} R$ *there is $r \in R$ such that $r\alpha$ is integral over R.*

Proof. Suppose that $c_n\alpha^n + c_{n-1}\alpha^{n-1} + \cdots + c_0 = 0$ with $c_i \in R$, $c_n \neq 0$. Multiplying by c_n^{n-1} we have $(c_n\alpha)^n + c_{n-1}(c_n\alpha)^{n-1} + \cdots + c_n^{n-1}c_0 = 0$, hence we may take $r = c_n$. □

Definition 5.1.3. *A domain R is called* **normal** *or* **integrally closed** *if every element of* $\operatorname{Frac} R$ *which is integral over R lies in R.*

For instance, $\mathbb{Z}$ is a normal domain. This follows from the more general fact

Theorem 5.1.4. *Every UFD is normal.*

Proof. Let A be a UFD and $K = \operatorname{Frac} A$. Suppose that an element $x/y \in K$, with $x, y \in A$ relatively prime, is integral over A, and let

$$\left(\frac{x}{y}\right)^n + a_{n-1}\left(\frac{x}{y}\right)^{n-1} + a_{n-2}\left(\frac{x}{y}\right)^{n-2} + \cdots + a_0 = 0, \qquad a_i \in A$$

be a monic equation for x/y. Multiplying by y^n we obtain

$$x^n = -a_{n-1}x^{n-1}y - a_{n-2}x^{n-2}y^2 - \cdots - a_0y^n$$

and hence x is a multiple of y. But since x and y are relatively prime, we must have that y is a unit in A and hence $x/y \in A$, as required. □

Integral elements have a very useful intrinsic "polynomial-free" characterisation:

Theorem 5.1.5 (Characterisation of integral elements)**.** *Let $S \supset R$ be an extension of rings and $\alpha \in S$. The following are equivalent:*

(i) α is integral over R.

(ii) α is an element of an R-algebra $A \subset S$ which is finitely generated as an R-module.

Proof. Suppose that α is integral over R, root of a monic polynomial in $R[x]$ of degree n. Then $A = R[\alpha]$ is a finite R-module, generated by $1, \alpha, \ldots, \alpha^{n-1}$ over R, and $\alpha \in A$. Conversely, suppose that α belongs to a ring A which is finitely generated as an R-module, say by ω_i, $1 \leqslant i \leqslant n$. We use the so-called **determinant trick**: for $i = 1, \ldots, n$, we can write

$$\alpha\omega_i = \sum_{1 \leqslant j \leqslant n} m_{ij}\omega_j, \qquad m_{ij} \in R$$

In matrix notation, we can rewrite the previous system as $(\alpha I - M) \cdot \omega = 0$, where $M = (m_{ij})$, $\omega = (\omega_1, \ldots, \omega_n)^T$ and I denotes the $n \times n$ identity matrix. Since this homogeneous linear system in ω has a non-trivial solution, $\det(\alpha I - M) = 0$. Therefore α is a root of the monic polynomial $p(x) = \det(xI - M)$ in $R[x]$, i.e., α is integral over R. □

Using the above characterisation we easily obtain

Theorem 5.1.6. *Let $S \supset R$ be an extension of rings. The elements of S integral over R form a subring of S containing R.*

Proof. Let $\alpha, \beta \in S$ be roots of monic polynomials in $R[X]$ of degrees m and n, respectively. Then $R[\alpha, \beta]$ is a finite R-module generated by $\alpha^i\beta^j$, $0 \leqslant i \leqslant m-1$, $0 \leqslant j \leqslant n-1$. Therefore $\alpha + \beta, \alpha\beta \in R[\alpha, \beta]$ are integral over R. □

The subring of S consisting of all elements which are integral over R is called **integral closure** or **normalisation** of R in S. The name integral closure stems from

Theorem 5.1.7 (Transitivity of integrality)**.** *Let $T \supset S \supset R$ be extensions of rings. If T is integral over S and S is integral over R, then T is integral over R. In particular, if R is a domain and K is a field containing R then the integral closure of R in K is an integrally closed ring.*

Proof. Let $t \in T$ be integral over S with $t^n + s_{n-1}t^{n-1} + \cdots + s_0 = 0$, $s_i \in S$. Since each s_i is integral over R, $R[s_0, \ldots, s_{n-1}]$ is a finite R-module, hence so is $A = R[t, s_0, \ldots, s_{n-1}]$, and $t \in A$. Thus t is integral over R. □

Theorem 5.1.8 (Norms and Traces of integral elements)**.** *Let R be a domain and L be a finite separable field extension of $K = \operatorname{Frac} R$. Let S be the integral closure of R in L. If R is normal, then $T_{L/K}(s)$ and $N_{L/K}(s)$ are elements of R for all $s \in S$.*

Proof. Let $n = [L : K]$ and $\sigma_1, \dots, \sigma_n \colon L \to K_{sep}$ be the K-embeddings of L into K_{sep}. Since s is the root of a monic polynomial in $R[X]$, so are $\sigma_i(s)$ for all i. Hence $\sigma_i(s)$ are integral over R, and thus so are the elements $N_{L/K}(s) = \prod \sigma_i(s)$ and $T_{L/K}(s) = \sum \sigma_i(s)$ of $K = \operatorname{Frac} R$. Since R is normal, they must lie in R. □

Example 5.1.9. *The sequence of* **Fibonacci numbers** *is defined by $F_0 = 0$, $F_1 = 1$, and $F_{k+2} = F_{k+1} + F_k$ for $k \geqslant 0$. The first terms are thus $0, 1, 1, 2, 3, 5, 8, 13, \dots$ It is easy to show by induction that*

$$F_k = \frac{\alpha^k - \beta^k}{\alpha - \beta}, \qquad \text{where} \quad \alpha = \frac{1+\sqrt{5}}{2} \quad \text{and} \quad \beta = \frac{1-\sqrt{5}}{2}$$

Notice that α and β are integral over $\mathbb{Z}$.
We now prove that if $m \mid n$ then $F_m \mid F_n$. Write $n = dm$ with $d \in \mathbb{Z}$. We just perform the division:

$$\frac{F_n}{F_m} = \alpha^{(d-1)m} + \alpha^{(d-2)m}\beta^m + \cdots + \alpha^m \beta^{(d-2)m} + \beta^{(d-1)m}$$

This shows that $F_n/F_m \in \mathbb{Q}$ is integral over the normal domain $\mathbb{Z}$, hence $F_n/F_m \in \mathbb{Z}$ and we are done.

Example 5.1.10. *Let d be a square-free integer. Let $K = \mathbb{Q}(\sqrt{d})$ and denote by A the integral closure of $\mathbb{Z}$ in K. We have the following explicit description of A:*

$$A = \mathbb{Z} + \mathbb{Z}\omega, \qquad \text{where} \quad \omega = \begin{cases} \sqrt{d} & \text{if } d \equiv 2 \text{ or } d \equiv 3 \pmod 4 \\ \frac{1+\sqrt{d}}{2} & \text{if } d \equiv 1 \pmod 4 \end{cases}$$

Let us prove the case $d \equiv 3 \pmod 4$; the other ones are similar and are left to the reader. First, it is clear that $\mathbb{Z} + \mathbb{Z}\sqrt{d} \subset A$. Now let $\alpha \in A$ and write $\alpha = r + s\sqrt{d}$, $r, s \in \mathbb{Q}$; we need to show that $r, s \in \mathbb{Z}$. Observe that $T_{K/\mathbb{Q}}(\alpha) = 2r$ and $T_{K/\mathbb{Q}}(\alpha\sqrt{d}) = 2ds$ are integers, hence we may write $r = m/2$ and $s = n/2d$ for some $m, n \in \mathbb{Z}$. But $N_{K/\mathbb{Q}}(\alpha) = r^2 - ds^2 = (m^2 d - n^2)/4d$ is also an integer. Since $d \equiv 3 \pmod 4$, both m and n must be even, hence $r \in \mathbb{Z}$ and $ds^2 \in \mathbb{Z}$. Using the fact that d is square-free, we conclude that $s \in \mathbb{Z}$ as well.

5.2 Valuations

Definition 5.2.1. *An* **abelian ordered group** *$(G, +, \leqslant)$ is an abelian group $(G, +)$ together with a partial order $\leqslant$ such that $a \leqslant b \implies a + g \leqslant b + g$ for all $a, b, g \in G$. Examples of such groups are the additive groups $\mathbb{Z}$ and $\mathbb{R}$ with the usual order.*

Definition 5.2.2. *Let $(G, +, \leqslant)$ be an ordered group and ∞ be a formal symbol satisfying $g + \infty = \infty$ and $g \leqslant \infty$ for all $g \in G \cup \{\infty\}$. A* **valuation** *v on a field K with values in G is a map $v \colon K \to G \cup \{\infty\}$ such that*

(i) $v(xy) = v(x) + v(y)$;

(ii) $v(x+y) \geqslant \min\{v(x), v(y)\}$;

(iii) $v(x) = \infty \iff x = 0$.

If v is identically zero on $K^\times$, we say that v is **trivial**. *If G is a discrete subgroup of $\mathbb{R}$ (for example $\mathbb{Z}$), we say that v is a* **discrete valuation**. *We say that a discrete valuation v is* **normalised** *if $G = \mathbb{Z}$ and v is surjective.*

Valuations are intimately connected with prime factorisations in UFDs, as in the following

Example 5.2.3. *Let p be a prime number. Given any nonzero rational number r, we may write it as $r = p^n \cdot \frac{a}{b}$ with $a, b \in \mathbb{Z}$ both not divisible by p. Setting $v_p(r) = n$, we obtain a normalised discrete valuation on $\mathbb{Q}$.*
Similarly let k be any field and let $p(x) \in k[x]$ be an irreducible polynomial. Then we may write any nonzero $r(x) \in k(x)$ as $r(x) = p(x)^n \cdot \frac{f(x)}{g(x)}$ with $f(x), g(x) \in k[x]$ not divisible by $p(x)$. Setting $v_{p(x)}\big(r(x)\big) = n$ also gives a normalised discrete valuation on $k(x)$.

Let v be a valuation on a field K. Setting

$$|x|_v \stackrel{\text{def}}{=} 2^{-v(x)}$$

(where 2 denotes your favourite real number bigger than 1 and $2^{-\infty} = 0$ by definition) we obtain analogous "multiplicative" properties

(i) $|xy|_v = |x|_v \cdot |y|_v$;

(ii) $|x+y|_v \leqslant \max\{|x|_v, |y|_v\}$;

(iii) $|x|_v = 0 \iff x = 0$.

turning K into a **normed metric space**. The presence of a valuation thus allows us to employ techniques and ideas from Analysis and Topology. A word of caution: as a consequence of the "strong triangle inequality" $|x+y|_v \leqslant \max\{|x|_v, |y|_v\}$, some things behave a little differently from the real or complex world. For instance, one has the **super strong triangle inequality**

$$|x|_v \neq |y|_v \implies |x+y|_v = \max\{|x|_v, |y|_v\}$$

In fact, using 1 one can easily show that $|-y|_v = |y|_v$ for all y. Thus if $|x|_v > |y|_v$, say, then by 2 one has $|x|_v \leqslant \max\{|x+y|_v, |-y|_v\} \implies |x|_v \leqslant |x+y|_v$ while $|x+y|_v \leqslant |x|_v$, again by 2, proving the super strong triangle inequality. In particular amongst $|x|_v$, $|y|_v$ and $|x-y|_v$ there are always two equal values, showing that in a discretely valued field all triangles are isosceles!
A concept from Topology that will be quite useful to us is that of a complete metric space. We say that K is **complete** with respect to $|-|_v$ (or v) if every **Cauchy sequence** converges: given a sequence $\{x_n\}_{n\geqslant 0}$ in K such that

$$\forall \epsilon > 0 \quad \exists n_0 = n_0(\epsilon) \quad \text{such that} \quad n, m \geqslant n_0 \implies |x_n - x_m|_v < \epsilon$$

then $\lim_{n\to\infty} x_n$ exists in K, namely there is an element $x_\infty \in K$ such that

$$\forall \epsilon > 0 \quad \exists n_0 = n_0(\epsilon) \quad \text{such that} \quad n \geqslant n_0 \implies |x_n - x_\infty|_v < \epsilon$$

Definition 5.2.4. *Two valuations v and w on a field K are* **equivalent** *if they define the same topology on K.*

More explicitly, we have the following

Theorem 5.2.5 (Equivalence of valuations)**.** *Two valuations v and w on a field K are equivalent if and only if there exists $c > 0$ such that $v(x) = c \cdot w(x)$ for all $x \in K$.*

Proof. Clearly if v is a multiple of w then they define the same topology on K. Conversely, if v and w define the same topology then $w(x) > 0 \implies v(x) > 0$ since both conditions express the fact that $x^n \to 0$ when $n \to \infty$. We may assume that both valuations are nontrivial and therefore there exists $\pi \in K^\times$ such that $w(\pi) > 0$. Set $c = \frac{v(\pi)}{w(\pi)}$. Then $v(x) = c \cdot w(x)$ whenever x is a power of π. In general, for an arbitrary element $x \in K^\times$, we "approximate" x by a power of π, taking a rational approximation $\frac{n}{m}$ of $\frac{w(x)}{w(\pi)}$ such that $\frac{n}{m} < \frac{w(x)}{w(\pi)} \leqslant \frac{n+1}{m}$ where m, n are integers and $m > 0$. Then $m \cdot w(x) > n \cdot w(\pi) \iff w(\frac{x^m}{\pi^n}) > 0$ and therefore

$$v\Big(\frac{x^m}{\pi^n}\Big) > 0 \implies v(x) > \frac{n}{m} \cdot v(\pi) = \frac{n}{m} \cdot \big(c \cdot w(\pi)\big) \geqslant \frac{n}{n+1} \cdot \big(c \cdot w(x)\big)$$

Since we may take n arbitrarily large, we have that $v(x) \geqslant c \cdot w(x)$. Replacing x by x^{-1} we obtain the opposite inequality $v(x) \leqslant c \cdot w(x)$, finishing the proof. $\square$

The above proof shows that if $w(x) > 0 \implies v(x) > 0$ holds for all $x \in K$ then the two valuations are actually equivalent. In other words we have

Lemma 5.2.6 (Independency of Inequivalent Valuations)**.** *Let v and w be inequivalent valuations on a field K. Then there exists $x \in K$ such that $v(x) > 0$ and $w(x) \leqslant 0$.*

Every valuation v on a field K defines a subring

$$O_v \overset{\text{def}}{=} \{x \in K \mid v(x) \geqslant 0\} = \{x \in K \mid |x|_v \leqslant 1\}$$

called **valuation ring** of v. It is a local ring with maximal ideal

$$\mathfrak{m}_v \overset{\text{def}}{=} \{x \in O_K \mid v(x) > 0\} = \{x \in O_K \mid |x|_v < 1\}$$

and unit group

$$O_v^\times = \{x \in O_K \mid v(x) = 0\} = \{x \in O_K \mid |x|_v = 1\}$$

Observe that in O_v we have that $a \mid b \iff v(a) \leqslant v(b)$.

Theorem 5.2.7 (Discrete valuation rings)**.** *Let R be a domain with field of fractions $K = \operatorname{Frac} R$. Then the following conditions are equivalent*

(i) R is the valuation ring of a discrete valuation on K.

(ii) R is a local PID different from K;

(iii) R is a noetherian normal local ring with Krull dimension 1 (that is, the prime ideals of R are just (0) and its maximal ideal);

(iv) R is a UFD with a single irreducible element π up to units.

Any R satisfying the above conditions is called a **discrete valuation ring** *or* **dvr** *for short. Any generator of the maximal ideal of R is called a* **uniformiser**.

Proof. To show that $1 \implies 2$, observe that if v is a normalised valuation on K such that $R = O_v$ then given any ideal I of R with $n = \min\{v(r) \mid r \in I\}$ then I can be generated by any $r \in R$ such that $v(r) = n$.

Clearly $2 \implies 4 \implies 3$. We also have $4 \implies 1$ since we may define a valuation v on K by setting $v(a) = n$ where $a = u \cdot \pi^n$, $u \in R^\times$.

Finally, we show that $3 \implies 2$. Let $\mathfrak{m}$ be the maximal ideal of R. First observe that any nonzero ideal of R contains some power of $\mathfrak{m}$. This follows by noetherian induction: if there is a non-zero ideal I that does not contain any power of $\mathfrak{m}$, then we may choose I maximal with this property. Since R has Krull dimension 1 and $I \neq \mathfrak{m}$ and $I \neq (0)$, we have that I is not prime and hence there exist $a, b \notin I$ such that $ab \in I$. Then both $I + (a)$ and $I + (b)$ properly contain I, hence $I + (a) \supset \mathfrak{m}^i$ and $I + (a) \supset \mathfrak{m}^j$ for some i and j and therefore $I \supset \big(I + (a)\big) \cdot \big(I + (b)\big) \supset \mathfrak{m}^{i+j}$, a contradiction.

Next consider the R-module

$$\mathfrak{m}^{-1} \stackrel{\text{def}}{=} \{a \in K \mid a \cdot m \in R \text{ for all } m \in \mathfrak{m}\}$$

Let $a \in \mathfrak{m}$ be any nonzero element and let i be the smallest positive integer such that $(a) \supset \mathfrak{m}^i$. If $b \in \mathfrak{m}^{i-1} - (a)$ then $b \cdot \mathfrak{m} \subset (a)$ but $b \notin (a)$, i.e., $\frac{b}{a} \in \mathfrak{m}^{-1}$ but $\frac{b}{a} \notin R$, showing that $\mathfrak{m}^{-1} \not\subset R$. This implies that $\mathfrak{m} \cdot \mathfrak{m}^{-1} = R$. In fact, suppose not. Then $\mathfrak{m} \cdot \mathfrak{m}^{-1} \subset \mathfrak{m}$ and if $a \in \mathfrak{m}^{-1} - R$ then a is integral over R by the determinant trick, hence $a \in R$ since R is normal, a contradiction.

Now we show that for any nonzero ideal I we have that $I = \mathfrak{m}^n$ for a uniquely determined n. Uniqueness follows immediately from $\mathfrak{m} \cdot \mathfrak{m}^{-1} = R$ by multiplying two distinct representations by some power of $\mathfrak{m}^{-1}$. To prove existence, we use noetherian induction: let I be a nonzero ideal which is maximal among those that cannot be written as a power of $\mathfrak{m}$. Then $I \subset \mathfrak{m}$ and hence $\mathfrak{m}^{-1} \cdot I$ is an ideal of R which properly contains I since multiplying $I = \mathfrak{m}^{-1} \cdot I$ by $\mathfrak{m}$ and using $\mathfrak{m} \cdot \mathfrak{m}^{-1} = R$ we conclude that $I \cdot \mathfrak{m} = I$ and hence $I = (0)$ by Nakayama's lemma. Therefore $\mathfrak{m}^{-1} \cdot I = \mathfrak{m}^n$ for some n and hence $I = \mathfrak{m}^{n+1}$.

By Nakayama's lemma $\mathfrak{m} \neq \mathfrak{m}^2$. Let $\pi \in \mathfrak{m} - \mathfrak{m}^2$. Since (π) is a power of $\mathfrak{m}$, the only possibility is $\mathfrak{m} = (\pi)$. Hence $\mathfrak{m}$ is principal, and therefore so are all ideals of R. Therefore $R \neq K$ is a local PID. $\square$

5.3 Limits

5.3.1 Direct Limits

Definition 5.3.1. *A partially ordered set $(I, \geqslant)$ is said to be* **directed** *if given any pair $i, j \in I$ there exists $k \in I$ such that $k \geqslant i$ and $k \geqslant j$.*

Definition 5.3.2. *Let $(I, \geqslant)$ be a directed set. A* **direct system** *of rings (or groups, topological spaces, and so on) is a family of rings $(R_i)_{i \in I}$ (or groups, topological spaces, and so on), together with a family of morphisms of rings (or of groups, topological spaces, and so on) $\phi_{ij} \colon R_i \to R_j$, $i \leqslant j$, such that*

(i) $\phi_{ii} = id_{R_i}$ *for all* $i \in I$*;*

(ii) $\phi_{ik} = \phi_{jk} \circ \phi_{ij}$ *for any triple* $i \leqslant j \leqslant k$ *in* I*.*

Given a direct system (R_i, ϕ_{ij}), we may consider its **direct limit**, which consists of a ring (or group...)

$$R = \varinjlim_{i \in I} R_i$$

together with morphisms $\phi_i \colon R_i \to R$ which are compatible with the ϕ_{ij} in the sense that for all $i \leqslant j$ the diagram

$$\begin{tikzcd} R_j \arrow[r, "\phi_j"] & R \\ R_i \arrow[u, "\phi_{ij}"] \arrow[ur, "\phi_i"'] & \end{tikzcd}$$

commutes. Intuitively R can be viewed as a "generalised union" of the R_i, with "generalised inclusion maps" $\phi_i \colon R_i \to R$ (which are not necessarily injective). The direct limit is characterised by the following universal property, which distinguishes it as the "smallest" ring "containing" all the R_i: given a test ring T and maps $f_i \colon R_i \to T$ which are compatible with the ϕ_{ij}, namely $f_j \circ \phi_{ij} = f_i$ for all $i \leqslant j$, then there exists a unique $f \colon R \to T$ such that $f_i = f \circ \phi_i$ for all i.

We may construct the direct limit by setting

$$R = \frac{\coprod_{i \in I} R_i}{\sim}$$

the disjoint union of the R_i modulo the equivalence relation $\sim$ which identifies $r_i \in R_i$ and $r_j \in R_j$ if there exists $k \in I$ with $k \geqslant i$ and $k \geqslant i$ such that $\phi_{ik}(r_i) = \phi_{jk}(r_j) \in R_k$. One may perform addition and multiplication of the classes of $r_i \in R_i$ and $r_j \in R_j$ by replacing them by representatives in a common ring R_k where $k \geqslant i$ and $k \geqslant j$ (such k exists since the index set I is directed). It is easy to check that everything is well-defined. The maps $\phi_i \colon R_i \to R$ are the natural ones.

As a concrete example, let k be a field and R be a k-algebra. Consider the directed set given by the finite subsets of R, ordered by inclusion. Given a finite set S in this system, let R_S be the k-subalgebra of R generated by the elements of S. If $S \subset S'$, let $\phi_{SS'} \colon R_S \hookrightarrow R_{S'}$ be inclusion map. Then one has

$$R = \varinjlim_{S} R_S$$

5.3.2 Projective Limits

Let $(I, \leqslant)$ be a directed set. A **projective system** of groups (or rings, topological spaces, and so on) is a family of groups $(G_i)_{i \in I}$ and maps $\phi_{ji} \colon G_j \to G_i$, $i \leqslant j$, such that

(i) $\phi_{ii} = \mathrm{id}_{G_i}$ for all $i \in I$;

(ii) $\phi_{ki} = \phi_{ji} \circ \phi_{kj}$ for any triple $i \leqslant j \leqslant k$ in I.

The **projective limit** of the above projective system is a group (ring, topological space, ...)

$$G = \varprojlim_{i \in I} G_i$$

together with "projection" maps $\phi_i \colon G \to G_i$ such that

$$\begin{tikzcd} G \arrow[r, "\phi_j"] \arrow[dr, "\phi_i"'] & G_j \arrow[d, "\phi_{ji}"] \\ & G_i \end{tikzcd}$$

commutes for all $i \leqslant j$. The projective limit is characterised by the following universal property: given a test group T and morphisms $g_i \colon T \to G_i$ such that $g_i = \phi_{ji} \circ g_j$ for all $i \leqslant j$, then there is a unique $g \colon T \to G$ such that $g_i = \phi_i \circ g$ for all $i \in I$.
We can construct G as the subgroup of the product $\prod_{i \in I} G_i$ consisting of "coherent tuples", namely

$$G = \left\{ (\sigma_i) \in \prod_{i \in I} G_i \;\middle|\; \phi_{ji}(\sigma_j) = \sigma_i \text{ for all } i \leqslant j \right\}$$

with $\phi_i \colon G \to G_i$ given the i-th projection map.
Projective limits usually arise by "stacking" quotient maps. As a concrete example, consider the directed set of finite Galois extensions l of a field k in some separable closure k_{sep}, ordered by inclusion. Set $G_l = \mathrm{Gal}(l/k)$ and let $\phi_{ml} \colon G_m \twoheadrightarrow G_l$ be the natural quotient (or projection) maps for all $m \supset l \supset k$ in this directed set. Then

$$G = \varprojlim_{l} G_l$$

is just the absolute Galois group $\mathrm{Gal}(k_{sep}/k)$ of k, since to give an automorphism $\sigma \in \mathrm{Gal}(k_{sep}/k)$ is the same as to give a family $\sigma_l \in \mathrm{Gal}(l/k)$ of compatible automorphisms for all finite Galois extensions l of k.

5.4 Central simple algebras and Brauer group

We recall without proof some basic facts about central simple algebras, referring the reader to Gille-Szamuely's book for proofs.
Let K be any field. A **central simple algebra** A over K is a finite dimensional associative K-algebra which satisfies the following equivalent conditions:

(i) A has no trivial two sided ideals and the centre of A is K;

(ii) $A \cong M_n(D)$ for some n and some division algebra D with centre K (here $M_n(D)$ denotes the ring of $n \times n$ matrices with entries in D, and D is uniquely determined by A up to isomorphism);

(iii) there exists a finite Galois extension $L \supset K$ that "splits" A, i.e., $A \otimes_K L \cong M_n(L)$ for some n.

We say that two central simple algebras A and B over K are **Brauer equivalent** if there is an isomorphism $A \otimes_K M_m(K) \cong B \otimes_K M_n(K)$ for some m and n. The **Brauer group** $\mathrm{Br}(K)$ of K is the group consisting of all the (Brauer) equivalence classes $[A]$ of central simple algebras A over K. The product in $\mathrm{Br}(K)$ is induced by the tensor product: $[A] \cdot [B] = [A \otimes_K B]$. One verifies that this is well-defined, and turns $\mathrm{Br}(K)$ into a torsion abelian group. The identity element of $\mathrm{Br}(K)$ is $[K]$ and $[A]^{-1} = [A^{op}]$, where A^{op} is the opposite algebra of A, defined by inverting the order of the multiplication in A.

By 2 of the above definition, we have that the elements of the Brauer group of K are in 1-1 correspondence with the isomorphism classes of division algebras over K. In other words, $\mathrm{Br}(K)$ is a "directory" of all division algebras over K. One can also view $\mathrm{Br}(K)$ as a measure of the arithmetic complexity of K (i.e., of its absolute Galois group) due to the following important

Theorem 5.4.1. *For any field K, one has an isomorphism*

$$\mathrm{Br}(K) = H^2(G_K, K_{sep}^\times)$$

Furthermore, if $L \supset K$ is an arbitrary field extension, restriction $\mathrm{res}\colon H^2(G_K, K_{sep}^\times) \to H^2(G_L, L_{sep}^\times)$ *in cohomology corresponds to restriction* $\mathrm{Br}(K) \to \mathrm{Br}(L)$ *of Brauer groups, defined by* $[A] \mapsto [A \otimes_K L]$.

The above isomorphism can be described as follows. Let L be a finite Galois extension of K and set $G = \mathrm{Gal}(L/K)$. Given a 2-cocycle $f\colon G \times G \to L^\times$ representing an element of $H^2(G, L^\times)$ we can build a central simple algebra A_f (the so-called **crossed product**) which, as a vector space over L, is given by

$$A_f = \bigoplus_{\sigma \in G} L \cdot e_\sigma$$

Multiplication on A_f is defined by

$$(a_\sigma e_\sigma) \cdot (b_\tau e_\tau) = a_\sigma \sigma(b_\tau) f(\sigma, \tau) \cdot e_{\sigma\tau} \qquad \text{for} \quad a_\sigma, b_\tau \in L^\times, \quad \sigma, \tau \in G$$

One obtains a map $H^2(G, L^\times) \to \mathrm{Br}(K)$ given by $[f] \mapsto [A_f]$. Taking the direct limit over all L we obtain the desired isomorphism.

The "simplest" central simple algebras are the cyclic ones:

Definition 5.4.2. *A central simple algebra A over K is* **cyclic** *if there exists a cyclic extension $L \supset K$ that splits A (as in definition 3).*

Now we are ready to show the main result about central simple algebras over local fields:

Theorem 5.4.3. *Every central simple algebra A over a local field K is cyclic.*

Proof. Recall that we have a commutative diagram (see **??** and **??**)

$$\begin{tikzcd}
\mathrm{Br}(L) \arrow[r, "\mathrm{inv}_L", "\approx"'] & \mathbb{Q}/\mathbb{Z} \\
\mathrm{Br}(K) \arrow[u, "\mathrm{res}"] \arrow[r, "\mathrm{inv}_K", "\approx"'] & \mathbb{Q}/\mathbb{Z} \arrow[u, "{[L:K]}"']
\end{tikzcd}$$

Hence if $[A]$ has order n in $\mathrm{Br}(K)$ then **any** cyclic extension of degree n will split A, for instance the unramified extension of degree n. $\square$

As we saw, the Brauer group of a local field played a prominent role in the proof of the local reciprocity theorem. It also plays a central role in the proof of the **global** reciprocity theorem. We just state the important

Theorem 5.4.4 (Brauer group of global fields)**.** *Let K be a global field. One has an exact sequence*

$$0 \to \mathrm{Br}(K) \xrightarrow{\sum_v \mathrm{res}_v} \bigoplus_v \mathrm{Br}(K_v) \xrightarrow{\sum_v \mathrm{inv}_{K_v}} \mathbb{Q}/\mathbb{Z} \to 0$$

Here K_v denotes the completion of K with respect to the valuation v, $\mathrm{res}_v\colon \mathrm{Br}(K) \to \mathrm{Br}(K_v)$ and $\mathrm{inv}_{K_v}\colon \mathrm{Br}(K_v) \to \mathbb{Q}/\mathbb{Z}$ are the usual restriction and invariant maps, and the direct sum runs over all valuations of K, including the archimedean ones.

The proof of the above theorem is the main step of the proof of the global reciprocity law. In terms of central simple algebras, it describes the **local-global** or **Haße principle** for division algebras: a division algebra D over a global field K is completely determined by its restrictions $D \otimes_K K_v$ to the local fields K_v. In particular, D is trivial if and only if $D \otimes_K K_v$ is trivial for all v.

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